

Replacement Engine Control Unit

Complete documentation: the machine, the circuits, the specification, and the research all of it rests on.

This replaces the **Engine Control Unit** block on page 1 of the Kirloskar troubleshooting wiring diagram, manual 08-GP3-60-001. Every pin, signal and colour in these pages was read out of that document or measured in simulation; where something is inferred rather than confirmed, it says so on the line.

The engine has not been inspected yet. Of twelve items on the unknowns register, two are answered by inference and none is confirmed — every confirmation needs the machine. The CPCB IV+ certification position in Memo 06 is a decision for the machine's owners, not an engineering conclusion. **Read that memo before fitting anything.**

CONTENTS

1. Field Handbook

The machine, its interface, the circuits in narrative, and how to run it

2. Sensor Reference

Every sensor: what it measures, how we read it, what we do with it

3. Circuit Reference

All 17 circuit blocks: schematic, reasoning, simulated result, limits

4. Reconciled Specification

Binding source of truth -- pin map, power, deviations, unknowns

5. Memo 01 -- Engine Identity

Which engine this actually is, established from five hardware constraints

6. Memo 02 -- KG640C Controller

What the existing genset controller does, and what it needs from us

7. Memo 03 -- ECU Connector

The 94-pin connector: candidates, sourcing, and what stays unconfirmed

8. Memo 04 -- Injector Drive

Bank sharing proved safe, and the drive envelope it has to hit

9. Memo 05 -- MCU Selection

Why the S32K148, and the timing argument that actually binds

10. Memo 06 -- Standards & Law

ISO, CISPR and the CPCB IV+ certification position

11. Memo 07 -- BOM & Sourcing

Costed bill of materials, India sourcing, single-source flags

12. Memo 08 -- Engine Inspection Brief

The measurements to take at the machine, grouped by machine state

13. Memo 09 -- Resistive Sensor Front-Ends

How DSE and SEDEMAC build temperature inputs, and what transfers

14. Phase 1 Research Plan

The plan the eight memos were written against

Genset ECU Field Handbook

Everything known about replacing the engine control unit on a 25 kVA diesel genset — the machine, the 94-pin interface, every circuit with its simulated result, and the two risks that decide whether this ever runs.

Rev 1 13 Sep 2026 17 circuits verified 10 open unknowns Source: 08-GP3-60-001

[Circuit reference](#) → [Sensor Reference](#) →

01	The machine	02	The 94-pin interface
03	The circuits	04	Stopping a runaway
05	The two real risks	06	What we don't know
07	Running the simulations		

01 The machine

The engine is a **Kirloskar 3GK550ETA 4SR1** — three cylinders, turbocharged and aftercooled, common-rail injection, cooled EGR with a diesel oxidation catalyst and no SCR. **26.5 kW / 36 hp at 1500 rpm**, governing at constant speed to make 50 Hz. Application code GK3.8703, built 12 September 2025.

That constant speed matters more than it first appears, and it comes up twice in this handbook: once because it makes the timing problem easier than an automotive engine's, and once because the genset's own output frequency is rigidly tied to shaft speed, which turns out to be what is left of the runaway protection.

The panel controller is a **Deep Sea Electronics DSE4522 MKII AMF**, India special — *not* the Kirloskar KG640C the OEM wiring diagram shows. That distinction is not academic; \$04 explains what it cost.

PARAMETER	VALUE	STATUS
Engine	3GK550ETA 4SR1 · app code GK3.8703	Confirmed — rating plate
Rated output	26.5 kW / 36 hp @ 1500 rpm · ISO 3046	Confirmed — rating plate
Type approval	ARAI/MoEF/DGTA/IGES4/KOEL-P25/2825/24	Confirmed — rating plate
Genset	25 kVA, 415 V 3-phase, 50 Hz, 4-pole alternator	Confirmed — controller config
Fuel system	Common rail, metering unit + 3 injectors	Confirmed
Aftertreatment	Cooled EGR + DOC — no SCR, no DEF, no DPF	Confirmed
Emissions tier	CPCB IV+ (IGES4) · GSR 804(E), 3 Nov 2022	Confirmed
Electrical	12 V, 75 Ah	Confirmed
Panel controller	DSE4522 MKII AMF (India SP) · 4522-001-01	Confirmed — photographed
ECU connector	Bosch 1 928 405 192 / 194 · code C · 94-way	Confirmed — markings
Our MCU	NXP FS32K148HAT0MLQT, 144-LQFP	Selected

ON THE NAME "GK550" — WE GOT THIS WRONG

The engine was first described to us as a **GK550**. An earlier revision of this handbook said no Kirloskar document used that string, that it was a garbled run-together of the genset prefix and the engine family number, and that it "will find nothing" when ordering parts. The rating plate reads 3GK550ETA 4SR1. **The original description was correct and our inference was wrong.** The reasoning that produced it was sound — five hardware constraints read off the wiring diagram, all of which match the real engine — but it converged on a different Kirloskar engine that happens to satisfy the same five. Use 3GK550ETA 4SR1 and GK3.8703.

02 The 94-pin interface

The OEM ECU presents one 94-pin connector. The troubleshooting diagram traces **44 of those pins**; the other 50 — power grounds, unused positions, whatever drives the intake throttle — are simply not on a document written for field technicians. A genuine drop-in replacement needs the connector part number to get the rest.

Every wire in the diagram is colour-coded by signal class, and that code is load-bearing rather than cosmetic: it is how we established that the discrete inputs are switched-battery rather than switched-ground, and that the fuel metering unit is driven low-side. The same colours are used below.

- DC positive · sensor supply · injector high side
- DC negative · sensor ground · injector low side
- Analog in · PWM out
- Digital in · frequency
- CAN H
- CAN L

PIN	SIGNAL	CLASS	FRONT-END
21	Battery + feed	● supply	EMI filter → fuse → TVS → reverse-battery FET
04 / 06	V_V_BAT_1R / 2R	● supply	120 k / 10 k divider → ADC · 10 A fused each
09	Synchronization ground	● ?	Function unresolved
11 / 41 / 79 / 34	Boost pressure + air temp	● analog	Combo sensor, 4-pin, shared supply and ground
33	Coolant temperature	● analog	NTC, pull-up divider · ground splices to 34
32 / 35 / 08	Rail pressure	● analog	Ratiometric · dedicated sensor ground
39 / 80	Oil pressure	● analog	Ratiometric · ground splices to 36
15 / 37 / 36	EGR position	● analog	Potentiometer, shares EGR connector
52 / 74 / 30	Crankshaft speed	● frequency	Variable reluctance, differential, twisted pair

PIN	SIGNAL	CLASS	FRONT-END
45 / 46 / 44	Camshaft position	● frequency	Hall effect, already digital
43 / 23	Coolant switch · droop switch	● digital	Switch-to-return + debounce
20 / 24 / 71	Audio abort · override · ignition	● switched batt	Active high — zener clamp, see §03
01 / 02	G_G_B AT1 / AT2	● ground	Switch returns, or main power grounds — unresolved
88	Fuel metering unit	● PWM low side	Low-side driver + current sense
59 / 81	EGR high / low	● PWM	H-bridge, position feedback on pin 37
03	Injector high side — bank A	● high side	Cylinders 1 and 3, joined at splice V5
05	Injector high side — bank B	● high side	Cylinder 2
73 / 07 / 29	Injector low side — cyl 1 / 3 / 2	● low side	Current-controlled peak-and-hold
50 / 69	Main relay · buzzer relay	● low side	N-FET + flyback diode
55 / 77	CAN channel 1	● CAN H ● CAN L	To the KG640C
87 / 86	CAN channel 2	● CAN H ● CAN L	Also wired in the OEM system

INJECTORS SHARE THEIR HIGH SIDES

Pin 03 feeds cylinders 1 *and* 3. Only the low sides are individual. That is five pins, not six, and two boost-fed bank switches rather than three — which matters, because the boost stage is the expensive part of the board. It is safe because on a three-cylinder four-stroke any two cylinders are 240 crank degrees apart, or 26.7 ms at 1500 rpm, against an injection lasting perhaps 3 ms. That separation holds under either candidate firing order, so the unknown firing order stops mattering.

Signals that are not on this connector

The fuel level sender and the water-in-fuel switch both appear in the wiring diagram, and neither reaches the ECU — they route through the XC11/XC1 harness to the panel. They belong to the GCU. Earlier design passes had built front-ends for both.

Also absent from the entire system: any magnetic pickup, any DEF or urea sensor, any LIN bus, any SCR, and any separate overspeed switch.

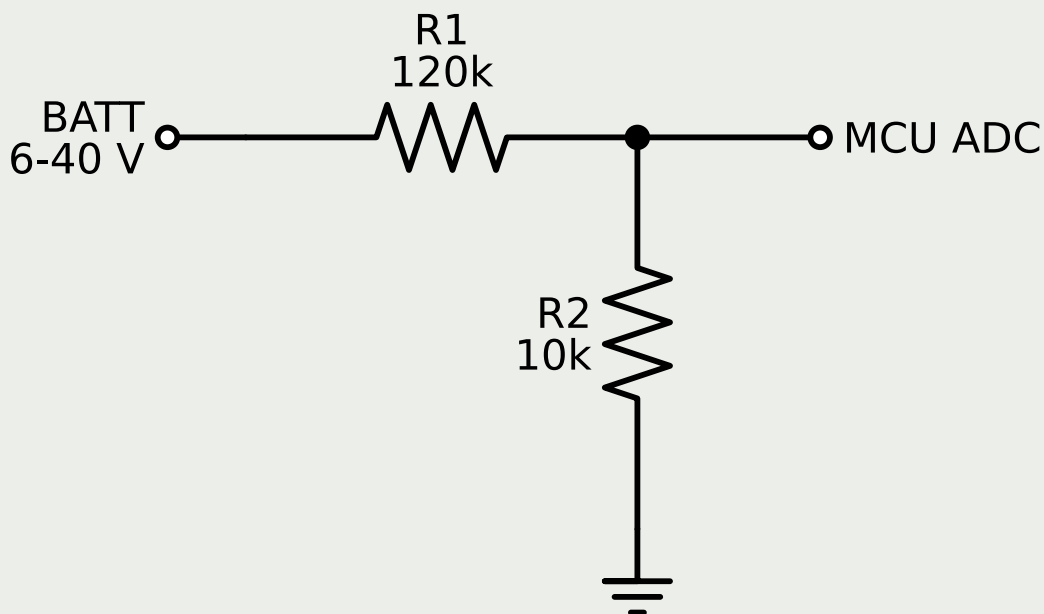
What the diagram labels OVERSIDE SWITCH on pin 24 traces to -S7

Override SS SW — one component, not two.

03 The circuits

Every circuit below exists as a SPICE netlist that runs, with checks that are claims about the circuit rather than about the simulator. A failing check means the design is wrong. All six currently pass — but two of them only after simulation caught errors that had survived four rounds of document review.

Battery voltage sense pins 04, 06

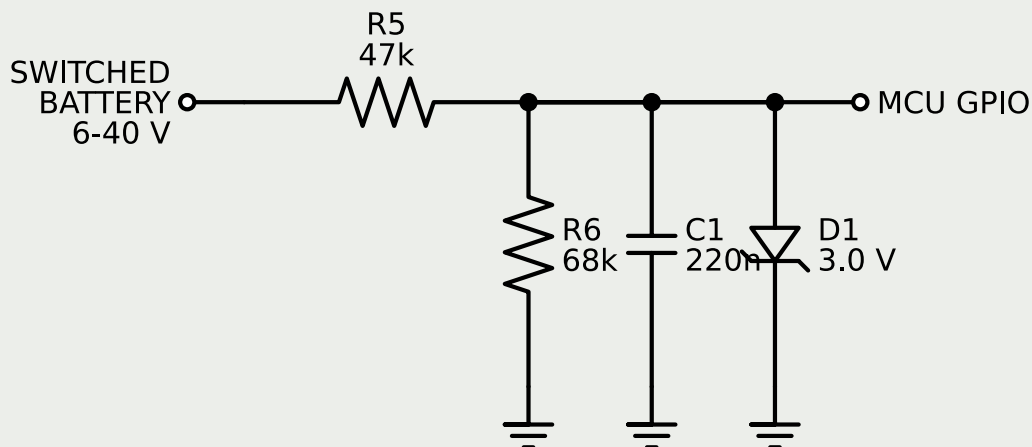


A plain resistive divider scaling the 6–40 V rail into the ADC. Two channels, one per fused rail, which doubles as fuse-integrity monitoring: voltage present downstream means the fuse is intact.

The inherited value was 75 k / 10 k, and it would have destroyed an ADC pin. That ratio was sized for a 5 V ADC; this MCU's analog supply is 3.3 V. At the top of the input range it delivers 4.69 V into a 3.3 V pin. Corrected to 120 k / 10 k.

PASS	Peak output at 40 V in	3.069 V
PASS	Headroom below the 3.3 V rail	0.231 V
PASS	Reading at the 6 V cranking dip	0.462 V · 573 LSB
PASS	Linearity residual	< 1 μ V

Discrete switch input pins 20, 24, 71 · active high



These are switched-battery inputs: the switch connects the pin to battery, not to ground. Earlier documents had them as pull-up-to-5 V with the switch closing to ground — the opposite sense, with a front-end that cannot survive battery voltage.

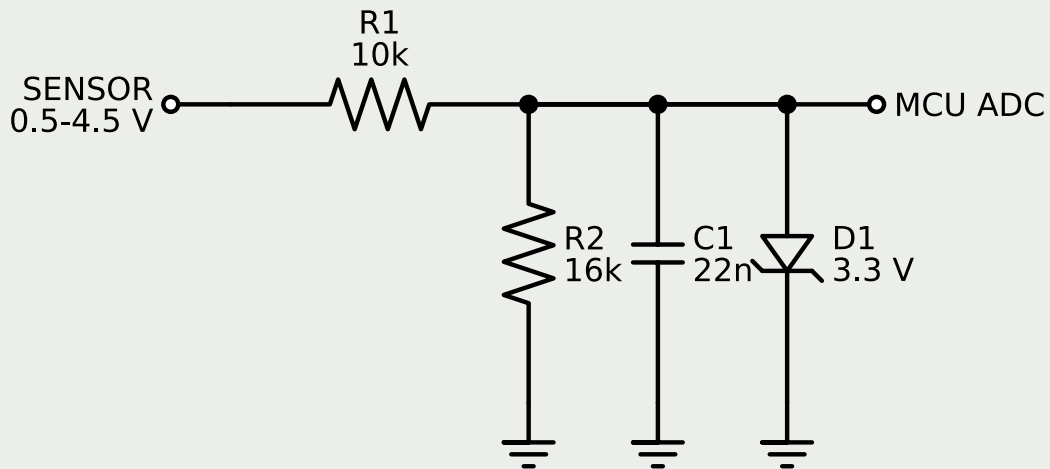
This one cannot be built as a divider at all. Reading logic high at a 6 V cranking dip needs a ratio above $2.31/6 = 0.385$. Staying under 3.3 V at 40 V needs it below $3.3/40 = 0.0825$. Those do not overlap, so no fixed ratio works at both ends — the original was impossible, not merely mis-valued.

A 3.0 V zener sets the clamp level and the divider only biases into it, so the logic level stops tracking the battery and goes flat.

PASS	Level at 6 V cranking dip	2.863 V
PASS	Level at 40 V load dump	2.954 V
PASS	Spread across the whole range	92 mV
PASS	Threshold crossings during a 5-bounce contact	0
PASS	Debounce time constant	6.11 ms

Debounce capacitor went from 100 nF to 220 nF after simulation showed the smaller value let two threshold crossings through a bouncing contact — which firmware would have read as two button presses.

Ratiometric sensor front-end pins 41, 35, 80, 37

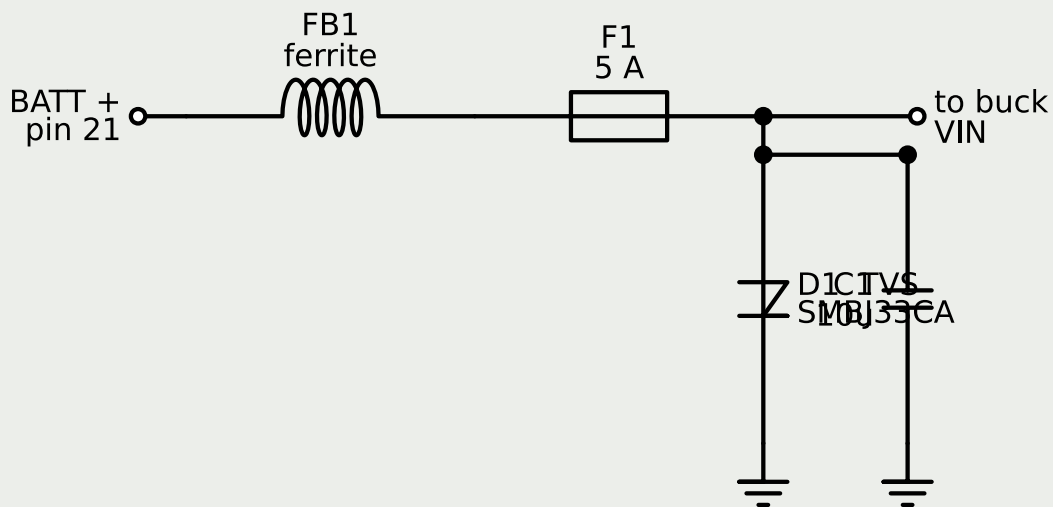


Boost pressure, rail pressure, oil pressure and EGR position are all 0.5–4.5 V ratiometric sensors — their output is a fraction of their own 5 V supply, so the ADC reference has to track that same rail or a drifting supply reads as a pressure change.

Full scale is deliberately placed at 2.77 V, below the 3.3 V clamp rather than hard against the ADC ceiling. That costs about 4% of span and buys a front-end that a 40 V harness short cannot damage.

PASS	Sensor zero (0.5 V) maps to	0.308 V
PASS	Sensor full scale (4.5 V) maps to	2.769 V
PASS	ADC span used	74.6 %
PASS	Low-pass corner	1175 Hz
PASS	Loss at the sensor's own 100 Hz	0.03 dB
PASS	Attenuation at 100 kHz switching noise	38.6 dB
PASS	Pin voltage on a 40 V harness short	3.31 V · 3.7 mA

Transient clamp battery input · ISO 7637-2 pulse 2a



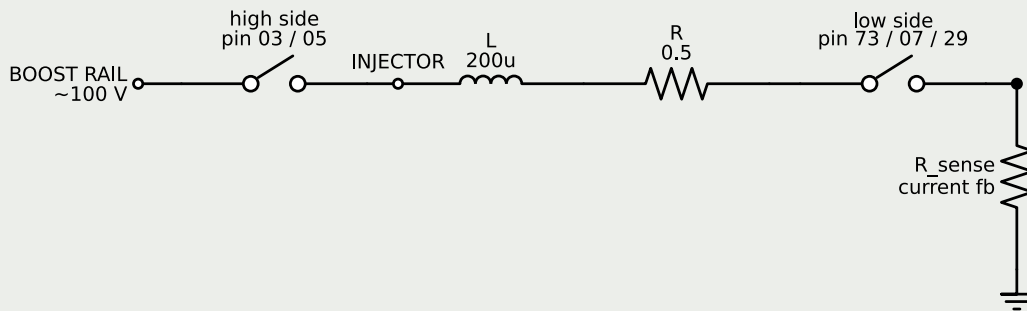
This is the circuit that decides whether the board survives its electrical environment, so it gets simulated rather than assumed. Pulse 2a models a sudden current interruption in another device sharing the harness: **+112 V through 2 Ω for 50 μ s**, at Level IV severity.

PASS	Clamped voltage at the buck input	50.0 V
PASS	Margin inside the LM5164's 100 V rating	2.0 ×
PASS	Recovery after the pulse	12.98 V

A SPEC TARGET THAT SHOULD BE DROPPED

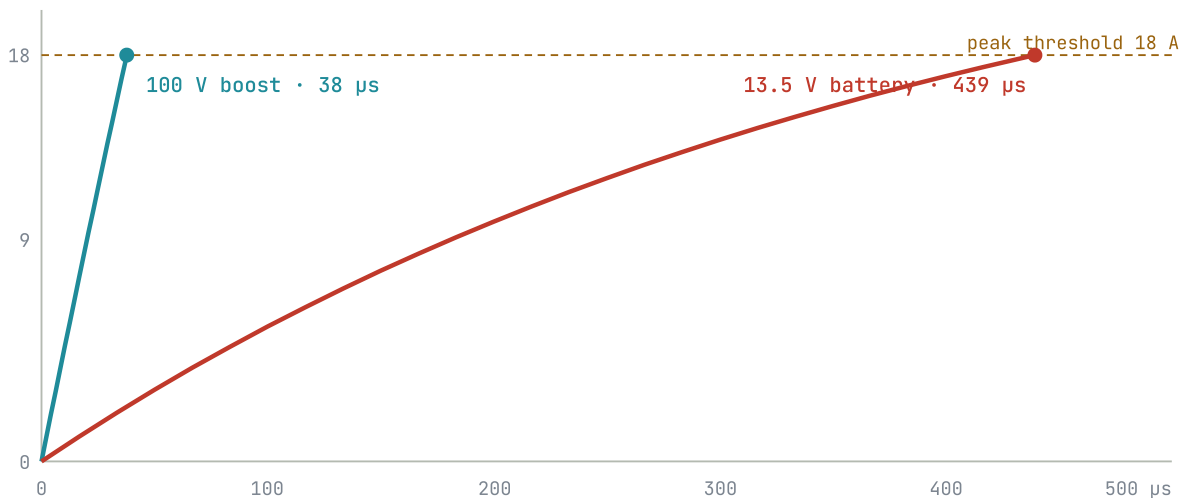
Earlier documents asked for a clamp "under 42 V". An SMBJ33CA cannot do that — the real part is specified at 53.3 V clamping at rated pulse current, so 50 V here is the component behaving correctly. The 42 V figure was never derivable from a datasheet. Chasing it with a lower-voltage TVS would make the clamp conduct during a legitimate suppressed load dump and force it to absorb that energy. Drop the target; keep the part.

Injector drive — why the boost stage exists pins 03, 05 · 73, 07, 29



An injector is an inductor with resistance, so current through it cannot change instantly — $di/dt = V/L$. The needle does not lift until current reaches its peak threshold, so every microsecond spent getting there is dead time in series with every injection event.

This is the single most expensive stage on the board. Here is its justification as a measurement rather than an assertion.

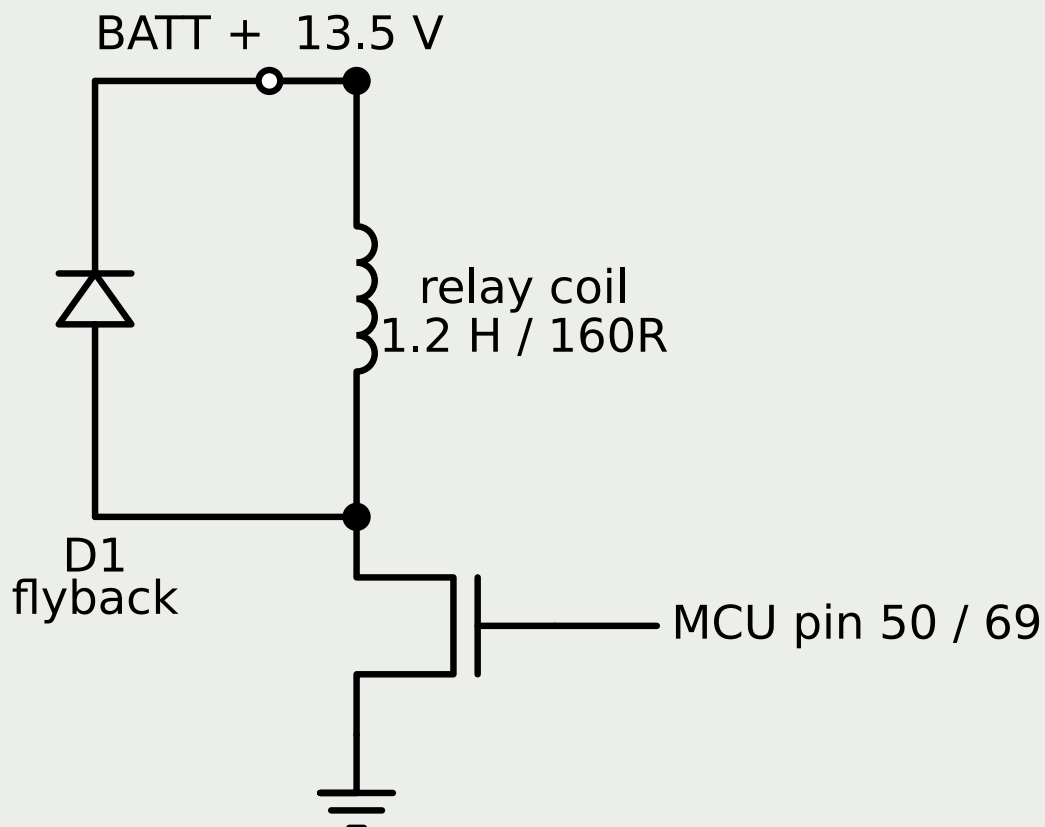


Injector current reaching its 18 A peak threshold. Modelled at 200 μ H / 0.5 Ω , the mid-range of the envelope for solenoid common-rail injectors of this class.

PASS	Time to 18 A from a 100 V boost rail	37.8 μ s
PASS	Time to 18 A from the 13.5 V battery	439.5 μ s
PASS	Boost is faster by	11.6 $\times$
PASS	Boost drive, as a share of a 1 ms injection	3.8 %
PASS	Battery drive, as a share of a 1 ms injection	44.0 %

Without the boost stage, nearly half of the shortest injection event is spent merely lifting the needle. That is why no production common-rail ECU drives injectors from the battery rail, and why the boost converter is not optional.

Relay driver pins 50, 69



A low-side N-FET switching a relay coil. The flyback diode is the entire point: a coil is an inductor, and interrupting its current generates $-L \cdot di/dt$ across it. The diode gives the collapsing field somewhere to circulate.

PASS Drain peak with the flyback diode

14.3 V

PASS Drain peak without it

unbounded

The simulator reports hundreds of kilovolts for the undiode case. Read that as a model artifact – an ideal switch opening instantly across an inductor implies an unbounded voltage. A real MOSFET avalanches at its breakdown rating and turns the coil's stored energy into heat in the die, once per relay operation, until it fails. The magnitude is fictional; the conclusion is not.

04 Stopping a runaway

A 25 kVA set in runaway is a mechanical hazard to anyone near it. The failure that matters is a controller hanging with fuel still flowing — and a controller cannot be its own protection against that.

This section has now been wrong twice, in opposite directions, and the second time is worth understanding before trusting the third.

We first read the system as sending engine speed to the panel over CAN only, which would leave overspeed protection depending on the very microcontroller and bus most likely to have failed. We then found the panel controller had an independent alternator-frequency speed path, concluded the pessimism was unwarranted, and relaxed the requirement for a standalone trip module. **That second reading was about the wrong device.**

The OEM wiring diagram shows a Kirloskar K6640C. The controller actually fitted, photographed front and back, is a **Deep Sea Electronics DSE4522 MKII AMF**. And DSE's own manual for its siblings says plainly that with a CAN engine, *"compatible ECU's receive the start command via CAN and transmit the engine speed to the DSE controller"*.

So engine speed reaches this controller from us. Its configured 1710 rpm overspeed shutdown reads a number our ECU sends, and a hung ECU transmitting a stale, plausible speed defeats it completely. The first, pessimistic reading was closer to the truth than the second.

WHAT THIS CHANGED

The standalone magnetic-pickup trip module — a GAC SSW675 or a Murphy HD9063, both with Indian distributors — goes back to **mandatory**. The relaxation to "recommended" was bought entirely by the independent-speed claim, and that claim described a controller that is not on this machine.

One independent path does survive, and it is worth having — but it is a different mechanism and a more conditional one. The DSE4522 measures **generator frequency** directly at its terminals 21–24, straight off the alternator's three phases, without touching CAN at all.

Generator frequency

**Over-frequency
shutdown**

Fuel relay, terminal 3

Measured directly at terminals 21–24. On a 4-pole alternator with no slip, 1500 rpm *is* 50 Hz.

→

Configured to trip at 56.0 Hz — that is **1680 rpm**, below the CAN-dependent 1710 rpm trip.

→

DC Output A. 5 A continuous — a coil driver, consistent with operating the 70 A -13RB1 contactor.

→

Fuel system stays powered

Measured 18 Sep 2026: the relay signals, it does not remove power.

The chain ends here, and it ends broken.

No link before the last one touches this ECU, so a hung ECU does not defeat them. But note what is being relied on: a *generator* protection doing the work of an *engine* protection. It inherits that mechanism's dependencies — the alternator must be excited and producing measurable output. An excitation failure removes it. That is a narrower guarantee than a magnetic pickup watching the flywheel, and it should not be described as equivalent to one.

THE GATE CLOSED, AND IT CLOSED BADLY

Every protective action this controller can take ends at the same place: it de-energises the fuel relay on terminal 3. The question was whether those contacts remove supply, or merely signal. **Measured at the machine on 18 September 2026: they signal.**

So the chain terminates in the failure it was meant to catch. Read together with the CAN finding above, the two compound into one sentence worth being blunt about: **the controller either cannot see the fault or cannot act on it.** Overspeed it reads from our CAN, so a hung ECU hides it. Over-frequency it measures for itself — and then acts through a relay that only signals.

Nothing on this machine currently interrupts fuelling without the cooperation of the ECU being protected against. That is what the standalone trip module now exists to fix, and it is why its contacts have to sit in series with the fuel metering unit's own battery feed through fuse 8F1 — not in the GCU relay chain, which has just failed this exact test.

One caveat kept for honesty: the contact-type check that disambiguates a normally-closed relay is not confirmed to have been done first. It is still wanted. Nothing above changes if it comes back the other way, because the unsafe reading is already the one being designed to.

AND THE ONE NOBODY CONSIDERED

A diesel that starts burning its own lubricating oil — through a failed turbo seal, or crankcase fumes drawn into the intake — has a fuel supply that **none of the protections above can touch**. Every one of them cuts diesel or electrical power. The standard countermeasure is a mechanical **air-intake shutoff valve**, and no such device appears anywhere in this design.

It was never considered and rejected. It was never considered. Whether to fit one is a decision for the machine's owners — the part is mechanical, not electronic — but it should be a decision rather than an omission.

05 The two real risks

Neither is a hardware problem, and neither is solved by better circuit design. They are the two things that decide whether this project ends with a working board on a bench or a running genset.

You cannot buy the injector calibration

The drive stage can be designed, built, and fully verified against a known inductive load. That is real engineering progress and it should proceed.

What it cannot do is deliver the *right amount of fuel*. Producing an electrically correct current pulse is not the same as knowing how much fuel that pulse injected — and that mapping, fuel mass against pulse width across the rail pressure range, is proprietary flow-bench data tied to a specific injector's hydraulic geometry. Bosch and Denso ship it with their own ECUs and not otherwise.

It gets harder in the **ballistic region**, where pulses are too short for the needle to reach full lift and delivery becomes steeply non-linear — which is exactly where a genset at light load spends its time. And empirical tuning on a running engine is not a fallback: at 19.7:1 compression, wrong fuelling risks combustion damage and turbocharger overspeed long before trial and error converges.

ROUTE TO THE DATA	COST	LEGALITY
Supplier licence	Structured around OEM production volumes. Realistically out of reach at this scale.	The sanctioned route.
Measure it yourself	Buy injectors, measure delivery directly on a rate-of-injection rig — Bosch tube or Zeuch method. Bounded and controllable with access to a fuel-injection lab.	Entirely legal. First-party measurement of purchased hardware.
Extract the OEM calibration	Cheapest in engineering time if a donor ECU exists, but the binary is undocumented and likely protected.	Contested. Implicates the manufacturer's copyright. Not recommended without legal clearance.

THE ONE WORTH PURSUING

Route two. It is legal, the cost is bounded, and a university with a fuel-injection lab is a far more reachable thing than a Bosch development agreement.

The regulatory position

This is an emissions-certified genset in service. CPCB IV+ type approval attaches to an engine family at the manufacturer level — it is not a per-unit certificate that travels with the machine regardless of what is done to it.

READ THIS BEFORE PLANNING DEPLOYMENT

CPCB's own RECD System & Procedure states that ECU characteristics governing injection timing, air-mass metering and emission-reduction strategy **must not be altered** — and that is under CPCB's own sanctioned retrofit scheme, which separately excludes modifications limited to engine control. General Condition 3 of GSR 804(E) states that no person shall use a genset lacking a valid Type Approval and CoP certificate. No approval pathway for a third-party ECU replacement was found.

This does not touch the engineering. Designing the board, building it, and bench-testing it against simulated crank and cam signals are all unaffected. It bears on putting it into service on a genset that is in use.

And it compounds the calibration problem in a way neither risk shows alone: this engine meets its emissions limits *partly through calibration* — EGR scheduling and injection timing — not through hardware alone. A replacement ECU can therefore move it out of compliance without a single component changing.

This is a decision for the project's owners to take to qualified counsel or to CPCB/ARAI directly. It is recorded here because it is material and had not previously been examined — not to make the decision for anyone.

06 What we don't know

Nothing on this list may be guessed into copper. Most of it needs somebody at the machine with a multimeter and a camera rather than more research — which is worth knowing, because it means one site visit closes most of it.

Four closed on 18 September, from photographs and one manufacturer's manual. Three of the four closed *against* something this project had inferred, which is a fair measure of how much the desk research was carrying.

#	CLOSED	ANSWER
U11	Engine identity	3GK550ETA 4SR1, app code GK3.8703 — rating plate. Our inferred model number was wrong.
U13	Which controller is fitted	DSE4522 MKII AMF (India SP), serial 11021794 — not the KG640C the diagram shows. Cost us §04's argument.
U1	The connector	Bosch 1 928 405 192 / 194, code C keying. 94 ways: power 1-8, then 9-28, 29-50, 51-72, 73-94.
U3	Does the controller have a speed pickup?	No — DSE's terminal list runs 1-35 with no pickup input.

#	STILL OPEN	BLOCKS	HOW IT CLOSES
U12	Does the fuel relay remove power, or only signal?	CLOSED 18 Sep 2026 — it only signals. §04's safety gate, resolved against us	Closed by measurement. The trip module becomes the only ECU-independent shutdown.
U4	Exact J1939 PGN set	CAN firmware	Mostly answered — addresses 234 / 44, and the controller wants oil pressure, coolant temp, coolant level and speed from us. Log the bus for the rest.
U2	Boost air temp — NTC or ratiometric?	One sensor front-end	One resistance reading across the sensor, engine cold

#	STILL OPEN	BLOCKS	HOW IT CLOSES
U5	Injector part number	Boost rail target, not the driver itself	Read the injector body
U6	What pin 09 "synchronization ground" does	Ground plan	Continuity, battery disconnected
U7	Are pins 04/06 sense-only or load feeds?	Power sheet current rating	Measure current at the fuse
U8	Is the 6-pin intake throttle on the ECU?	Whether a throttle driver is in scope	Follow the cable by hand
U9	Where the catalyst temp sensor goes	Whether an EGT front-end is needed	Trace on the engine
U10	Which signal uses SENT	Possible digital sensor front-end	Inspect the harness
U14	Pre-heat / post-heat — controller or ECU?	Whether a heater output is in scope. We have none.	Trace the heater supply. The controller config enables both at 50 °C.
U15	What "CRS-878" denotes	Possibly bears on the injector	Ask KOEL, or find DSE's engine-profile list
–	Are pins 01/02 switch returns or power grounds?	Ground plan	Continuity to battery negative

U12 NOW STANDS ALONE

With the controller identified, every protective action it can take ends at one relay. U12 asks what that relay's contacts actually carry, and the whole of §04's second line of defence rests on the answer. It is a multimeter and ten minutes, and nothing else on this list substitutes for it.

07 Running the simulations

You already have a SPICE engine. KiCad ships `libngspice`, and the harness in the repository drives it directly, so the circuits run and check themselves without installing anything further.

```
.venv/bin/python sim/run_sim.py          # every block
.venv/bin/python sim/run_sim.py injector  # blocks matching "injector"
```

Output is one line per claim, green or red. Because the claims are about the circuit — "reads logic high at a 6 V cranking dip", "never exceeds the MCU's absolute maximum" — a red line means the design is wrong, not that the simulator is unhappy.

The netlists are plain SPICE and are not tied to this harness. They open directly in **KiCad**'s built-in simulator (Tools → Simulator), in **LTspice**, or in a standalone **ngspice**. Each carries its own `.control` block, so it runs without further setup.

Read the comments inside each netlist before the netlist itself. They carry the reasoning — why the discrete input is not a divider, why the sensor full scale sits below the clamp, why the 42 V clamp target should be abandoned — which the schematic alone cannot tell you.

Sensor Reference

Every sensor in the system — what it measures, how its signal reaches the processor, and what the engine actually does with the number. Including the ones that belong to the panel rather than the ECU, and the ones we would add.

14 on the ECU 6 panel-side 7 proposed Rev 1 · 13 Sep 2026

[Circuit reference](#) → ← [Field Handbook](#)

00	How to read an entry	01	Position & speed	02	Fuel system
03	Air path	04	Engine health	05	Electrical
06	Operator inputs	07	Panel-side	08	What we would add
09	Channel budget				

00 How to read an entry

Every sensor below has the same five rows, because those are the five things you need to know before you can either build its front-end or write its firmware:

- **Measures** — the physical quantity and its useful range.
- **Signal** — the electrical form it arrives in, its ECU pins, and its wire class from the Kirloskar diagram's own colour legend.
- **Acquisition** — the front-end circuit, which MCU peripheral reads it, and how often.
- **Used for** — what the engine does with the number. This is the row that matters and the one most sensor documentation leaves out.
- **Fault detection** — how firmware knows the reading is lying.

Sampling rates are proposals, not measurements — they follow from each signal's own bandwidth and from what the control loop using it needs. Marked **inferred** where that is the case.

A NOTE ON SAMPLE RATE

The instinct is to sample everything fast. Resist it. Coolant temperature has a thermal time constant measured in tens of seconds, so sampling it at 1 kHz buys nothing and costs ADC bandwidth that rail pressure genuinely needs. Rate should follow the physics, and the physics here spans four orders of magnitude — from crank edges every 667 μs to a fuel level that moves over hours.

01 Position & speed

Everything angle-related derives from these two. Get them wrong and nothing else in the system can be right, because injection timing is meaningless without knowing where the crankshaft is.

Crankshaft speed pins 52 / 74 / 30		ON ECU
MEASURES	Crank angular position and speed, from a toothed wheel. Nominal running speed 1500 rpm ; must also resolve cranking speed, roughly 100–250 rpm.	
SIGNAL	Variable reluctance , differential. ● Frequency I/P High on pin 52, ● Frequency I/P Low on pin 74, ● sensor ground on pin 30. Twisted pair. Output is AC whose amplitude scales with speed — near-zero while cranking, several volts at idle.	
ACQUISITION	Zero-cross adaptive-threshold conditioner producing one clean digital edge per tooth, into a FlexTimer input capture channel. Every edge is timestamped; at 1500 rpm with a 60-tooth wheel that is an edge every 667 µs . The missing-tooth gap gives the angular reference.	
USED FOR	Speed governing — the primary feedback for the metering unit loop. Injection timing — every injection event is scheduled against crank angle derived from here. Overspeed shutdown — the ECU's own first line of defence. Crank sequencing — the starter is disengaged the moment speed crosses the catch threshold. Reported over CAN in EEC1, PGN 61444.	
FAULT DETECTION	Missing edges, implausible tooth intervals, or loss of the sync gap. Loss of crank signal on a running engine is a shutdown, not an alarm — the ECU can no longer time injection.	

MEASURES	Which half of the four-stroke cycle the crank angle refers to. One pulse per two crank revolutions.
SIGNAL	Hall effect , already a clean digital square wave — which is why it needs no conditioning IC and has no Low pin. ● Switch Battery 5 V on pin 45, ● Frequency I/P on pin 46, ● sensor ground on pin 44.
ACQUISITION	Pull-up to the protected sensor rail, straight into a second FlexTimer capture channel. One event per 80 ms at running speed.
USED FOR	Cylinder phase , and only that. The crank sensor says the engine is at 15° before top dead centre; the cam sensor says whether that is cylinder 1's compression stroke or its exhaust stroke. Without it you can measure speed perfectly and still inject into the wrong cylinder.
FAULT DETECTION	Absent pulse, or a pulse at the wrong crank angle relative to the sync gap. Loss after sync is survivable for a while — the engine can run on the last known phase — but it must not be allowed to re-sync blind.

02 Fuel system

Rail pressure pins 32 / 35 / 08 ON ECU	
MEASURES	Common-rail fuel pressure. Typically 250–1800 bar on an engine of this class — inferred , the exact range needs the sensor part number.
SIGNAL	Ratiometric 0.5–4.5 V . ● Switch Battery 5 V on pin 32, ● Analog I/P on pin 35, ● sensor ground on pin 08 — a dedicated ground, not spliced with any other sensor. That is deliberate: this is the most accuracy-critical analog channel on the engine, and a shared return would inject other sensors' currents into its reference.
ACQUISITION	Ratiometric front-end into the ADC. Sampled fast — 1 kHz or better, inferred — because it is the feedback for a closed loop that has to reject the pressure ripple every injection event causes.
USED FOR	Closed-loop rail pressure control : the error between measured and target pressure drives the fuel metering unit's PWM duty. Also sets the injection quantity calculation — the same pulse width delivers more fuel at higher rail pressure, so quantity is always a function of both. Over-pressure protection is a hard shutdown.
FAULT DETECTION	Out-of-range reading, or pressure that fails to follow a metering unit command — which distinguishes a dead sensor from a stuck metering unit. Loss of rail pressure feedback means injection quantity becomes unknown; treat as shutdown.

Fuel metering unit current pin 88, sensed on-board

ON ECU

MEASURES	Current through the metering unit solenoid — the actuator's own feedback, not an external sensor.
SIGNAL	Shunt and amplifier on the ECU's low-side driver. The metering unit itself is only 2-pin: ● battery via the 20 A fuse, and ● PWM low side to pin 88.
ACQUISITION	ADC channel synchronised to the PWM cycle, so current is sampled at a consistent point in the switching period rather than at random phase.
USED FOR	Inner loop of the rail pressure cascade. Rail pressure sets a target current; current control sets PWM duty. Two loops because solenoid force follows current, and current follows duty only loosely once coil temperature and battery voltage move. Also gives open and short circuit detection on the actuator.
FAULT DETECTION	Zero current with duty commanded means an open coil or blown fuse. Full current at zero duty means a shorted driver — which is dangerous, since it drives the rail to maximum pressure.

MEASURES	Current through each injector solenoid during its firing event. Peak 12–24 A , hold 8–13 A .
SIGNAL	Sense resistor in each low-side leg, read by the driver IC's own current comparators.
ACQUISITION	The MC33816 regulates current in hardware against programmed thresholds — the MCU sets the profile, the driver enforces it microsecond by microsecond. Far too fast for firmware to close.
USED FOR	Peak-and-hold profile enforcement. A high current opens the needle, then a lower current holds it, because holding at peak would cook the solenoid. Also injector diagnosis : a coil that fails open or short shows up immediately in the current trace.
FAULT DETECTION	Failure to reach peak within the expected time means an open coil or a collapsed boost rail. The handbook's injector chart shows why that timing is diagnostic — the rise time is set by the rail voltage and the coil, so a slow rise means one of them is wrong.

03 Air path

Boost pressure pins 11 / 41 / 34

ON ECU

MEASURES	Absolute pressure in the intake manifold, downstream of the turbocharger and aftercooler. Typically 1–3 bar absolute on a turbocharged engine of this size — inferred .
SIGNAL	Ratiometric 0.5–4.5 V. One half of a four-pin combination sensor shared with manifold air temperature: ● Switch Battery 5 V on pin 11, ● Analog I/P Pressure on pin 41, ● sensor ground on pin 34 shared with both legs and with the coolant sensor.
ACQUISITION	Ratiometric front-end into the ADC. 100 Hz is ample — manifold pressure changes with turbo speed, which has real inertia. Inferred .
USED FOR	Air mass estimation , with air temperature, via the ideal gas law — and air mass is what sets the smoke-limit fuelling cap . A diesel that injects more fuel than its air charge can burn produces visible smoke and heat rather than torque, so the fuelling limit tracks boost. Also EGR control , since EGR displaces fresh air, and turbo overspeed protection .
FAULT DETECTION	Range check, plus plausibility against barometric pressure at rest — with the engine stopped, manifold pressure must read close to ambient. A sensor that fails high would silently allow over-fuelling, so this one defaults conservative on fault.

MEASURES

Intake charge temperature after the aftercooler. Roughly **–20 to +100 °C**.

SIGNAL

● Analog I/P Temperature on pin 79, sharing the combination sensor's supply and ground. **Whether it is an NTC or a conditioned ratiometric output is unresolved.** The wire colour class cannot distinguish them — the coolant NTC uses the same magenta class. Industry convention for combination pressure-and-temperature sensors is an NTC element, so that is the working assumption.

ACQUISITION

Pull-up divider if NTC, ratiometric front-end if not. The board carries a **DNP pull-up footprint** so either works without a respin — the cheapest possible insurance against a question we cannot currently answer.

USED FOR

Air density correction. Boost pressure alone does not give air mass; the same pressure at 20 °C and at 80 °C is meaningfully different charge. Feeds the same smoke-limit calculation. Also **intercooler diagnostics** — a rising delta over ambient means the aftercooler is fouling.

FAULT DETECTION

Range check, and plausibility against coolant temperature at cold start: on a machine that has stood overnight, intake air, coolant and ambient should all agree within a few degrees. That check is free and catches a surprising number of wiring errors.

EGR valve position pins 15 / 37 / 36

ON ECU

MEASURES	Actual position of the EGR valve, 0–100% travel.
SIGNAL	Ratiometric potentiometer. ● Switch Battery 5 V on pin 15, ● Analog I/P wiper on pin 37, ● sensor ground on pin 36. Shares its five-pin connector with the actuator's own ● EGR High and ● EGR Low drive pins.
ACQUISITION	Ratiometric front-end into the ADC, 100 Hz , matched to the valve's mechanical response. Inferred.
USED FOR	Closed-loop EGR position control — the H-bridge drives until measured position matches demand. EGR rate is one of the two things this engine uses to meet its NOx limit, the other being injection timing, which is exactly why the certification question in the handbook bites so hard. Also stuck-valve detection .
FAULT DETECTION	Position failing to follow demand within a timeout means a stuck valve, a failed motor, or a broken position sensor — distinguishable by whether current is flowing. A stuck-open EGR valve at load is a derate condition.

Intake throttle position 6-pin connector, routing unknown

UNRESOLVED — U8

MEASURES	Intake throttle plate position. On a CPCB IV+ diesel the throttle is not a power control — it creates intake depression to drive EGR flow when the pressure differential is otherwise insufficient.
SIGNAL	Six pins , which is two more than a motor pair plus a three-wire position sensor accounts for. The extra two are unexplained. It does not appear on any ECU pin traced in the wiring diagram — it may belong to a variant, or route somewhere we have not found.
ACQUISITION	Not designed. Blocked until the connector is traced on the engine.
USED FOR	If present: EGR flow authority at low load , and a clean shutdown — closing the throttle starves the engine of air and stops it smoothly.
FAULT DETECTION	Not designed.

04

Engine health

These are the sensors that stop the engine before it destroys itself. They earn their place in an ECU that never optimises anything.

Coolant temperature pin 33		ON ECU
MEASURES	Engine coolant temperature, -40 to +130 °C . Normal running is roughly 80–95 °C.	
SIGNAL	NTC thermistor, two wires only — no excitation pin, unlike the ratiometric sensors. ● Analog I/P on pin 33; its ● sensor ground does not get its own ECU pin but splices into the boost sensor's return at pin 34.	
ACQUISITION	Pull-up divider into the ADC, linearised in firmware against the sensor's resistance-temperature table. 2 Hz is plenty — an engine block's thermal time constant is tens of seconds. The pull-up is chosen to put maximum sensitivity around 85 °C, where the decisions are.	
USED FOR	Cold-start fuelling enrichment — a cold diesel needs more fuel and earlier injection to light off. Overheat derate then shutdown . EGR inhibit when cold , since EGR on a cold engine fouls the intake. Reported over CAN in ET1, PGN 65262.	
FAULT DETECTION	Open circuit reads full pull-up voltage and short reads zero — both outside any physical temperature, so both are unambiguous. Critically, an open circuit must not read as "very cold" , which would trigger enrichment on a hot engine. The independent coolant switch below exists precisely because this channel can fail.	

Coolant temperature switch pin 43, return on 01 / 02

ON ECU

MEASURES	A single over-temperature threshold. Not a measurement — a trip.
SIGNAL	● Digital I/P on pin 43, switch-to-return. Its return leg is one of the G_G_B pins.
ACQUISITION	Pull-up and RC debounce into a GPIO, polled at 10 ms .
USED FOR	A second, independent path to overheat shutdown. This is the same design instinct as the overspeed discussion in the handbook: the analog coolant channel and this switch fail in different ways, so requiring only one of them to trip means a single sensor failure cannot leave the engine cooking. It is cheap redundancy on the failure that quietly destroys engines.
FAULT DETECTION	Wired normally-closed where possible, so an open harness reads as tripped rather than as healthy.

Lubricating oil pressure pins 39 / 80

ON ECU

MEASURES	Engine oil gallery pressure, typically 0–10 bar . Normal running above roughly 2 bar; idle above roughly 1 bar. Inferred.
SIGNAL	Ratiometric 0.5–4.5 V, three pins: ● Switch Battery 5 V on pin 39, ● Analog I/P on pin 80, and a ● sensor ground that splices with the EGR sensor's into pin 36.
ACQUISITION	Ratiometric front-end into the ADC at 10 Hz . Inferred — oil pressure responds within a revolution, but no decision needs faster.
USED FOR	Low oil pressure shutdown , the single most important protection on any engine — bearing damage from oil starvation happens in seconds. The threshold is speed-dependent : pressure that is normal at idle is a fault at rated speed, so the limit is a function of the crank sensor's reading. Also start inhibit until pressure builds, and reported over CAN in EFL/P1, PGN 65263.
FAULT DETECTION	Range check. The dangerous failure is reading high when pressure is actually low, so a plausibility check against engine state helps: with the engine stopped, oil pressure must read near zero. A sensor that reads 3 bar on a stopped engine is lying.

Battery voltage, ×2 pins 04 / 06		ON ECU
MEASURES	Voltage on two separately fused battery rails, 6–40 V covering cranking sag through clamped load dump. Nominal 12 V, charging 13.5–14.4 V.	
SIGNAL	Two ● DC positive feeds, V_V_BAT_1R and V_V_BAT_2R, each behind its own 10 A fuse.	
ACQUISITION	120 kΩ / 10 kΩ divider into the ADC — no active conditioning, this is a slow rail rather than a delicate signal. 10 Hz . The divider value was corrected from an inherited 75 kΩ / 10 kΩ that would have put 4.69 V on a 3.3 V pin.	
USED FOR	Low and high battery alarms. Charging system diagnosis — voltage that never rises above resting means the alternator is not charging. Injector timing compensation , since boost converter recharge time depends on supply voltage. And because the two rails are separately fused, comparing them infers a blown fuse without a dedicated fuse-status circuit. Reported over CAN in VEP1, PGN 65271.	
FAULT DETECTION	The two channels check each other: they should track within a volt. A persistent divergence means a fuse, a corroded connection, or a failed divider.	

Ignition / run request pin 71

ON ECU

MEASURES	Whether the panel is commanding the engine to run.
SIGNAL	● Switched battery, active high. Downstream of the GCU's 70 A ignition relay.
ACQUISITION	Zener-clamped bias network into a GPIO — not a divider, for the reason given in the handbook. Polled at 10 ms .
USED FOR	Run enable and orderly shutdown. Losing this input is the normal stop request: the ECU stops injection, closes the throttle if fitted, and reports the stop over CAN. It is also, per the handbook's safety section, the signalling half of a path whose other half is physical — the same relay that drops this input is believed to remove power from the fuel system entirely, which is what makes a hung ECU survivable.
FAULT DETECTION	Not detectable as an open circuit — an open wire and a genuine stop request are electrically identical. That is an accepted limitation of matching the OEM harness, and it is recorded in the FMEA rather than engineered around, because changing the sense would break compatibility.

06 Operator inputs

Not sensors in the strict sense, but they arrive on the same connector and firmware treats them identically — debounced digital state with a defined meaning.

INPUT	PIN	SENSE	WHAT IT DOES	STATUS
Audio abort	20	● active high	Silences the buzzer without clearing the underlying fault. The fault stays active, the annunciator stays lit, and only the noise stops — so an operator can think without losing the alarm.	Confirmed
Override SS	24	● active high	Function genuinely unconfirmed. Labelled OVERSIDE SWITCH at the ECU and -S7 Override SS SW at the panel — one switch, not two , despite earlier documents listing an overspeed switch and an override switch separately.	Open
Droop select	23	● switch to return	Selects droop versus isochronous governing. In droop, speed falls slightly as load rises, which lets multiple sets share load without fighting. Isochronous holds speed exactly, which is right for a set running alone.	Confirmed

DO NOT WIRE CONTROL LOGIC TO PIN 24 YET

The override switch's actual function is a guess. Read its state, log it, report it over CAN — but do not let it change what the engine does until somebody has confirmed what the panel intends by it. A switch whose meaning you have assumed is worse than a switch you ignore.

The fitted DSE4522's own configuration takes **oil pressure**, **coolant temperature**, **coolant level** and **engine speed** from the ECU over CAN rather than from its own analogue inputs, on source addresses 234 (engine) and 44 (instrumentation). Its analogue inputs A and B are configured *Not Used* for exactly that reason.

Two consequences for this design. Those four values are no longer optional instrumentation — the controller's protections act on them, at 0.99 bar oil pressure and 200 °C coolant. And **ECU Data Fail** is **configured to shut the set down**, so our CAN transmission is itself a watchdog held by another device. Engine speed is the uncomfortable one: the controller's overspeed trip reads the number we send, which is why handbook §04 now calls a standalone trip module mandatory.

07 **Panel-side sensors**

These exist in the machine and appear in the wiring diagram, but they do not reach the ECU. They belong to the panel controller — a Deep Sea DSE4522 MKII AMF, identified 18 Sep 2026; earlier revisions of this page said KG640C, which is what the OEM wiring diagram shows but not what is fitted. They are documented here because earlier design passes built ECU front-ends for two of them, and because knowing what the panel already measures tells you what the ECU does not need to.

SENSOR	GOES TO	SIGNAL	USED FOR
Fuel level	Panel via XC pin 22	Resistive float sender, 2-pin	Tank level display and low-fuel alarm. Not an ECU signal — earlier documents wrongly gave it an ECU front-end.
Water in fuel	Panel via XC pin 16/1	Conductivity probe, 3-pin with built-in electronics	Water-in-fuel warning. Water in a common-rail system destroys injectors, so this is a genuine alarm. Not an ECU signal .
Catalyst temperature	Unresolved — U9	2-pin, one sensor only	DOC inlet or outlet temperature. Earlier documents specified three EGT channels; the diagram shows one.

SENSOR	GOES TO	SIGNAL	USED FOR
Alternator voltage & frequency	GCU terminals 27–30	3-phase AC sense, sampled at 5 kHz	Output metering, and — per the handbook's safety section — the GCU's only engine speed source. On a 4-pole alternator 1500 rpm is rigidly 50 Hz, which is what makes overspeed detection possible without an ECU.
Load current	GCU terminals 35–42	Current transformers	Load metering, overload protection, load-dependent start/stop.
Charging alternator D+	GCU	Excitation sense	Charge-failure alarm, and a secondary indication the engine is turning.

08 What we would add

The OEM sensor set is the minimum to run the engine and meet emissions. These are the additions worth designing space for now, since a connector position or an unused ADC channel costs almost nothing at design time and cannot be retrofitted cheaply.

Ordered by how much each earns its place.

ADDITION	INTERFACE	WHY IT EARNS A PLACE	COST TO DESIGN IN NOW
Barometric pressure	On-PCB MEMS, I ² C	Altitude and weather change air density, and therefore the smoke-limit fuelling cap. Without it the engine either smokes at altitude or is derated everywhere to be safe. Needs no harness at all — barometric pressure is the same inside the enclosure as outside.	One IC, two I ² C pins, a vented enclosure port. Already in the design.
Magnetic pickup	VR, to a standalone trip module	The independent overspeed path discussed in the handbook. Currently recommended rather than mandatory, because the GCU's alternator-frequency path already bypasses the ECU — but it is the only option that also survives a GCU firmware hang.	A flywheel sensor, a trip module, and panel wiring. No ECU pins at all, which is the point.
Exhaust gas temperature, ×2	Type K thermocouple, CJC amplifier	DOC inlet and outlet. The difference across a catalyst tells you whether it is working; a single sensor tells you much less. Also protects the turbine from over-temperature.	Two amplifier ICs, two ADC channels, four connector positions.
Fuel temperature	NTC, 2-wire	Diesel density varies with temperature, so the same injected volume is a different mass. Matters most for fuel consumption reporting and for hot-restart behaviour, where return fuel has been heated.	One ADC channel, one pull-up, two connector positions.

ADDITION	INTERFACE	WHY IT EARNS A PLACE	COST TO DESIGN IN NOW
Crankcase pressure	Low-range ratiometric	The best single indicator of engine wear available. Rising blow-by past the rings shows here long before anything else notices, which turns an unplanned failure into scheduled maintenance.	One ADC channel and a sensor. Genuinely the highest-value addition for a machine expected to run for years.
Ambient air temperature	NTC, 2-wire	Separates engine-related temperature rise from a hot day. Makes the derate strategy fair, and makes the aftercooler diagnostic in §03 meaningful.	One ADC channel.
Accelerometer	MEMS, SPI or I ² C, on-PCB	Vibration signature as a condition-monitoring input — bearing wear, misfire, and mounting problems all have signatures. Speculative for a genset, but nearly free if the board is being laid out anyway.	One IC and a few pins. Firmware effort is the real cost, not hardware.

WHAT FUTURE-PROOFING ACTUALLY MEANS HERE

Not fitting every sensor on the list. It means **reserving the channels and connector positions** so that adding one later is a firmware change and a harness leg, not a board respin. Concretely: leave four ADC channels unrouted but broken out, keep two spare positions on the sensor connector, put the I²C bus on a header even if only the barometric sensor uses it now, and leave the second CAN channel populated.

The MCU has room for all of it — the channel budget below shows the design using well under half of what the part provides. Spending some of that headroom on unpopulated options is the cheapest decision available at this stage.

09 Channel budget

What the sensor set costs in MCU resources, against what the S32K148 provides. The headroom is the argument for designing the additions in now.

RESOURCE	USED NOW	WITH ALL ADDITIONS	S32K148 PROVIDES	HEADROOM
ADC channels	14	19	up to 64	ample
Timer capture	2	2	64 across 8 FlexTimers	ample
Timer output / PWM	8	8	shared with above	ample
Digital in	5	5	abundant GPIO	ample
CAN	2	2	3 FlexCAN	1 spare
I ² C	1	1	2 LPI2C	1 spare
SPI	1 (injector driver)	2	3 LPSPI	1 spare

The ADC count is the one to watch, and it is not close to a limit. The caveat from the MCU memo stands: how many of those 64 channels are actually broken out on the 144-pin package needs the reference manual's pin-mux table, which is a Phase 2 item.

Circuit Reference

Every circuit block that has been built and verified, with the schematic, the reasoning, the simulated result, and what each model leaves out. A dashed part is on the board but not in the netlist below it — its impedance is either folded into a lumped value elsewhere in the same circuit, or left out entirely. The checks are claims about the solid parts only.

19 blocks all checks **passing** generated from **sim/blocks/** ← Field handbook

Sensor reference →

01 Power chain

02 Sensor front-ends

03 Speed and position

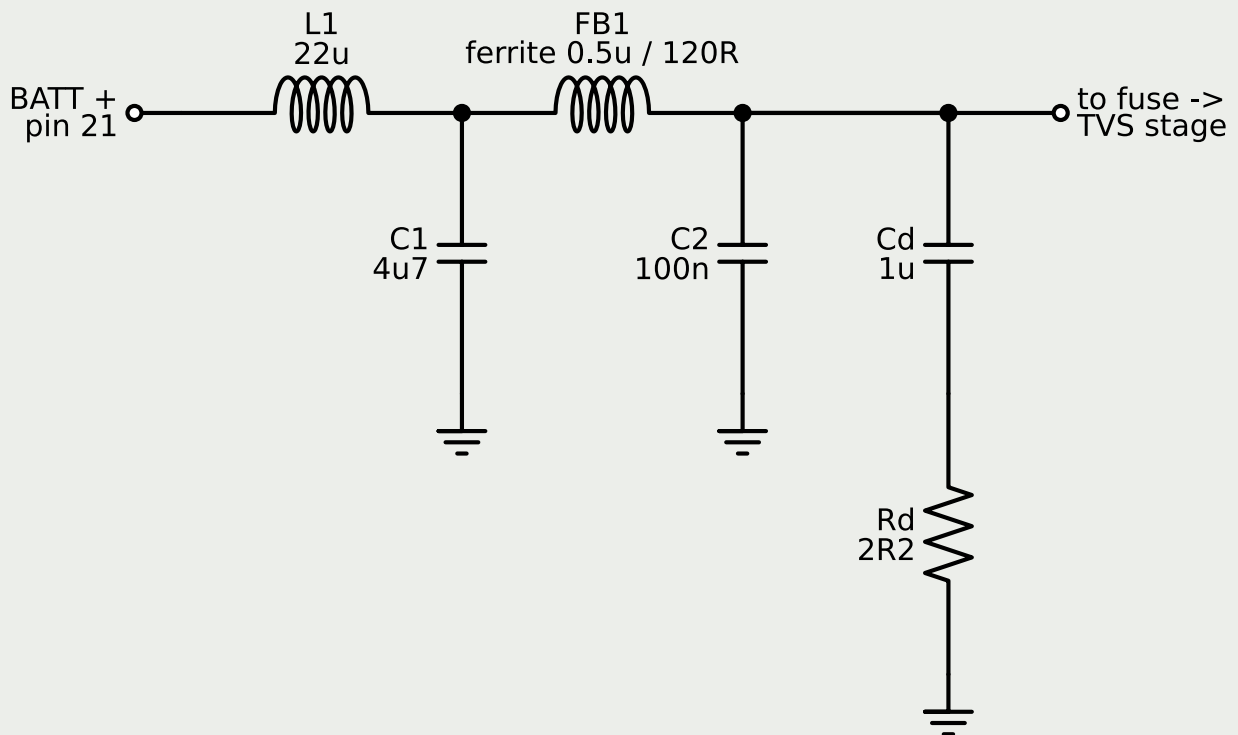
04 Injection

05 Actuators

06 Communications

01 Power chain

From the battery terminal to the MCU rail. Six stages, each one a different failure it has to survive.



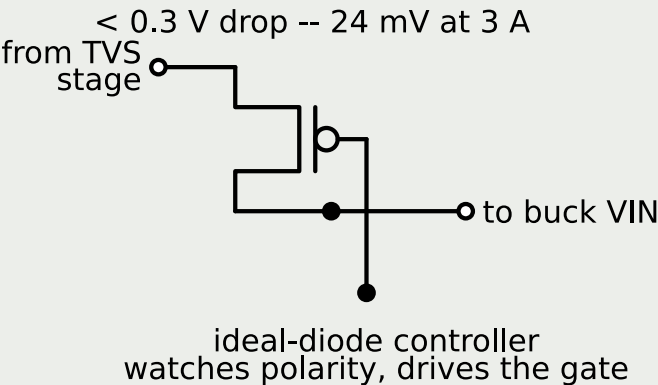
Two LC stages between the battery and everything else, measured 50 ohm to 50 ohm the way a LISN would see it.

Why it looks like that. One stage is not enough, and the reason is the capacitor: C1's own lead inductance puts its self-resonance near 1.9 MHz, and above that it stops being a capacitor. Stage two picks up there. The damper across C2 is not optional either – without it the ferrite and C2 ring at 717 kHz and hand back 12 dB right where the switching frequency's second harmonic sits.

What the model does not cover. Differential mode only.

Common-mode noise – both wires swinging together against chassis, which is most real automotive failures – needs the common-mode choke, and that cannot be modelled honestly without a layout to give it a return path.

PASS insertion loss at 150 kHz (band start)	-41.071 dB
PASS insertion loss at 400 kHz (switching f)	-51.546 dB
PASS insertion loss at 1 MHz	-74.812 dB
PASS insertion loss at 30 MHz	-141.697 dB
PASS insertion loss at 100 MHz (FM band)	-128.977 dB
PASS worst insertion loss anywhere above 150 kHz	-41.171 dB
PASS ... worst point sits at	151.5 kHz
PASS is 40 dB enough for Class 5?	unknown -- needs measured unfiltered emissions on hardware
Netlist: sim/blocks/emi_filter.cir	



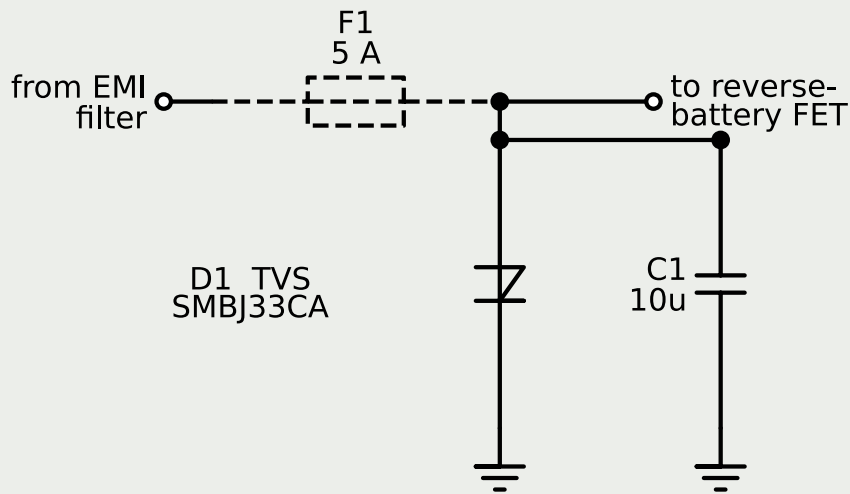
A P-FET held on by a controller that watches polarity, compared against the obvious Schottky.

Why it looks like that. Somebody will reverse the jump leads eventually. A Schottky survives it and charges you 0.46 V on every amp forever; the ideal-diode arrangement costs a controller IC and drops 24 mV. At 3 A that is 1.4 W of heat that simply never happens.

What the model does not cover. Steady state only. A real controller takes about a microsecond to turn the FET off, and current flows backwards through the body diode during that window. The switch model here turns off instantly, so it will always look perfect in reverse — that number is a datasheet question.

PASS Schottky drop at 3 A	0.465 V
PASS ideal-diode FET drop at 3 A	0.024 V
PASS FET drop stays under 0.3 V spec to 5 A	0.040 V
PASS ... Schottky does NOT meet it	0.52 V at 5 A -- fails the 0.3 V spec
PASS power burnt in the Schottky at 3 A	1.395 W
PASS ... and in the FET instead	0.072 W
PASS Schottky blocks -14 V (leak into 1k)	1.000 mV
PASS ideal-diode FET blocks -14 V	0.014 mV

Netlist: sim/blocks/reverse_battery.cir



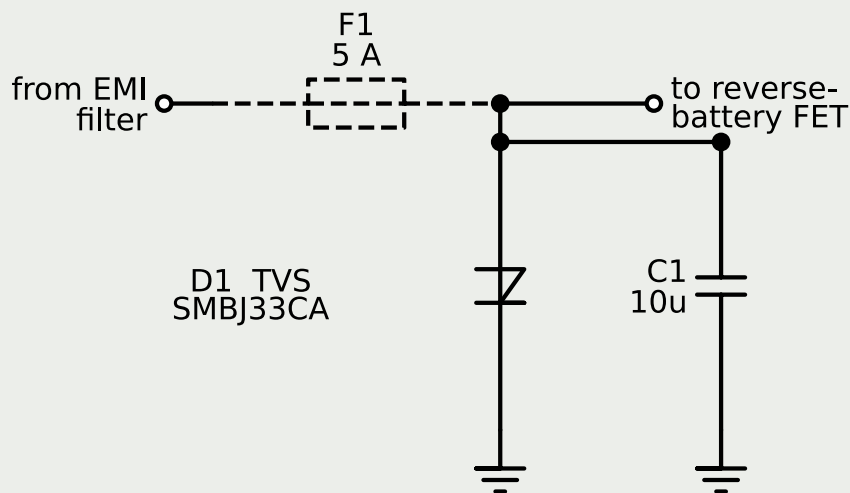
F1 dashed: on the board per spec sec.4, but this netlist folds all series impedance into Rsrc -- F1 is not modelled, and its survival under pulse 2a's repeated ~38 A stress is untested

A bidirectional TVS behind a 5 A fuse, against ISO 7637-2 pulse 2a: +112 V through 2 ohms for 50 us. Second and third stages of spec sec.4's pin 21 order — EMI filter, then fuse, then this TVS, then the reverse-battery FET.

Why it looks like that. Clamps to 50 V, which leaves 2x margin inside the buck controller's 100 V rating and keeps 60 V-class parts viable everywhere downstream.

What the model does not cover. It misses the 42 V figure the earlier documents asked for, and that figure should be dropped: it was arbitrary, with no standard behind it. The real constraints are the parts' ratings. The fuse (dashed in the schematic) is not modelled either — this netlist folds all series impedance into Rsrc — and its survival under pulse 2a's repeated ~38 A instantaneous stress has not been tested, unlike load_dump, which did check this for its own pulse.

PASS pulse actually applied (source peak)	125.500
PASS clamped voltage at buck input	49.998
PASS ... inside LM5164's 100 V rating	2.000 x
PASS ... 60 V-class parts still viable	49.998
PASS ... but misses the inherited 42 V target	50.0 V -- target was arbitrary
PASS recovers to nominal after pulse	12.981 V
Netlist: sim/blocks/transient_clamp.cir	



pulse 5b: $U_s^* = 40\text{ V}$, 0.5 R , 400 ms
 D1 absorbs $\sim 46\text{ J}$ at $\sim 116\text{ W}$ -- FAILS an SMBJ-class part's rating
 F1 dashed (drawn, not in this netlist) does not rescue it either:
 fault current is 3.04 A , $0.6\times$ F1's rating, and won't clear in 400 ms

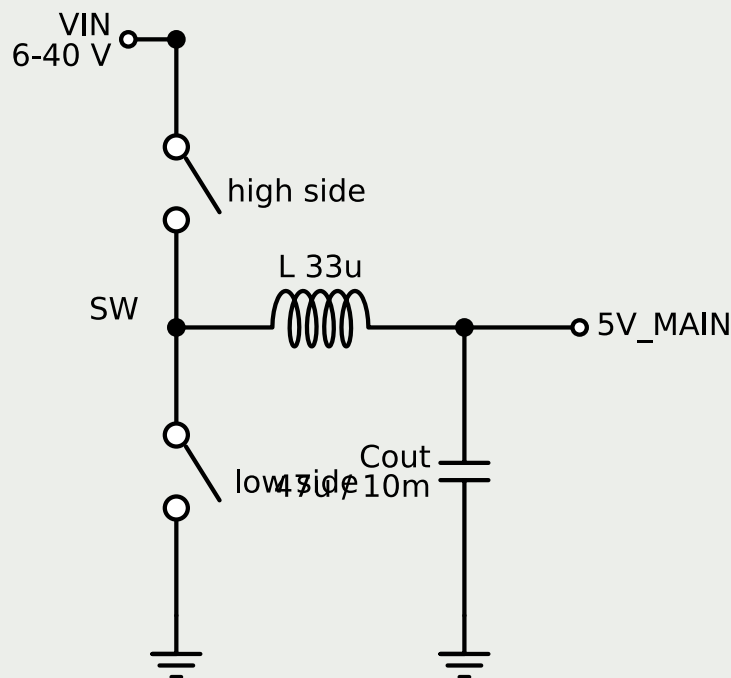
The same TVS and capacitor as transient_clamp, against ISO 7637-2
 pulse 5b: a suppressed load dump, 40 V through 0.5 ohms for 400 ms —
 four orders of magnitude longer than pulse 2a.

Why it looks like that. Clamps to 38 V , comfortably inside every voltage rating in the chain — the same story as pulse 2a. But 400 ms is long enough that the limit stops being clamp voltage and becomes dissipated energy: the TVS absorbs roughly 46 J at $\sim 116\text{ W}$ average, which is on the order of $100\times$ an SMBJ-class part's single-pulse energy rating and its continuous power rating both. This block FAILS on purpose — a TVS sized against pulse 2a's peak voltage was never sized against pulse 5b's energy, because nothing before this block asked the energy question. But the whole failure turns on one inequality: re-run at $U_s^* = 35\text{ V}$, the figure usually quoted for a 12 V suppressed dump, and the TVS never reaches its 36.7 V breakdown — 0.000 A , 0 J , no problem at all. So the finding is not that an SMBJ33CA cannot survive a load dump. It is that a TVS whose standoff sits BELOW the dump level gets forced to conduct during a normal event and then has to absorb it. The fix is a higher standoff (SMBJ40CA/45CA class), not a bigger energy rating: the TVS then stays off and the buck, rated 100 V , rides 40 V without complaint. Energy absorption only becomes unavoidable under pulse 5a.

What the model does not cover. Whether this pulse even applies: 5b assumes the alternator's rectifier is suppressed/clamping. That has not been confirmed for this genset — if it is not, the real event is pulse 5a (65–87 V, unclamped), a harder problem this block does not simulate. Also not modelled: the real pulse's exponential-decay shape (approximated here as a trapezoid, which is pessimistic in the right direction), and the TVS's actual junction thermal transient. The 5 A fuse (dashed in the schematic, not a netlist element) does not rescue this failing case either: the fault current is 3.04 A, 0.6x the fuse's rating, and it will not clear in 400 ms.

PASS pulse actually applied (source peak)	40.000 V
PASS clamped voltage through the 400 ms plateau	38.097 V
PASS ... inside LM5164's 100 V rating	2.625 x
PASS ... 60 V-class parts still viable	38.097 V
PASS TVS current through the plateau	3.044 A
PASS average power dissipated in the TVS	115.954 W
PASS total energy absorbed by the TVS over the pulse	46.440 J
FAIL energy stays within a single-pulse (~0.5 J) rating	46.440 J
FAIL average power stays within continuous (~3 W) rating	115.954 W
PASS ... exceeds the continuous rating by	38.651 x
PASS recovers to nominal after pulse	13.366 V
Netlist: sim/blocks/load_dump.cir	

400 kHz -- above CISPR 25's 150 kHz band start

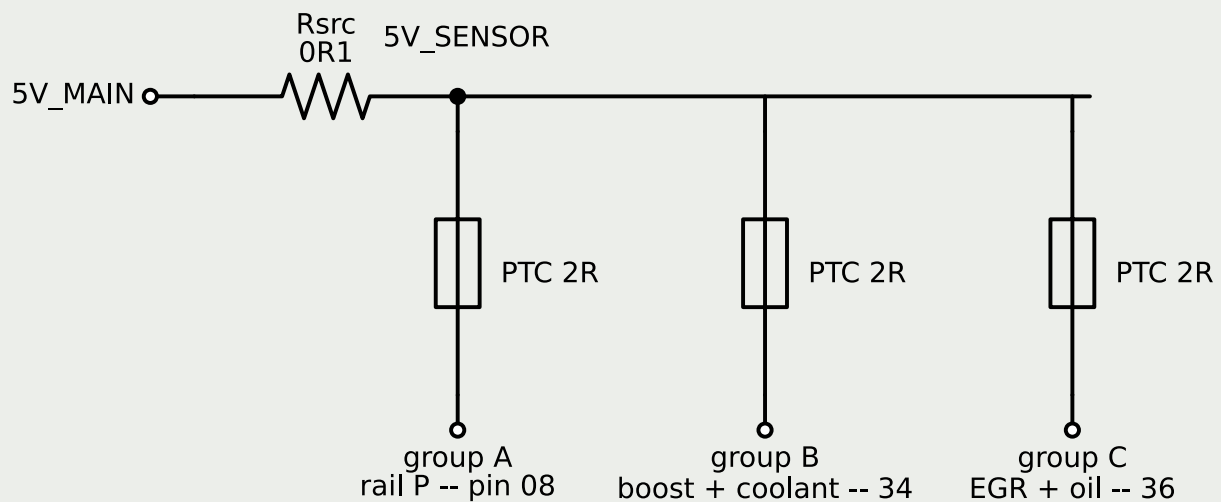


6-40 V down to 5 V at 400 kHz, run open-loop at fixed duty so the output filter is visible.

Why it looks like that. 400 kHz is chosen against CISPR 25, not for size: the conducted band starts at 150 kHz, so a fundamental below it would put every harmonic inside the measurement from the first one up. The inductor is sized at HIGH line, not low – ripple current grows with input voltage even though duty cycle falls.

What the model does not cover. Ideal switches with a fixed on-resistance. Good enough for ripple, which is set by L and C ; useless for efficiency, which is set by switching overlap, gate charge and reverse recovery, none of which this model has.

PASS output at 13.5 V in (open loop, D=0.370)	4.926 V
PASS output at 40 V in (open loop, D=0.125)	4.945 V
PASS inductor ripple at 13.5 V	0.256 A
PASS inductor ripple at 40 V	0.348 A
PASS ... worst case is HIGH line, not low	1.362 x
PASS worst-case ripple as fraction of 1 A load	34.840 %
PASS output ripple at 13.5 V	13.845 mV
PASS output ripple at 40 V	21.423 mV
Netlist: sim/blocks/buck_preregulator.cir	



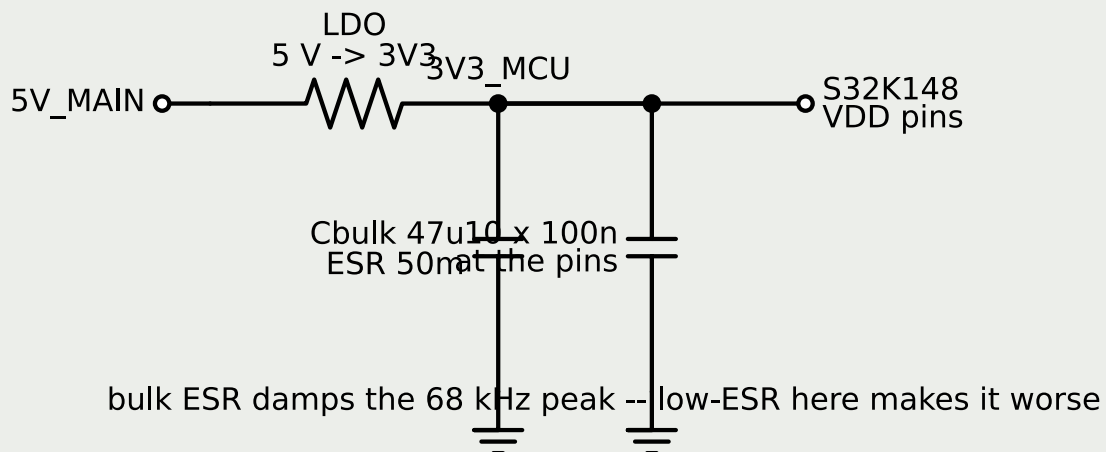
One resettable fuse per sensor group, and the same fault simulated without them.

Why it looks like that. A 5 V sensor supply shorted to engine ground is a routine fault, not an exotic one. The question is what it does to the OTHER sensors: with per-group protection the healthy groups keep 4.66 V, without it the whole rail collapses to 0.5 V and the ECU loses every analog channel at once — which means it cannot tell a wiring fault from an engine failing.

What the model does not cover. The polyfuse is modelled cold, which is the worst case for the healthy groups: the instant after the short, before the device has heated and tripped. Once tripped the healthy groups recover further. Firmware still has to notice — a tripped fuse and a dead sensor look identical from the ADC's side.

PASS healthy group with PTCs, group A shorted	4.661 V
PASS ... rail itself holds up	4.754 V
PASS healthy group WITHOUT PTCs, same fault	0.495 V
PASS ... so one harness short blinds the whole ECU	0.50 V -- every analog channel lost at once
PASS protection improves the healthy group by	9.409 x
PASS fault current drawn from the 5 V rail	2.458 A

Netlist: sim/blocks/sensor_rail.cir



The impedance the S32K148 sees looking into its own supply, from 1 kHz to 100 MHz.

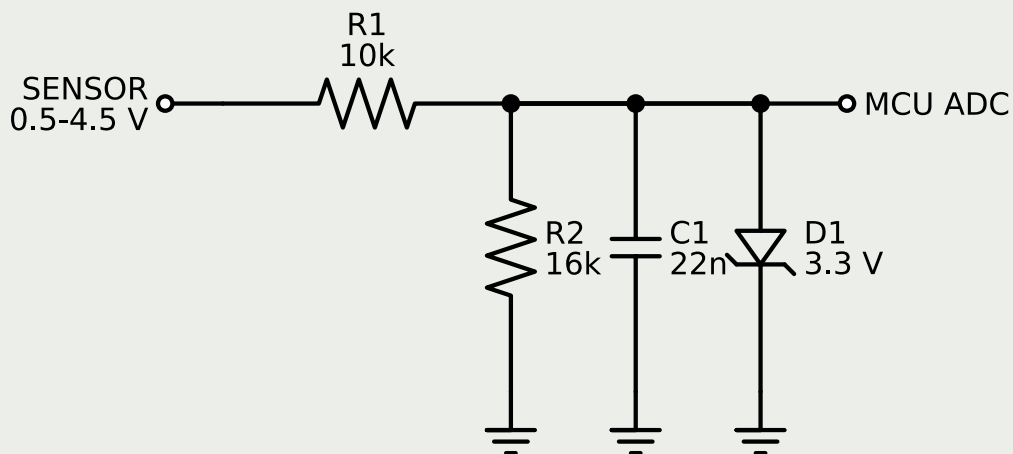
Why it looks like that. The bulk capacitor here is sized by a resonance, not by charge. The first version used 10 uF with low ESR and peaked at 3.4 ohms around 48 kHz – the regulator's own output inductance ringing against the bulk capacitance, decades away from the bulk/ceramic anti-resonance the design was drawn around. Bigger AND lossier fixes it, which is why a plain aluminium part beats a ceramic in that position.

What the model does not cover. Lumped: no planes, no via inductance, no spreading inductance, so it is optimistic above about 100 MHz – which is exactly where the sweep stops. An earlier version swept to 1 GHz and obediently reported ESL times omega at the last point as a 'peak'.

PASS peak impedance, 1 kHz to 100 MHz	74.197 m0hm
PASS ... and it sits at	67.608 kHz
PASS ... i.e. regulator L against bulk C, not bulk against ceramic	68 kHz -- decades below the ceramics
PASS ten 100 nF parts at 100 MHz	46.923 m0hm
PASS one 1 uF part at 100 MHz	357.419 m0hm
PASS ... same capacitance, this much worse	7.617 x
PASS ten-part net at the 3.5 MHz anti-resonance	55.468 m0hm
PASS ... one-part net is LOWER there (its ESR damps it)	34 m0hm vs 55 -- more ESR, better damping
Netlist: sim/blocks/mcu_pdn.cir	

02 **Sensor front-ends**

Everything that turns something on the engine into a number in the ADC.



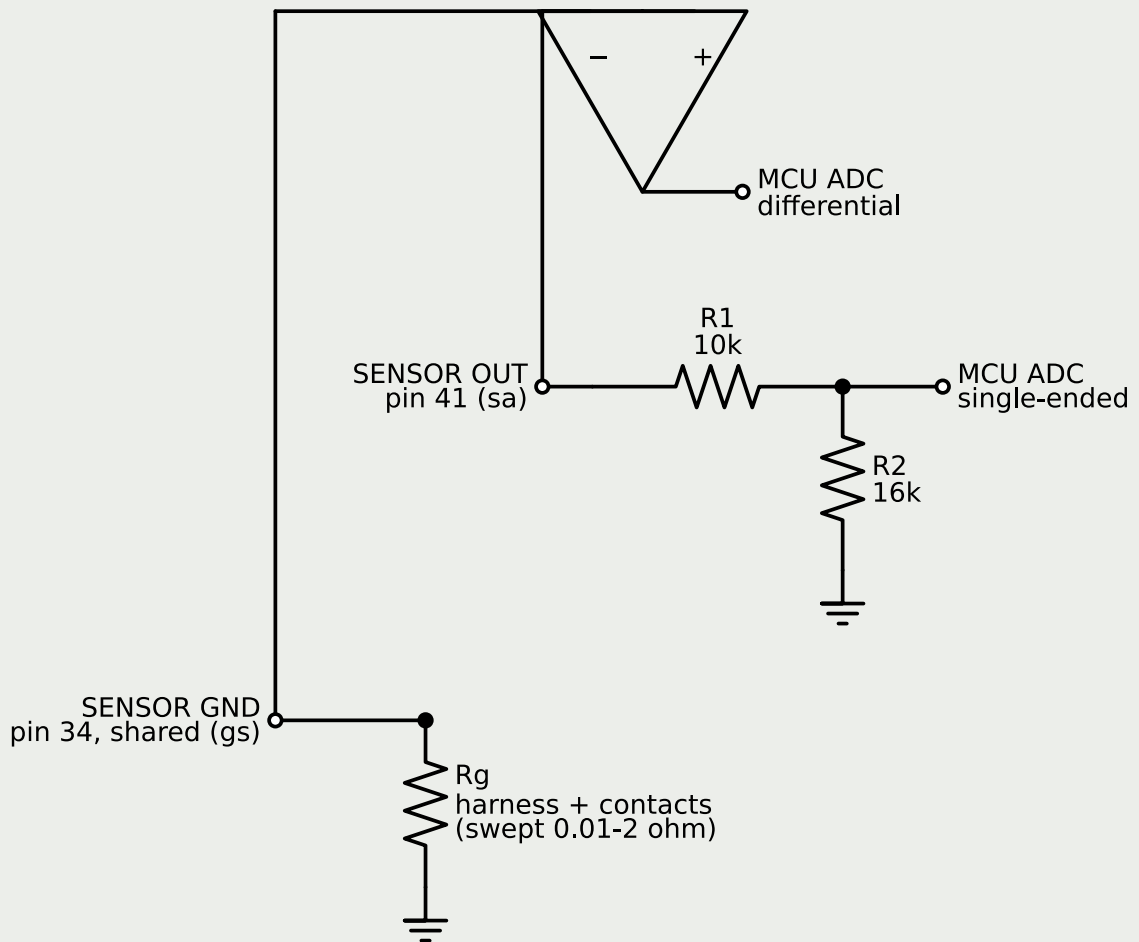
The 0.5–4.5 V pressure and position channels, scaled into a 3.3 V ADC.

Why it looks like that. Full scale deliberately lands at 2.77 V rather than hard against the ceiling, so the protection clamp can do its job on a harness fault without ever touching the signal in normal running. That costs a little span and buys a front-end 40 V cannot damage. The last four checks cover the ratiometric property itself, which this block was named for and did not model until the spec review caught it: a sensor's output is a fraction of its own supply, so a reading taken against an assumed 5.000 V carries the rail's whole error band — 10% here. Giving the rail its own ADC channel through an identical divider removes it, and the divider ratio cancels algebraically, so its tolerance drops out too.

What the model does not cover. Ground offset. Every network here measures against a perfect ground, and pins 34 and 36 are shared returns — which is why spec §4 now specifies a differential front-end for those channels and leaves this one for pin 08, the dedicated return. Also not modelled: divider tolerance matching, on which the rail correction depends, and ADC sampling time against the source impedance.

PASS sensor zero (0.5 V) maps to	0.308 V
PASS sensor full scale (4.5 V) maps to	2.769 V
PASS ... clears the 3.3 V clamp	0.531 V
PASS span used of ADC range	74.592 %
PASS low-pass corner	1,174.898 Hz
PASS loss at sensor's own 100 Hz	-0.031 dB
PASS attenuation at 100 kHz switching	-38.596 dB
PASS 40 V harness short clamps pin to	3.308
PASS ... fault current limited to	3.669 mA
PASS raw reading moves with a +/-5% rail drift	10.000 %
PASS ... so an assumed 5.000 V rail is a 10% error band	10.0% of reading, and nothing in the raw value reveals it
PASS corrected reading is flat	0.000 %
PASS correction recovers the true fraction	0.500
PASS ... divider ratio cancels, so its tolerance drops out too	measurand = V_{sig}/V_{rail} -- k appears in both and divides away
Netlist: sim/blocks/sensor_ratiometric.cir	

diff amp -- netlist's Bd: gain 16/26 (matches the single-ended ratio) plus a realistic 80 dB CMRR term



Not drawn: Ib (netlist, 20 mA into gs) -- the coolant sensor's own return current, sharing this ground via splice V3. It is what moves gs; Rg is what decides how far it moves.

The same channel measured sensor-to-sensor-ground instead of sensor-to-ECU-ground, swept against the harness return's resistance.

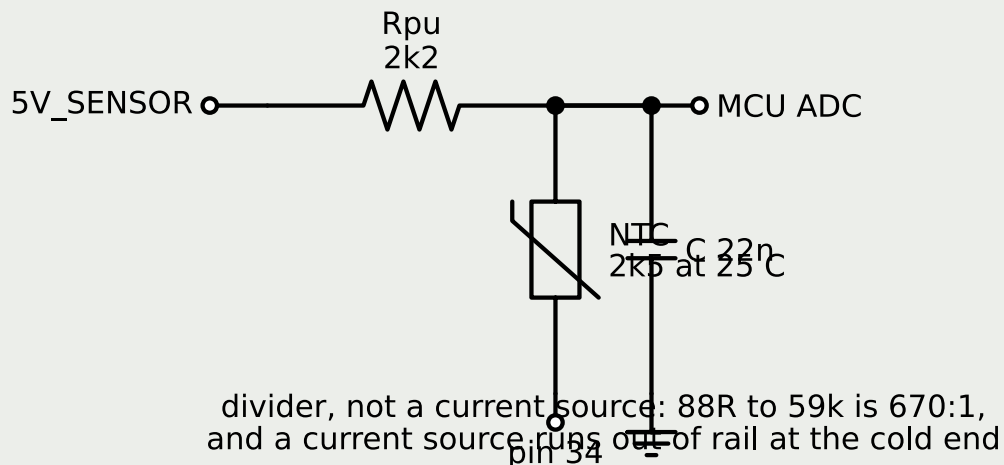
Why it looks like that. Pin 34 is spliced between the boost sensor and the coolant sensor, so it carries both sensors' return current. A single-ended input reads the resulting offset as signal. Research memo 09 found this is exactly what Deep Sea Electronics specifies against (± 2 V common mode on every sender channel) and what SEDEMAC's dedicated sensor common point exists for. Note what the OEM did with the channel that could not tolerate it: rail pressure gets pin 08, spliced with nothing. The schematic below draws one divider ($R1/R2$, the single-ended path) plus a generic amplifier for the differential path,

because that is what the netlist models — a behavioral gain block (Bd: 16/26 gain, matched to R1/R2's own ratio, plus an 80 dB CMRR term), not a second resistor divider. An earlier version of this drawing showed two dividers joined by a plain wire, which matched neither the netlist nor any circuit that would actually reject a ground offset.

What the model does not cover. The error is about half a percent of span at 1 ohm of return resistance, and that understates it. It is a bias rather than noise, so it never averages out; it grows as contacts corrode; and it is indistinguishable from a real reading — a boost pressure biased high permits more fuel, and the first symptom is smoke, not a fault code. Rg, the swept harness + connector resistance, is drawn; Ib, the 20 mA coolant-sensor current that actually moves gs, is not — it is a stimulus, not a part of this front end, and it is captioned rather than drawn for that reason.

PASS clean harness (50 mOhm), single-ended error	0.612 mV
PASS 1 ohm return, ground offset present	19.903 mV
PASS ... single-ended reading error	12.248 mV
PASS ... differential reading error	0.127 mV
PASS differential is better by	96.448 x
PASS single-ended error at 2 ohm (bad contact)	0.995 % span
PASS ... single-ended needs contact resistance below	~0.1 ohm forever, to hold 0.1% -- differential needs nothing

Netlist: sim/blocks/sensor_differential.cir

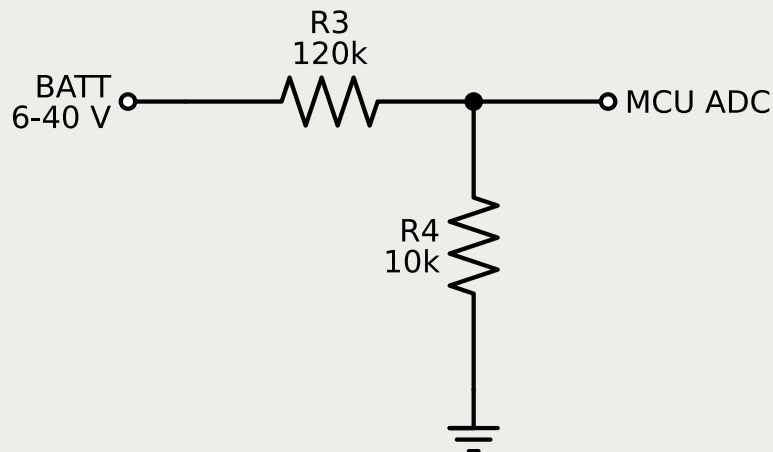


The thermistor branch of U2, as a pull-up divider and as a constant-current source, across the full -40 to 150 C range.

Why it looks like that. Memo 09 found DSE drives its temperature senders with a fixed 15 mA and measures differentially, and the obvious move is to copy it. Checked against an actual charge-air NTC it does not transfer: 88 ohms to 59 kilohms is a $670:1$ span against the $50:1$ a genset sender covers, and a current source sized to give useful signal at 150 C demands 5.9 V from a 5 V rail at -40 C. The divider's non-linearity is what saves it — it is bounded by the supply at both ends.

What the model does not cover. Which branch this engine actually needs is still U2, and one resistance measurement at the machine settles it. The self-heating figure should be revisited once the real sensor's dissipation constant is known.

PASS divider at 150 C (88 ohm)	0.192 V
PASS divider at 25 C (2.5k)	2.664 V
PASS divider at -40 C (59k)	4.820 V
PASS ... never leaves the 5 V rail	4.820 V
PASS 100 uA source needs this at -40 C	5.899 V
PASS ... but the rail is 5 V	5.9 V demanded -- out of compliance, reading clips
PASS 100 uA source at 150 C gives only	8.800 mV
PASS current that WOULD stay in compliance	84.746 uA
PASS ... leaving this at 150 C	7.458 mV
PASS divider self-heating power at 25 C	2.829 mW
PASS ... error at ~2 mW/C dissipation constant	1.414 degC
Netlist: sim/blocks/ntc_frontend.cir	



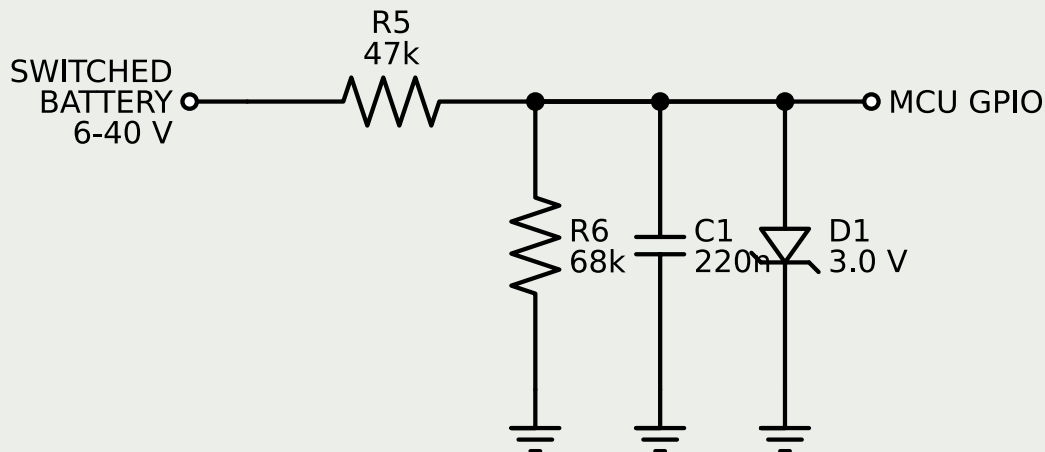
R3/R4: the adopted divider (spec sec.4, pins 04/06).
battery_sense.cir also carries a rejected R1/R2 75k/10k
divider, sized for a 5 V ADC -- not drawn here.

The 6–40 V system rail scaled into the ADC.

Why it looks like that. The inherited 75k/10k divider was sized for a 5 V ADC and reaches 4.69 V at 40 V — it would have damaged the input pin. 120k/10k peaks at 3.07 V and still resolves 573 counts at the 6 V cranking dip. The schematic below draws only the adopted 120k/10k divider, and it now carries the netlist's own designators for it, R3/R4 — spec §4 gives the same 120k/10k for pins 04/06. The rejected 75k/10k comparison is R1/R2 in battery_sense.cir; it is not drawn, since it is not the circuit on the board, and using its designators for the drawn divider (an earlier version of this drawing did) pointed a reader at the wrong resistors.

What the model does not cover. Nothing subtle. A resistive divider is exactly linear and the check confirms it, which is there to catch a wrong topology rather than a wrong value.

PASS inherited 75k/10k peak at 40 V	4.694 V
PASS ... and so exceeds a 3.3 V ADC	4.69 V > 3.3 V -- would damage input
PASS corrected 120k/10k peak at 40 V	3.069 V
PASS ... stays under the 3.3 V rail	3.069
PASS headroom below rail	0.231 V
PASS reading at 6 V cranking dip	0.462 V
PASS ... resolves to usable ADC counts	572.867 LSB
PASS divider is linear (max residual)	0.000 V
PASS slope matches 10/130	0.077
Netlist: sim/blocks/battery_sense.cir	



The switched-battery inputs, active high.

Why it looks like that. This one cannot be built as a divider at all. Reading logic high at a 6 V cranking dip needs a ratio above 0.385; staying under 3.3 V at 40 V needs it below 0.0825. No fixed ratio satisfies both, so the original was impossible rather than merely mis-valued. A 3.0 V zener sets the level and the divider only biases into it, which holds the output flat within 92 mV across the whole range.

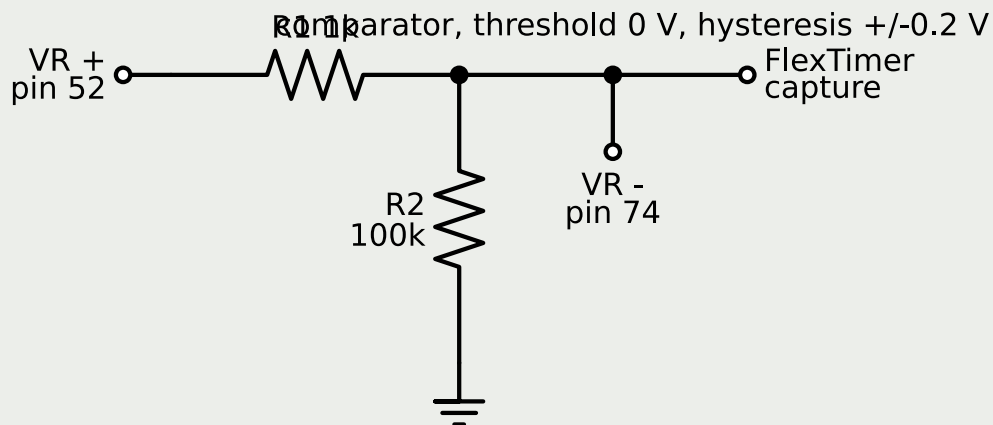
What the model does not cover. The debounce capacitor went from 100 nF to 220 nF after simulation showed the smaller value let two threshold crossings through a bouncing contact — which firmware would have read as two separate presses.

PASS never exceeds 3.6 V abs max over 6-40 V	2.954
PASS logic high at 6 V cranking dip	2.863
PASS logic high at 40 V load dump	2.954
PASS level is flat across 6-40 V (spread)	0.092 V
PASS settled high after bounce ends	2.910
PASS threshold crossings during bounce	0.000
PASS debounce time constant	6.114 ms

Netlist: sim/blocks/discrete_input.cir

03 **Speed and position**

The most safety-critical input on the board.



A variable-reluctance sensor into a zero-crossing comparator, run at both running speed and cranking speed.

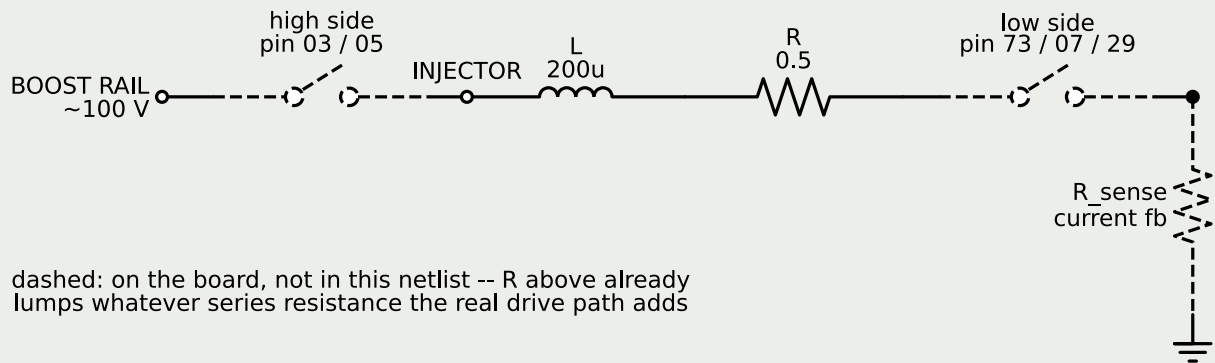
Why it looks like that. A VR sensor generates voltage by induction, so its amplitude is proportional to speed: roughly 20 V peak at 1500 rpm and 2 V at cranking. A fixed 5 V threshold, which looks perfectly sensible against a 20 V signal, produces all 30 teeth at running speed and NOTHING at cranking. That is the worst failure shape there is — it works on the bench and the engine never starts. Zero-crossing detection does not care about amplitude, and the crossing is also where the waveform is steepest and timing jitter is smallest.

What the model does not cover. A real VR signal is not a clean sine — tooth edges distort it and the missing-tooth sync gap produces one large excursion per revolution. Amplitude at true cranking speed on a cold engine with a weak battery can be under a volt, which is when starting matters most, and the hysteresis window needs checking against the real sensor rather than this 2 V estimate.

PASS zero-cross, running (1500 rpm, 30 teeth)	30.000 edges
PASS zero-cross, cranking (150 rpm, 3 teeth)	3.000 edges
PASS fixed 5 V threshold, running	30.000 edges
PASS fixed 5 V threshold, cranking	0.000 edges
PASS ... so a fixed threshold never works at 1500 rpm, blind at 150 rpm -- passes on the bench starts the engine	
PASS running tooth period	666.667 us
PASS ... jitter across the window	0.000 us
Netlist: sim/blocks/vr_conditioner.cir	

04 Injection

The most expensive stage on the board and the reason it exists.



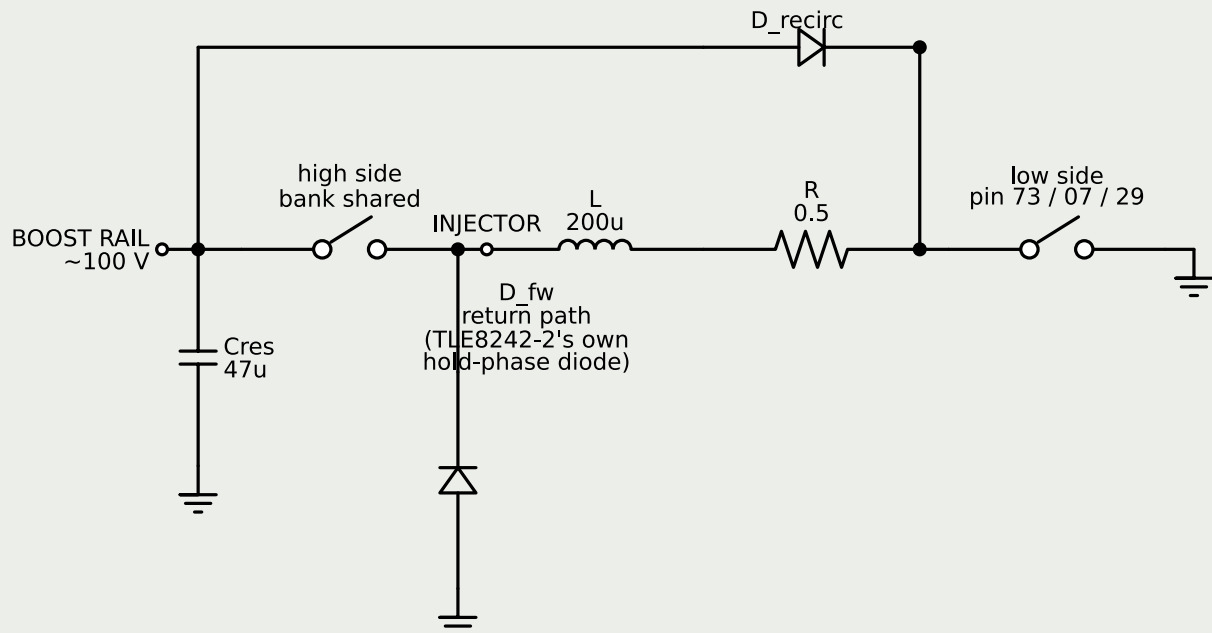
The same injector driven from the battery and from a 100 V rail, compared on time to reach the 18 A peak threshold.

Why it looks like that. 439 us from the battery, 38 us from the boost rail. On a 1 ms injection the battery drive spends 44% of the event just lifting the needle; the boost rail spends 3.8%. That ratio is the entire argument for the stage.

What the model does not cover. An RL model of the injector with no magnetic saturation and no moving needle. The envelope it sits in — 65–115 V, 12–24 A peak — is family-wide from memo 04, not this injector's measured values. That is U5. Also not modelled: the high/low-side switches and the sense resistor, dashed in the schematic — this is a turn-on-only step response straight onto L/R, and R already stands in for whatever series resistance the real drive path adds.

PASS time to 18 A peak from 13.5 V battery	439.496 us
PASS time to 18 A peak from 100 V boost	37.796 us
PASS boost is faster by	11.628 x
PASS battery drive eats this much of a 1 ms injection	43.950 %
PASS boost drive eats this much of a 1 ms injection	3.780 %

Netlist: sim/blocks/injector_boost.cir



18 A $\rightarrow$ ~0 A in ~33 μ s, rail bump ~+6.4 V/event (measured)
 memo 10 predicted ~36 μ s / ~+6.9 V ideal -- real diode and switch losses shave both, and the bump undershoots the 7.3 V on-phase droop it is paired with -- net loss, not gain

The low-side switch opening at 18 A, with a recirculation diode routing the coil's current back into the same boost cap (Cres, 47 μ F) instead of a clamp that just burns it — plus a second diode (D_fw) the loop needs to close, and a comparison against freewheeling to the battery instead.

Why it looks like that. Research memo 10 surveyed real injector-driver topologies and ruled out a Zener/active clamp (Nexperia AN50003 flags repetitive active clamp as reliability-“Questionable”) and a bare battery freewheel (too slow) in favor of recirculation to the existing boost rail — confirmed production diesel-injector practice via a 1988 Marelli Autronica patent, not a textbook flourish. Simulated: 33 μ s to collapse 18 A to zero (memo predicted ~36 μ s), 25x faster than freewheeling to the battery (830 μ s — slower than the memo's own simplified estimate, because a real RL circuit's exponential tail matters more at 13.5 V than at 100 V). The rail bump per event is +6.4 V, under the memo's +6.9 V ideal because real diode and switch losses are modelled here, and it undershoots the 7.3 V on-phase droop it is paired with — net loss, not gain, so three back-to-back events never push the rail over its 100 V setpoint. But that verdict hinges on which current is being cut off: boost_converter.cir's own arrangement A has the battery supply HOLD current, not the boost rail, and recirculating a hold-current cutoff through the same diode has no offsetting boost-side draw to net against

— modelled separately here, it overshoots to 102 V on a single event. boost_converter.cir's charge regulation (Bchg) has no bleed path either way; it just happens not to need one for the peak-only case.

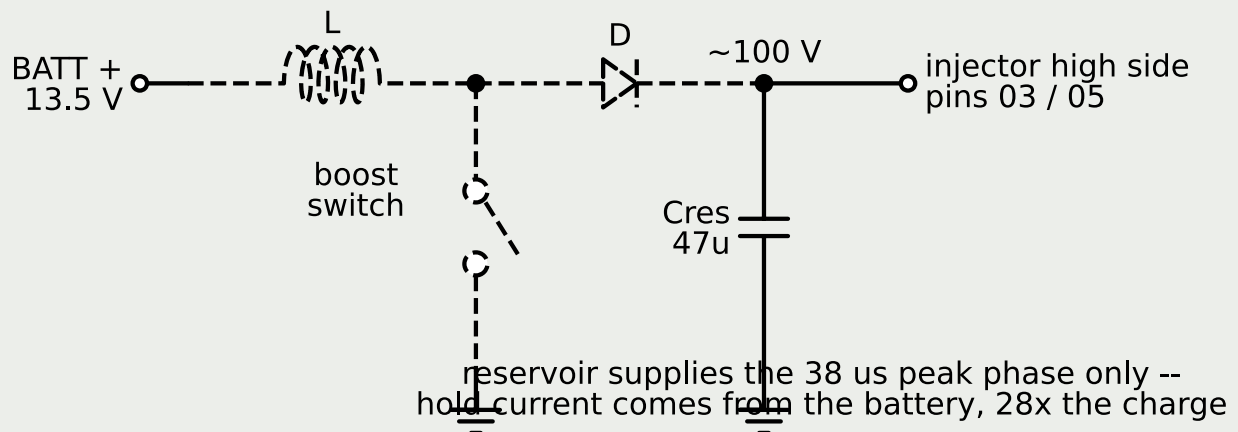
What the model does not cover. The differential-mode EMI edge (memo's own claim 5) is not modelled — the switches here are ideal with zero transition time, so any dV/dt read off this simulation would be a solver artifact, not a FET's real slew rate. Also not modelled: diode reverse recovery (Q_{rr}), the diode's thermal transient, and — inherited from injector_boost.cir — magnetic saturation and needle motion.

PASS turn-off time, 18 A → ~0 A on the boost rail	
PASS ... vs freewheel-to-battery baseline (comparison only)	
PASS ... boost recirculation is faster by	
PASS on-phase droop, pilot event	
PASS ... matches boost_converter.cir's own droop figure	
PASS recovery bump, pilot event (isolated)	
PASS ... undershoots the droop it is paired with (real losses)	
PASS rail never exceeds 100 V setpoint (3-event circuit)	
PASS rail recovered by the next cylinder (26.67 ms after post)	
PASS hold-current cutoff into the SAME cap, no offsetting draw	
PASS ... THIS is the scenario that overshoots 100 V	
PASS ... so the verdict hinges on whether hold-current cutoffs share the boost-recirculation diode	peak-only recirculation never overshoots; hold-current reci
PASS peak instantaneous power in D_recirc during turn-off	
PASS ... exceeds the ~3.6 W worst-case AVERAGE by	

PASS differential-
mode edge vs
emi_filter.cir

not modelled here (ideal switches) --

Netlist: sim/blocks/injector_turnoff.cir



The reservoir capacitor under a real injection event, with and without the hold phase drawn from the same rail.

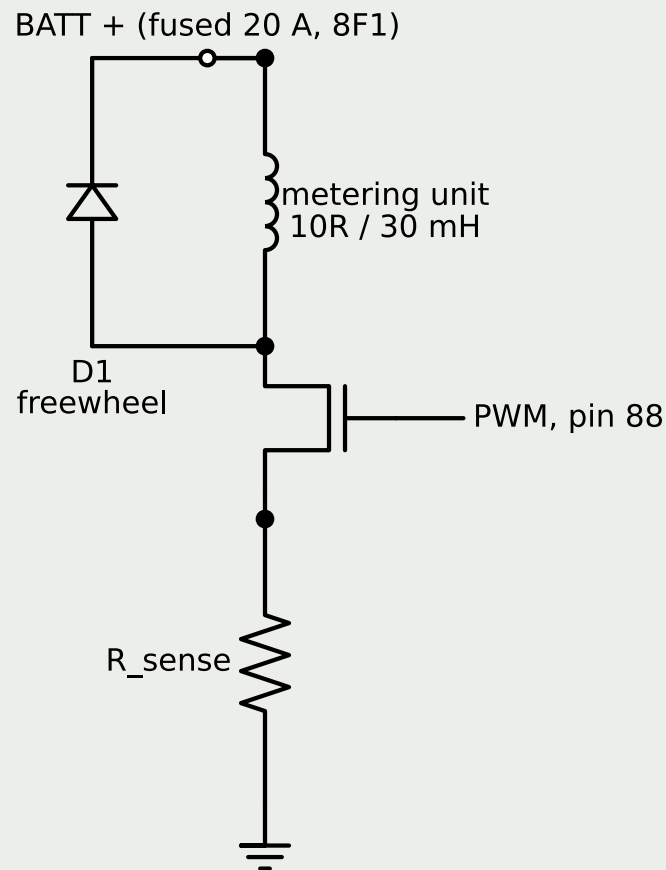
Why it looks like that. The converter cannot deliver 18 A — it delivers a trickle and the capacitor delivers the pulse, so the capacitor is the component that matters. The design question is where the HOLD current comes from, and the answer is not subtle: hold costs 29 times the peak phase's charge, so a rail supplying both collapses completely. Hold comes from the battery. 50 mA of average charging recovers the rail with room to spare in the 26.67 ms before the next cylinder fires.

What the model does not cover. Charge balance only. L, D and the switch are dashed in the schematic because the converter physically has them, but there is no inductor, no switch and no control loop in the netlist — just a current source standing in for the converter's average output. It says nothing about the converter's stability or its inrush, and the collapsed voltage in the second case should be read as 'far below usable' rather than as a number.

PASS droop, peak phase only (47 uF)	7.255 V
PASS ... rail stays usefully above battery	92.745 V
PASS droop, peak AND hold from the rail	210.913 V
PASS ... so the rail collapses	-111 V -- hold must come from the battery
PASS hold phase costs this much more charge	29.071 x
PASS recovered by the next injection (26.67 ms)	100.000 V
PASS ... 50 mA average charging is enough	100.000 V
Netlist: sim/blocks/boost_converter.cir	

05 **Actuators**

Outputs that move something.



Low-side PWM into the metering solenoid at three frequencies, all at the same duty.

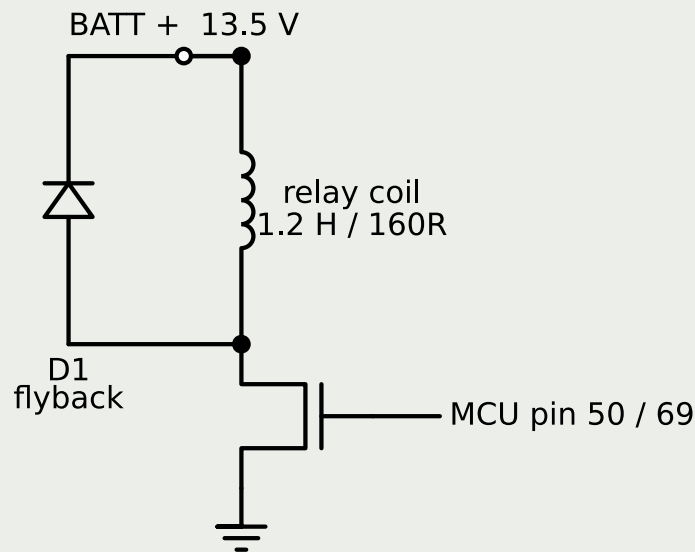
Why it looks like that. Rail pressure is controlled by the AVERAGE current through this coil, and all three frequencies produce the same average — which is what makes the control law independent of the PWM constant. What changes is ripple, and ripple is not purely a nuisance here: a proportional solenoid has stiction, and deliberate dither keeps it moving. At 100 Hz the period is comparable to the coil's own 3 ms time constant and the result is 152% ripple — that is not dither, that is chopping, and the valve follows the PWM rather than its average.

What the model does not cover. The correct dither amplitude is a property of the specific metering unit's mechanics and is manufacturer data tied to a part number — the same gap as U5, in a smaller form. Keeping the frequency a firmware constant

rather than a hardware one means it can be tuned on the machine. Coil inductance is also not constant: it falls as the armature moves and as the core saturates.

PASS mean current at 100 Hz	0.639 A
PASS mean current at 1 kHz	0.639 A
PASS mean current at 10 kHz	0.634 A
PASS ripple at 100 Hz	0.969 A
PASS ... as fraction of mean	151.786 %
PASS ripple at 1 kHz	0.118 A
PASS ripple at 10 kHz	0.012 A
PASS ripple scales as 1/f	9.974 x
PASS 100 Hz is not dither, it is chopping	152% ripple -- valve follows the PWM

Netlist: sim/blocks/metering_unit_pwm.cir



A low-side FET switching a relay coil, run with and without the flyback diode.

Why it looks like that. With the diode the drain stays at 14.3 V on turn-off. Without it the coil's stored energy has nowhere to go except through the FET.

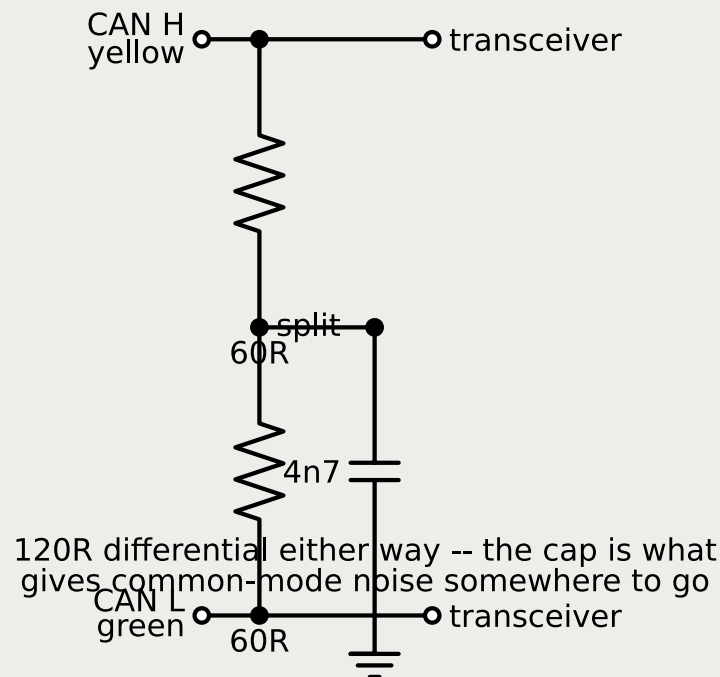
What the model does not cover. The simulated peak without the diode is hundreds of kilovolts, which is the ideal-switch model taken literally. A real MOSFET avalanches at its breakdown voltage and dissipates that energy in the die instead — once per relay operation, until it fails. The magnitude is a model artifact; the conclusion is not.

PASS drain peak WITH flyback diode	14.274 V
PASS drain peak WITHOUT diode	402,899.522 V
PASS ... diode suppresses the kick by	28,226.510 x

Netlist: sim/blocks/relay_driver.cir

06 Communications

The bus to the genset controller.



A split 60+60 termination with a capacitor to ground at the midpoint, against a single 120 ohm resistor.

Why it looks like that. Both are 120 ohms differential and a schematic label cannot tell them apart. The difference is common mode: a single resistor gives noise on both wires together nowhere to go, while the split version shunts it to ground without touching the differential signal, because the midpoint is a virtual ground for differential drive. That matters more here than on a truck. The fitted controller is a Deep Sea DSE4522, and DSE's manual is explicit that with a CAN engine the ECU transmits engine speed to it — so this bus carries a safety-relevant value, and it runs beside 415 V three-phase wiring and contactor coils in the same panel.

What the model does not cover. Termination is not the whole common-mode story; the choke is the other half and needs a layout. And this does not decide whether to fit termination at all: a 120 ohm terminator belongs at each END of the backbone, and if this ECU is not an end node, fitting one makes the bus worse. That is U4 — so the board should carry the parts with a link and let the machine decide. The DSE4522 terminates CAN on its terminals 18/19/20 and DSE specify 120 ohm cable, which settles the cable but not which nodes carry the terminators.

PASS split termination, differential at 250 kbit/s	119.993 ohm
PASS single 120R, differential at 250 kbit/s	119.993 ohm
PASS ... identical differentially	0.000 ohm
PASS split termination, common-mode at 10 MHz	30.189 ohm
PASS single 120R, common-mode at 10 MHz	500.000 kohm
PASS ... split shunts common mode better by	16,562.490 x
PASS split common-mode at the bit rate stays high	138.099 ohm
Netlist: sim/blocks/can_termination.cir	

Genset ECU — Reconciled Specification

Date: 2026-09-13 **Status:** Phase 0 — supersedes all four prior artifacts
Target: Functional replacement for the `ENGINE CONTROL UNIT` block of Kirloskar wiring diagram `08-GP3-60-001` , on a 25 kVA diesel genset with a 12 V / 75 Ah battery.

1. Why this document exists

Four artifacts precede this one:

Artifact	Rev	Role
Genset ECU Design Brief	J	Spec
ECU Circuit Guide	H	Per-block why/how/prove
ECU Circuit Schematics	H	15 topology sheets
PCB Design Roadmap	A	Brief → fab sequence

They were written by reverse-engineering the Kirloskar wiring diagram from memory of it, not from the document itself. Now that `GP3.8703.C4.pdf` is in hand, roughly a third of their content is wrong — not stylistically, but in ways that change the BOM, the driver topology and the pin count.

This document is the single source of truth from here. Where it contradicts an artifact, this document wins. The artifacts remain useful for the reasoning behind blocks that survived unchanged, and the Circuit Guide's "prove it" bench procedures remain valid for those blocks.

Raw extracted evidence: `refs/ecu-pinout-extracted.md` .

2. What changed, and why it matters

2.1 Injector drive is bank-shared

The diagram traces injector high-side pin `03` to both cylinder 1 and cylinder 3, joined at splice V5. Pin `05` serves cylinder 2 alone. Low-side selects are individual: `73` (cyl 1), `07` (cyl 3), `29` (cyl 2).

Five pins, two banks. The artifacts specify an independent high/low pair per cylinder — six pins, three banks — which would put three boost stages on a board that needs two.

This is the conventional arrangement: the boost converter and high-side switch are the expensive, thermally-significant parts, so they are shared across cylinders that never fire simultaneously. Cylinder selection happens on the low side, which is cheap.

Consequence: two high-side bank switches, one shared boost rail, three low-side current-controlled selects.

2.2 Fuel metering unit is low-side PWM, not an H-bridge

Metering unit pin 2 sits on fused battery (20 A, 8F1). Pin 1 goes to ECU pin 88 on a black wire — the low-side class per the page-2 legend. The ECU sinks current; it does not source it, and it never reverses polarity.

The Design Brief specifies a TLE9201SG H-bridge with one leg tied off. That works electrically but buys a reversing bridge for a solenoid that only pulls one way.

Consequence: one high-current low-side driver with current sense, closed-loop in firmware. One fewer part family to source.

2.3 Fuel level and water-in-fuel are not ECU signals

Both route through the XC11/XC1 36-pin harness to the panel — fuel level to XC pin 22 (wire 1C2), water-in-fuel to XC pin 16/1 (wire 4B). Neither reaches an ECU pin anywhere in the diagram. They belong to the GCU.

Consequence: two analog front-ends and two connector positions removed.

2.4 Discrete inputs are active-high, not active-low

AUDIO ABORT (20), OVERSIDE SWITCH (24) and IGNITION (71) are all red wires — switched battery. The switch connects the pin *to battery*, not to ground.

The artifacts specify pull-up to 5 V with the switch closing to ground. That is the opposite sense, and the front-end is wrong for it: these pins see battery voltage, not 5 V, so they need a divider rated for the full 6–40 V input range plus a clamp.

The fail-safe reasoning also inverts. Circuit Guide Group D argues that normally-closed wiring makes an open harness read as a fault. With switched-battery inputs an open wire reads "not asserted" — indistinguishable from a switch that simply isn't pressed. That argument does not hold here and should not be repeated in the new design. Detecting an open wire on these inputs requires either a pull-down and a diagnostic current source, or accepting that it is not detectable. **We accept it is not detectable** and note it in the FMEA, because changing the sense would break harness compatibility.

Pins 01 (G_G_B AT1) and 02 (G_G_B AT2) are ECU-provided return/ground legs for the switch circuits. The coolant switch (43) and droop switch (23) are cyan — digital-input class — and are conventional switch-to-return inputs.

2.5 Both CAN channels are real

Pins 55 / 77 and 87 / 86 are both drawn as twisted pairs with bidirectional markers. The artifacts treat CAN1 as a spare, DNP on the harness. In the OEM system both are wired.

Consequence: populate both transceivers and both connector positions.

2.6 Scope reductions confirmed

None of the following exist anywhere in this system: magnetic pickup unit, DEF/urea sensor, LIN bus, SCR, separate overspeed switch, barometric sensor in the harness. There is **one** catalyst temperature sensor connector (2-pin), not three EGT channels.

OVERSIDE SWITCH (pin 24) and Override SS SW -S7 are one component. The Design Brief lists them as two.

3. Safety position

Revised 18 Sep 2026 — second revision, and it moves in the pessimistic direction. The genset controller has been photographed and identified. It is a Deep Sea Electronics DSE4522 MKII AMF (India SP), part 4522-001-01 , serial 11021794 — **not the KG640C** that the OEM wiring diagram shows and that research memo 02 analysed. This section was built on memo 02. Its central claim is now known to be false for the device actually fitted, and the relaxation that claim bought is withdrawn.

What changed

The previous revision rested on this: *the controller senses speed from alternator frequency, not from CAN, so its overspeed trip cannot be defeated by a hung ECU.*

That is true of the KG640C. **It is not true of the DSE4522.** From DSE's own operator manual for the DSE4510/4520 MKII (document 057-260 , the 4522's siblings):

"If the unit has been configured for CAN, compatible ECU's receive the start command via CAN and **transmit the engine speed to the DSE controller.**"

and, in the engine-at-rest detection logic:

"Engine speed is zero as detected by the CAN ECU"

Engine speed reaches this controller from our ECU, over CAN. The configured overspeed shutdown at 1710 rpm therefore reads a number we send. An ECU that hangs while transmitting a stale, plausible speed defeats it completely — which is precisely the failure mode this section exists to guard against.

What is confirmed, and what replaces it

No magnetic pickup — now confirmed rather than inferred. The DSE4522's full terminal list runs 1–35 and contains no speed-pickup input of any kind: 1–2 DC supply, 3–4 outputs A (FUEL) and B (START), 5 charge fail/excite, 6–9 outputs C–F, 10 sensor common, 11–13 analogue sensors, 14–17 digital inputs, 18–20 CAN, 21–24 generator voltage sensing, 25–28 mains, 29–35 CTs, charge alternator and comms. Read from the manufacturer's own terminal table, not from a diagram's silence.

An independent path does still exist — but it is a different mechanism, and a more conditional one. The module measures generator frequency directly at terminals 21–24, straight off the alternator's three phases. It does not use CAN to do this. The configuration sets:

Protection	Trip	Source	Equivalent speed (4 poles)
Generator over-frequency shutdown	56.0 Hz (112 %)	Alternator terminals 21–24 — independent of CAN	1680 rpm
Engine over-speed shutdown	1710 rpm	CAN, from our ECU	1710 rpm

On a 4-pole alternator at no slip, speed and output frequency are rigidly linked. The over-frequency trip fires at 1680 rpm, before the CAN-dependent overspeed trip at 1710 rpm, and it cannot be defeated by anything our ECU does or fails to do.

Its conditions, stated plainly. It is a generator protection being relied on as an engine protection, and it inherits that mechanism's dependencies: the alternator must be excited and producing measurable output, and the AVR must be working. An excitation failure removes the protection. That is a narrower guarantee than a magnetic pickup watching the flywheel, and it should not be described as equivalent to one.

Where the chain terminates — U12 CLOSED 18 Sep 2026, and it closed badly

The measurement came back: the fuel relay cuts a signal. It does not remove power.

Reported from the machine alongside two other findings — the intake throttle and the pre-heat/post-heat heaters are all fed **from the battery**, not from the ECU.

This is the outcome the inspection brief told the reader to flag immediately rather than save for a writeup, and the reasoning below was written before the answer was known. It stands as written; what follows is what it now means.

Both independent lines of defence are gone, for two separate reasons.

The controller has no independent speed path — it takes engine speed from this ECU over CAN (U13, closed 18 Sep). And its every protective action terminates in a relay that only signals. So a hung ECU keeps the fuel metering unit energised, and the device that is supposed to catch that is asking the hung ECU how fast the engine is turning.

What survives: generator over-frequency at 56 Hz, measured from the alternator, which does not depend on this ECU for its *measurement* — but which acts through the same signalling relay. It can detect the runaway. It cannot stop it.

Confidence, stated honestly. Brief task 1.4 exists because a normally-closed relay produces the same voltage reading with the opposite meaning, and it is not confirmed that 1.4 was performed before this result. The wording reported ("cuts signal") is a statement about function rather than a raw voltage, which suggests tracing rather than a single measurement — but that is an inference about how the work was done, not evidence. Adopting the unsafe reading costs nothing if it is wrong; adopting the safe one and being wrong costs the machine. Confirmation is still wanted, and it does not gate anything, because nothing below changes if it comes back the other way.

Every protective action this controller can take ends at the same place: it de-energises **DC Output A, terminal 3, the FUEL relay** (rated 10 A for 10 s, 5 A continuous — a relay coil driver, consistent with it operating the 70 A -13RB1 contactor in the OEM diagram).

So the independent over-frequency path is only as good as what that relay's contacts carry. If they remove supply from the ECU and the fuel metering unit, the path works. If they merely signal a hung ECU, **the path terminates in the failure it was meant to catch**, and there is no independent protection at all.

That is U12, unchanged in substance and now the single load-bearing measurement in this section.

The position for this design

1. The ECU implements overspeed shutdown in firmware as a **primary** function, with the crank sensor as its input, at the highest scheduling priority. First line, and it is ours.
2. The controller's **generator over-frequency shutdown at 56 Hz**, measured directly from the alternator, is the **second line**. It is independent of this ECU but conditional on alternator excitation. We must not weaken it: the ECU must not latch the fuel metering unit on in a way that survives its supply being removed, and must fail safe when the ignition input at pin 71 drops.
3. A standalone MPU-driven overspeed trip module is **mandatory again**, not recommended. The relaxation in the previous revision was bought entirely by the claim that the controller had an independent *speed* path; that claim is false for the fitted device. What remains is a generator-frequency protection with an excitation dependency, terminating in a relay whose function is unverified. That is not enough to carry a 25 kVA set on its own. Two candidates are costed in memo 02 (GAC SSW675, Murphy HD9063) — that costing survives even though the memo's device analysis does not.
4. ~~The gate: U12 must be closed by measurement before any build takes injection authority.~~ Closed 18 Sep 2026, in the failing direction. The gate was written to admit the possibility that U12 came back clean. It did not. The relay signals, so line 2 detects a runaway without being able to act on it, and item 3's "mandatory" is no longer a precaution against an unverified relay — it is the only shutdown path on this machine that does not run through the ECU being protected against.
5. **The trip module must break the fuel metering unit's supply directly.** This follows from U12 and is now a requirement, not a preference. A trip module that signals the same GCU relay chain inherits exactly the defect U12 just exposed. Its contacts have to sit in series with the metering unit's own battery feed — the one through fuse 8F1 — so that tripping removes fuel authority regardless of what the ECU, the controller, or the GCU relay board is doing. Both costed candidates in memo 02 are contact-output devices and can be wired this way; the costing survives, the wiring assumption behind it does not.

The owners may of course revisit item 3 with evidence. What they should not do is inherit the previous revision's conclusion, because the fact it was built on turned out to describe a controller that is not on this machine.

What this section still does not address

A diesel that begins burning its own lubricating oil — through a failed turbocharger seal or crankcase fumes drawn into the intake — has a fuel supply that **none of the three lines above can interrupt**. Every one

of them cuts diesel or electrical power. The standard countermeasure is a mechanical or pneumatic **air-intake shutoff valve**, and no such device appears anywhere in this design, this register, or §9's out-of-scope list. It was never considered rather than considered and rejected.

This is recorded here, unresolved, because it is a decision for the machine's owners: the part is mechanical, not electronic, and fitting it is not within this PCB's scope. It should not be allowed to stay invisible.

4. ECU pin map

Confirmed pins from the wiring diagram. **44 of 94**. See §8 for the remaining 50.

Power

Pin	Signal	Front-end
21	Battery + feed	Main input: EMI filter → fuse → TVS → reverse-battery FET
04	V_V_BAT_1R (10 A fused)	120 k / 10 k divider → ADC; feeds high-current loads [?]
06	V_V_BAT_2R (10 A fused)	120 k / 10 k divider → ADC; feeds high-current loads [?]
09	SYNCHRONIZATION GROUND	Function unresolved — see §8

The battery-sense divider is **120 k / 10 k**, not the **75 k / 10 k** inherited from the earlier artifacts. That value was sized for a 5 V ADC and delivers 4.69 V at a 40 V input — above the S32K148's 3.3 V analog supply. The corrected divider peaks at 3.07 V and still resolves 573 ADC counts at the 6 V cranking dip. Verified in `sim/blocks/battery_sense.cir`.

Ratiometric analog — 10 k / 16 k divider, 22 nF shunt, 3.3 V clamp, rail measured and corrected in firmware

Revised 14 Sep 2026. This section previously specified *1 kΩ series, 100 nF shunt, ADC ref tracks 5V_SENSOR*, which contradicted the simulated front-end and cannot be built on this MCU. Both halves were wrong, and for the same reason the 75 k / 10 k battery divider was wrong — they were written for a 5 V part:

- **VREFH cannot track 5V_SENSOR**. The S32K148's ADC reference may not exceed its 3.3 V analog supply. A reference at 5 V is not an option on any 3.3 V device.
- **1 kΩ in series scales nothing**. With no division, a sensor at 4.5 V arrives at a 3.3 V pin. The series resistor is a fault-current limiter, not an attenuator.

So the signal must be divided, and the question becomes how to keep the *ratiometric* property after dividing it.

Decision: divide, then measure the rail and correct in firmware.

`5V_SENSOR` gets its own ADC channel. Every ratiometric reading is computed as a fraction of the measured rail rather than of an assumed 5.000 V, which is what the sensor's output actually represents. Cost: one ADC channel, zero analog parts. The channel budget has room — see the sensor reference.

Rejected alternative: dividing `5V_SENSOR` into `VREFH` by the same 16/26 ratio. It fits numerically ($5\text{ V} \times 16/26 = 3.08\text{ V}$, inside the 3.3 V limit) and gives true ratiometric conversion with no firmware arithmetic. It is rejected because `VREFH` is shared by every channel on that ADC, so it would silently make **battery sense and the NTC channels ratiometric to the sensor rail as well** — both of which need an absolute reference. It would also need a buffer, since a resistive divider is too noisy and too high-impedance to drive a reference input.

Single-ended or differential is decided by the ground, not by the channel. Where the sensor's return is shared, the other sensor's current develops an offset across the harness resistance that a single-ended input reads as signal: `sim/blocks/sensor_differential.cir` measures 12 mV of bias at 1 Ω of return resistance against 0.13 mV differential. Research memo 09 found both Deep Sea Electronics and SEDEMAC treat this as a first-class problem. So:

- **Shared ground (pins 34, 36) → differential front-end.**
- **Dedicated ground (pin 08) → single-ended is sufficient**, because there is no other sensor's current in that return. This mirrors the OEM's own ranking: they gave rail pressure a dedicated return precisely so it would not need one.

Pin	Signal	Group ground	Front-end
11 / 41	Boost: 5 V excitation / pressure signal	34	Differential
79	Boost: temperature signal — topology unresolved, see §8	34	Differential
34	Sensor ground — boost + coolant	—	—
33	Coolant temperature (NTC, pull-up divider)	34 (spliced)	Differential
15 / 37	EGR position: 5 V / wiper	36	Differential
36	Sensor ground — EGR + oil pressure	—	—
39 / 80	Oil pressure: 5 V / signal	36 (spliced)	Differential
32 / 35	Rail pressure: 5 V / signal	08	Single-ended
08	Sensor ground — rail pressure, dedicated	—	—

Pin	Signal	Group ground	Front-end
—	5V_SENSOR rail monitor	—	Divider → ADC, for the correction above

Sensor-ground grouping mirrors the OEM harness exactly: 34 boost+coolant, 36 EGR+oil, 08 rail alone, 30 crank alone, 44 cam alone. Rail pressure gets a dedicated return because it is the most accuracy-critical analog channel on the engine.

Speed

Pin	Signal	Front-end
52 / 74	Crank VR: Frequency I/P High / Low	Differential adaptive-threshold conditioner → timer capture
30	Crank sensor ground	Shield terminates here, ECU end only
45 / 46	Cam Hall: 5 V / Frequency I/P	Pull-up to 5V_SENSOR → timer capture
44	Cam sensor ground	—

Discrete inputs

Pin	Signal	Sense	Front-end
43	Coolant switch	Switch-to-return	Pull-up + RC debounce
23	Droop switch	Switch-to-return	Pull-up + RC debounce
01 / 02	G_G_B AT1 / AT2	—	Switch return legs — but see note below, these may be power grounds
20	Audio abort	Active-high	47 k / 68 k bias + 3.0 V zener clamp + 220 nF
24	Override SS	Active-high	same
71	Ignition	Active-high	same

The active-high front-end is not a divider, and cannot be. Reading logic high at a 6 V cranking dip needs a divider ratio above $2.31/6 = 0.385$; staying under 3.3 V at 40 V needs it below $3.3/40 = 0.0825$. Those constraints do not overlap, so no fixed ratio works at both ends. The zener sets the clamp level and the divider only biases into it, which makes the logic level flat to within 92 mV across the whole range instead of tracking the battery. Verified in [sim/blocks/discrete_input.cir](#).

Pins 01/02 need checking. Memo 03 infers from the connector drawing that pins 1, 2, 5 and 6 are large power contacts. Pins 05 and 06 fit that — injector high side and a battery rail. Switch return legs would not. The likelier reading is that G_G_B denotes battery ground and these are the ECU's main power grounds, which would also explain why no power-ground pin appears anywhere in the extraction. Queued as a continuity measurement.

Outputs

Pin	Signal	Driver
03	Injector high side — bank A (cyl 1 + 3)	Boosted high-side switch
05	Injector high side — bank B (cyl 2)	Boosted high-side switch
73 / 07 / 29	Injector low side — cyl 1 / 3 / 2	Current-controlled low-side, peak-and-hold
88	Fuel metering unit PWM	High-current low-side + current sense
59 / 81	EGR High / EGR Low	H-bridge + position feedback on pin 37
50	Main relay	Low-side FET + flyback
69	Buzzer relay	Low-side FET + flyback

CAN

Pin	Signal
55 / 77	CAN channel 1 — H / L
87 / 86	CAN channel 2 — H / L

5. MCU selection

Recommendation: NXP S32K148. This resolves the contradiction in the artifacts — Design Brief §03 specifies MPC5744P while its own §02 block diagram and §06 sensor table still say S32K148 and FTM.

The argument for MPC5744P was angle-synchronous injection timing needing the eTPU2.

Corrected 2026-09-13 after research memo 05 challenged the original reasoning and was right to. This section first argued that constant 1500 rpm operation relaxes the timing problem because a crank degree is 111 μ s and a 60-tooth wheel gives an edge every 667 μ s. That checks the wrong variable. Crank-edge *capture* rate was never the binding constraint in either case — at 6000 rpm a 60-tooth wheel still only produces an edge every 28 μ s, which is over 2000 CPU cycles at 80 MHz against an NVIC latency of roughly 12. The comparison was never close, so constant speed cannot be what makes it comfortable. The reasoning below replaces it.

What the eTPU2 actually buys. It is a co-processor that autonomously schedules and re-schedules many angle-domain output-compare events across a wide and *rapidly changing* speed range, without consuming the main CPU's interrupt budget and without the core's scheduling jitter — cache effects, higher-priority ISRs, RTOS latency — reaching the injection timing chain. That matters when rpm sweeps 800–6000 with tight transient response and pilot/main/post multi-pulse strategies all landing within microseconds of a moving target angle.

Why we don't need it. This set governs to a fixed 1500 rpm, and the pin map in §4 shows a single main injection pulse per cylinder per cycle — one bank high-side plus one low-side select, with no multi-pulse rate shaping in scope. The constraint that does bind is **injector pulse-width resolution**, which sets fuel-quantity metering accuracy and is independent of crank speed. FlexTimer output-compare runs off the bus clock: one tick is 12.5 ns at 80 MHz, 8.9 ns at 112 MHz HSRUN. Against a single-pulse metering requirement on the order of 1–4 µs, that is roughly 100–450× headroom.

The S32K148 also carries 2× Programmable Delay Blocks, which trigger ADC conversions or timer events at a defined delay from a capture event in hardware. That recovers part of what the eTPU2 offers — jitter isolation on the timing chain — without the eTPU2.

Caveat carried forward: the 1–4 µs resolution figure is a general diesel-injector engineering figure, not this injector's datasheet. Revisit if U5 closes with a part demanding materially finer control, or if multi-pulse injection ever enters scope.

What the S32K148 buys in exchange:

- Free S32 Design Studio; GCC ARM toolchain; cheap CMSIS-DAP/J-Link probes instead of a Power Architecture Nexus pod
- Stock KiCad symbol availability, or trivially built from NXP's own EVB files
- Real availability in small quantities in India
- 3× FlexCAN (we need 2); dual 12-bit ADC against a demand of ~14 channels — 8 sensor inputs, 2 battery-rail senses, and 4 current-sense returns from the injector, metering and EGR drivers; FlexTimer instances well beyond the ~10 timing channels required (crank capture, cam capture, 3 injector low-side, 2 injector high-side, metering PWM, 2 EGR bridge PWM)
- No eTPU2 microcode, which is a specialist skill and a hiring risk on a small team

Confirmed in Phase 1 — see <docs/research/05-mcu-selection.md>.

Orderable part **FS32K148HAT0MLQT**, 144-pin LQFP, 80 MHz RUN / 112 MHz HSRUN grade, –40...+125 °C. In stock at DigiKey India at roughly ₹1,900–2,000/unit. The S32K146 fallback was checked and is in worse supply shape (backordered, 12–52 week spread), so it is not needed.

Two items deferred to Phase 2: confirming how many ADC channels the 144-pin package actually breaks out (needs the reference manual's pin-mux table, not the datasheet), and pad-by-pad verification of the SOT486-2 package drawing before the footprint is locked.

6. Power architecture

Unchanged from the artifacts in topology; the reasoning there is sound.

Stage	Function
EMI input filter	Common-mode choke, CISPR 25 conducted margin
Reverse-battery	Ideal-diode controller + P-FET, < 0.3 V drop
Transient clamp	Bidirectional TVS, ISO 7637-2 pulses incl. load dump 5b
Buck pre-regulator	6–40 V → 5 V, wide-input synchronous
5V_SENSOR	PTC-protected per sensor group, isolated from MCU rails
3V3_MCU	LDO off 5 V
Supervisor	Windowed watchdog + brown-out reset

Input range 6–40 V: 6 V covers the cranking dip on a 12 V system, 40 V bounds the clamped load-dump transient.

Injector boost rail is new relative to the power chain as drawn: a separate boost stage generating the injector opening voltage, with its own reservoir bank sized for the pulsed draw of two banks. Target voltage is set by the injector part — see §8.

7. Deliberate deviations from the OEM design

Recorded so they are choices, not drift:

1. **Connector split.** The OEM uses one 94-pin connector. We use functionally-split sealed connectors. This simplifies harness build and field diagnosis, at the cost of needing an adapter harness to mate with an unmodified Kirloskar loom. That adapter is a deliverable, not an afterthought.
2. **Onboard barometric sensor.** Not present in the OEM harness. Added on-PCB behind a vented port; costs nothing in harness terms and improves fuelling correction.
3. **Independent overspeed path.** Added, per §3. The OEM does not have one.

8. Unknowns register

Nothing in this list may be guessed into copper. Each needs an answer before the sheet that depends on it is drawn.

Status updated at phase 1 exit, 14 Sep 2026. Three states, and the middle one carries weight: **Open** means no answer yet; **Answered** means phase-1 research produced an answer that is good enough to design

against but is *inferred*, not confirmed, and still carries the upgrade path in its memo; **Closed** means confirmed outright. Nothing here is Closed yet — every confirmation on this list needs the machine.

#	Unknown	Status	Blocks	How to close
U1	ECU connector identity	CLOSED 18 Sep 2026 — Bosch , part numbers 1 928 405 192 and 1 928 405 194 , code C mechanical keying. 94 cavities: power pins 1–8 in their own chamber, then rows 9–28, 29–50, 51–72, 73–94	—	Closed by markings on the part. Note: an adapter housing must be code C or it will not mate
U2	Boost air-temp (pin 79): NTC or ratiometric?	Open	Sheet 3b front-end	One resistance reading, sensor cold — brief task 1.3, closes it outright. Mitigation: a front-end that works either way
U3	Controller MPU input — does it exist?	CLOSED 18 Sep 2026 — No. The fitted controller is a DSE4522 MKII ; DSE manual 057-260 's terminal table runs 1–35 with no speed-pickup terminal	§3	Closed from the manufacturer's own terminal list
U4	Controller J1939 expectations	Largely answered — CAN source addresses 234 (engine) / 44 (instrumentation); J1939-75 instrumentation on, alarms off; the controller takes oil pressure, coolant temperature, coolant level and ENGINE SPEED from our ECU over CAN; ECU Data Fail → Shutdown	Firmware, CAN sheet	Remaining: the exact PGN set. Log the live bus, or read DSE publication 057-004
U5	Injector part number and drive profile	Partly answered — memo 04 gives a family envelope (65–115 V, 12–24 A peak, 8–13 A hold, 40–100 A/ms) sufficient to design the driver	Boost rail <i>target</i> voltage, not the driver sheet itself	Photograph the injector body for a Bosch 0445 1xx xxx or Kirloskar F6.xxx.xx.x.pr . Separately: the calibration data is a different problem — see §9 and memo 04
U6	Pin 09 SYNCHRONIZATION GROUND function	Open	Power/ground plan	OEM pin list; drawn red despite the name
U7	Pins 04/06 — sense only, or load feeds?	Open	Power sheet current rating	Measure on the live engine, or OEM pin list
U8	6-pin intake throttle — on the ECU at all?	Partially answered 18 Sep 2026 — reported battery-fed , so the ECU does not supply it	Whether a throttle <i>driver</i> is in scope. Supply and control are separate questions and only supply is answered	Still needs brief task 3.11: follow the cable and see whether it terminates at the 94-way connector. A battery-fed actuator can still be ECU-commanded

#	Unknown	Status	Blocks	How to close
U9	Catalyst temp sensor — connected where?	Open	Whether an EGT front-end is in scope	Trace on the engine
U10	SENT in the colour legend — which signal?	Open	Possible digital sensor front-end	Inspect harness; SENT is SAE J2716
U11	Engine identity	CLOSED 18 Sep 2026 — rating plate reads 3GK550ETA 4SR1, app code GK3.8703, 26.5 kW at 1500 rpm, type approval ARAI/MoEF/DGTA/IGES4/KOEL-P25/2825/24 . Memo 01's inferred 3R550ETA 4G1 was wrong ; the original "GK550" was right	—	Closed by plate
U13	Which controller is fitted	CLOSED 18 Sep 2026 — DSE4522 MKII AMF (India SP), part 4522-001-01 , serial 11021794. Photographed front and rear. Not a KG640C . Memo 02 superseded; spec §3 rewritten	—	Closed by photograph
U14	Pre-heat / post-heat: driven by the controller or by the ECU?	Partially answered 18 Sep 2026 — reported battery-fed	Whether a heater <i>output</i> is in scope. As with U8, supply is not control	Who switches that battery feed is still open. The DSE4522 config enables both at 50 °C, so the controller is the likely switch — photograph the heater relay and trace its coil
U15	What does CRS-878 denote in the controller's engine profile?	NEW 18 Sep 2026	Possibly bears on U5 — "CRS" plausibly identifies the common-rail system	Ask KOEL, or find DSE's engine-profile list
U12	Does the 70 A ignition relay -13RB1 remove power from the ECU and fuel metering unit, or only signal the ECU?	CLOSED 18 Sep 2026 — it only signals. Reported from the machine. The relay does not remove power; a hung ECU keeps the fuel metering unit energised	§3's safety gate, resolved against us. The trip module is mandatory, and §3 item 5 now requires its contacts to break the metering unit's own battery feed rather than signal the same relay chain	Closed. Task 1.4's normally-closed check is still wanted as confirmation, but nothing depends on it — the unsafe reading is already adopted

~~U12 now stands alone as the load-bearing unknown.~~ U12 closed 18 Sep 2026, and the answer was the bad one. U13 cost this project its safety argument: the fitted DSE4522 takes engine speed from our ECU over CAN, so the controller's overspeed trip is not independent of the thing it is meant to protect against. What independence remained was the generator over-frequency shutdown at 56 Hz — and every protective action the controller can take terminates in the same fuel relay on terminal 3.

That relay signals. It does not remove power. So the second line of defence can detect a runaway and cannot stop one. The two findings compound: the controller either cannot see the fault (overspeed, read from our CAN) or cannot act on it (over-frequency, acting through a signalling relay).

Nothing on this machine currently interrupts fuelling without the cooperation of the ECU being protected against. The trip module in §3 item 3 is what closes that, and §3 item 5 now specifies where its contacts have to sit — in series with the metering unit's own battery feed, not in the GCU relay chain that just failed this test.

Worth stating plainly, because it is the pattern of this whole phase: **four of the five unknowns closed by field evidence closed against what this project had reasoned.** U11 (engine identity), U13 (which controller), U1 (connector — this one closed *for us*), and now U12. Desk inference has a measured track record here, and it is poor.

U1, U3, U11 and U13 all closed on 18 September from photographs and one manufacturer's manual. Three of the four closed *against* an inference this project had made, which is the honest measure of how much desk research was carrying.

U3 and U11 were the other two on that list and are now answered, both by inference from primary documents rather than by confirmation. Neither answer is fragile — but both are the kind of finding that a single photograph at the machine would settle permanently, which is why they stay on the register rather than leaving it.

U5 has moved off the critical list for a reason worth stating plainly: memo 04's family envelope means the **injector driver can be designed now, without the part number.** What the part number buys is a boost-rail target instead of a range. The thing that genuinely gates running this engine is not on this register at all — it is the injector calibration data, recorded in §9 and memo 04.

9. Out of scope

- Injection calibration: timing maps, quantity maps, rail pressure setpoint schedules. The hardware will be capable; the maps are a dyno programme, not a PCB deliverable.
- Emissions certification. **But read this before planning deployment** — research memo 06 found, in primary sources, that the regulatory position is worse than "we simply won't certify it." CPCB's RECD System & Procedure states that ECU characteristics governing injection timing, air-mass metering and emission-reduction strategy must not be altered even under CPCB's own sanctioned retrofit scheme, and excludes

changes limited to engine control from that scheme entirely. General Condition 3 of GSR 804(E) states that no person shall use a genset lacking a valid Type Approval and CoP certificate. No approval pathway for a third-party ECU replacement was found.

This does not block the engineering: designing the board, building it, and bench-testing it against simulated signals are unaffected. It bears on **putting it into service on a genset that is in use**, and it compounds the calibration risk in memo 04 — this engine meets its limits partly through calibration, so a replacement ECU can move it out of compliance without any hardware changing at all.

This is the project owners' decision to take to qualified counsel or to CPCB/ARAI directly. It is recorded here because it is material and was not previously examined, not to make the decision for them. - The panel controller. We talk to the **DSE4522 MKII AMF**; we do not replace it. (Earlier revisions of this line said KG640C, which is what the OEM wiring diagram shows and not what is fitted — see §3.) - Panel wiring, ATS, alternator metering.

10. Phase sequence

Phase	Output	Status
0	This document	Done
1	Research memos: engine ID, controller, MCU confirmation, injector and connector sourcing	Done — nine memos
1.5	Block-level simulation: topology and component values verified before capture	In progress — 17 blocks, all passing
2	KiCad hierarchical schematic capture	Not started
3	4-layer PCB layout, DRC, fab outputs	Not started
4	Firmware skeleton: HAL, J1939, crank sequencing, fault manager	Not started

Phase 1.5 was added on 18 Sep 2026 to name work that was already happening. The simulated blocks are neither research memos nor schematic capture, and leaving them unnamed meant the roadmap did not describe the most productive activity on the project — the one that caught a divider that would have destroyed an ADC pin, a discrete input that was impossible as specified, and a decoupling network thirty times over its impedance target. It also gives the two gaps the circuit review found — load dump, and injector turn-off energy — a phase to belong to instead of floating as review findings.

Phase 2's blocked sheets are now fewer: **U1 is closed** (Bosch 1 928 405 192 / 194 , code C), so the connector and adapter-harness sheets are unblocked. U2 blocks one sensor front-end, U8 blocks whether a

throttle driver exists at all, and U5 sets the boost rail's target voltage but not the driver itself. Power, CAN, MCU support, speed inputs, the relay outputs and the analog front-ends are all fully specified above.

Research Memo 01: Engine Identity

Task: U11 (engine make/model/displacement/cylinder count/rated speed/emissions tier). Informs U8, U9. **Date:** 2026-09-13 **Author:** Phase-1 research (task 1)

Verdict

SUPERSEDED 18 Sep 2026 — READ THIS FIRST

The engine's rating plate has been photographed. It reads 3GK550ETA 4SR1 , application code GK3.8703 , 26.5 kW / 36 hp at 1500 rpm, type approval ARAI/MoEF/DGTA/IGES4/KOEL-P25/2825/24 , manufactured 12 Sep 2025.

This memo's model-number conclusion below is wrong. It identified the engine as 3R550ETA 4G1 and explicitly rejected "GK550" as a garbled recollection of "KG4" + "550", advising that GK550 "should not be used as a search term or part-lookup key going forward". The original designation given to this project was right; this memo's inference was not.

What survives: every hardware constraint in the Step 3 cross-check — three cylinders, common rail, turbocharged-aftercooled, cooled EGR with DOC and no SCR, 1500 rpm, CPCB IV+, 26.5 kW — matches the plate exactly. The reasoning from the wiring diagram was sound. What failed was the model-number identification built on top of it, which converged on a real Kirloskar engine that is not this one.

Also resolved: this memo recorded as a negative result that "GP3 cannot be resolved to a specific engine model or family". It is a project/application code — the wiring diagram is GP3.8703.C4.pdf and the engine's serial number begins GK3.8703/. The 8703 matches.

Evidence: refs/field-evidence-2026-09-18.md . U11 is closed, by plate rather than by inference.

Verdict (as originally written, superseded above)

The engine is almost certainly the Kirloskar 3R550ETA 4G1 (Kirloskar "R550" family, 3-cylinder, 1.65 L, turbocharged-aftercooled, common-rail, cooled-EGR + DOC), as fitted to the Kirloskar KG4-25WS1 25 kVA CPCB IV+ genset. Every hardware constraint from the wiring-diagram cross-check (Step 3) is satisfied, and the genset controller named in the OEM brochure (KG640C) is the exact same controller named in the OEM wiring-diagram extraction. This is treated as **inferred with high confidence** rather than **confirmed**, because no single primary source explicitly states "manual 08-GP3-60-001 / file GP3.8703.C4.pdf belongs to engine 3R550ETA 4G1" — the identification is a convergence of independent documents, not a direct statement. A rating-plate photograph (see "What would close this," below) would upgrade this to confirmed.

"GK550" as reported by the user does not match any Kirloskar Oil Engines (KOEL) designation found in searching. It is most plausibly the user's own conflation of the genset model prefix **KG4** and the engine family number **550** — but this is speculation on my part, not a sourced fact, and is flagged as such.

Step 1: What does "GK550" designate?

Searched: Kirloskar GK550 , Kirloskar Green GK 550 25 kVA genset , KOEL GK550 25 kVA diesel genset specification , "GK550" Kirloskar engine OR genset OR generator , Kirloskar "GK" series genset air-cooled OR water-cooled kVA range .

Result: no primary source uses the string "GK550." No Kirloskar Oil Engines product page, brochure, distributor listing, or spare-parts catalogue found in this search returned that exact designation.

(Negative result — searched via general web search, no domain restriction, September 2026.)

One lead was run down and ruled out: - ManualsLib lists a document titled "KIRLOSKAR GK SERIES INSTRUCTION ON INSTALLATION, OPERATION AND MAINTENANCE" (<https://www.manualslib.com/manual/1698613/Kirloskar-Gk-Series.html>). Search-result snippets describe this as covering Kirloskar pump type GK(P) — a Kirloskar Brothers Limited (pumps) product, not a Kirloskar Oil Engines (engines/gensets) product. These are different companies under the Kirloskar Group umbrella. **Ruled out** — different Kirloskar Group entity, wrong product category. (Confirmed via search-result description; the page itself returned HTTP 403 to direct fetch and could not be read in full.)

What KOEL genset/engine naming actually looks like, established from primary sources fetched directly: - Genset models use a `KG<n>-<kVA><suffix>` pattern, e.g. KG4-25WS1 (25 kVA), KG4-30WS1, KG4-40WS1, KG4-58.5WS (all from the same brochure) — source: Kirloskar Oil Engines, "CPCBIV+ COMPLIANT — INDIA'S LARGEST FLEET OF GENSETS 25-58.5 kVA," https://www.kirloskaroilengines.com/documents/541738/2079942/2_A4_CPCB+IV+25-58.5+kVA.pdf (downloaded and text-extracted directly; doc ref 20240610/REV01/A4 CPCB IV+25-82.5 kVA , footer code IBG.V.045 Rev0). **Confirmed.** - Engine models use an `<cyl>R<displacement-code><aspiration><generation>` pattern, e.g. 3R550ETA 4G1 (3-cylinder, "550" family, Electronic-governed, Turbocharged-Aftercooled, generation suffix "4G1") for the 25 kVA genset in the same brochure. Source: same PDF, table row "ENGINE / Engine Model." **Confirmed.** - A related, non-genset-specific brochure from Kirloskar Americas (<https://www.kirloskaramerica.com/documents/5928996/5929154/Kirloskar+R550+G+Drive+Brochure+25052022+Rev0.pdf> — downloaded directly) lists the bare engine-platform naming: 2R550NA, 3R550NA, 3R550TC, 3R550TA — i.e., "R550" is the engine platform name, and the suffix encodes cylinder count and aspiration. **Confirmed.**

Working hypothesis (inferred, not sourced): the user's "GK550" is most likely a garbled or telescoped memory of "KG4" (the genset-model prefix) and "550" (the engine-family number that actually appears in the engine model string, 3R550ETA), possibly by transposing K/G. I found no evidence this is an official SKU, a genset model, or a Kirloskar Green product code in its own right. It should not be used as a search term or part-lookup key going forward — use KG4-25WS1 (genset) and 3R550ETA 4G1 (engine) instead.

Step 2: Working backwards from the manual number

Manual number 08-GP3-60-001, file GP3.8703.C4.pdf, served from <https://ikonnnect.kirloskar.com/KOEL/Services/TroubleShootingWiringDiagram?manualName=08-GP3-60-001> (confirmed present in the PDF's own footer on every page, verified via `pdftotext` extraction of the source file).

Attempts made: - Fetched <https://ikonnnect.kirloskar.com/KOEL/Services/TroubleShootingWiringDiagram> (no query string) directly — connection failed (no response body; the endpoint requires the `manualName` parameter and/or an authenticated session). - Fetched <https://ikonnnect.kirloskar.com/KOEL> (portal home) — retrieved (22 KB), but it is a login/marketing shell with no manual index; no list of manual-family codes is exposed to an unauthenticated fetch. - Searched Kirloskar "GP3" engine family diesel, Kirloskar "GP3" ECU OR "engine control unit" wiring diagram manual, "08-GP3-60-001" Kirloskar, Kirloskar "GP3" platform OR series diesel engine common rail EGR CPCB — **no result anywhere maps "GP3" to a specific engine model or family.** (Negative result.) - Read the wiring diagram itself in full (`pdftotext -layout` extraction of </home/mohan/workws/ecu25kva/refs/GP3.8703.C4.pdf>, all 5 pages) — it does not state an engine model number anywhere in its body text. It is a browser print-to-PDF of the ikonnnect portal page (page metadata shows

a print timestamp), consistent with the note already recorded in `ecu-pinout-extracted.md` that the diagram is troubleshooting-oriented and does not carry a full connector/engine cross-reference.

Conclusion: "GP3" cannot be resolved to a specific engine model or family from public sources. (*Negative result — this closes off Step 2 as originally scoped.*) A plausible but **unconfirmed and unsourced** guess: KOEL's CPCB IV+ genset range (7.5–800 kW) uses the same control architecture (common rail, cooled EGR, DOC, KG640C controller) across several engine displacements (R550, R1040/R1190, R810, per the CPCB IV+ 25-58.5 kVA brochure), so "GP3" may denote a shared ECU/wiring *platform generation* spanning multiple engine sizes rather than one engine. This would explain why the diagram itself carries no engine-specific model text — it may be reused, with only the pin count and sensor complement varying by engine. This is **speculation, explicitly not a finding**, and should not be treated as established.

Step 3: Cross-check against the hardware evidence

Per the brief, a valid candidate must match **all five** of: 3 cylinders; common-rail; EGR fitted; catalyst temp sensor present but no SCR/DEF; 12 V electrical system. Evidence for each column below is drawn directly from `/home/mohan/workws/ecu25kva/refs/ecu-pinout-extracted.md` (hardware) and the KOEL CPCB IV+ 25-58.5 kVA brochure (candidate engine spec, downloaded and extracted directly from https://www.kirloskaroilengines.com/documents/541738/2079942/2_A4_CPCB+IV+25-58.5+kVA.pdf).

Constraint	Wiring-diagram evidence	3R550ETA 4G1 (candidate)	Match
3 cylinders	Injector bank wiring: cyl 1/2/3, pins 03/05/73/07/29 (pinout doc, "Injectors" section)	"No. of cylinder: 3" (brochure table)	Match — confirmed
Common-rail	Rail pressure sensor (pins 32/35/08) + PWM-driven fuel metering unit (pin 88) (pinout doc, "Analog sensors" and "Outputs")	CRDi (Common Rail Direct Injection); brochure narrative: "High pressure common rail system employed on Kirloskar CPCB IV+ Gensets"	Match — confirmed
EGR fitted	Dedicated EGR position sensor (pins 15/37/36) and EGR H/L bridge-driven actuator output (pins 59/81) (pinout doc, "Analog sensors" and "Outputs")	"EGR System used to reduce the level of NOx emitted by Engine" (brochure feature callout)	Match — confirmed
Catalyst temp sensor present, no SCR/DEF	Catalyst Temperature Sensor connector present in connector appendix (2-pin, one only); pinout doc's "Absent from the entire system" section: "No DEF/urea sensor... No SCR"	Brochure lists DOC ("DOC system sets off the reaction to meet the CPCB norms") for this kVA family; no SCR or DEF mentioned anywhere in the 25–58.5 kVA brochure	Match — confirmed
12 V electrical system	Battery+ feed pins 04/06/21, all via fused red supply; no 24 V evidence anywhere in diagram	"Electrical Battery Start in R Voltage: 12 Volts-DC" (brochure); user-reported 12 V/75 Ah battery	Match — confirmed

All five constraints are satisfied simultaneously by only one engine in the brochure's four-model 25–58.5 kVA lineup (the other three — 3R1190ENA, 3R1190ETA, 4R810ETA — are 30/40/58.5 kVA respectively, not 25 kVA, and one is naturally aspirated, so they fail on rated output alone even before considering hardware detail). **This is the strongest evidence in the memo** and is why the identity is called "inferred with high confidence" rather than a bare guess.

Independent corroboration found during this cross-check (not requested by the task, but material): the pinout extraction names the genset control unit as **Kirloskar KG640C** (`ecu-pinout-extracted.md`, "Other system components named"). The CPCB IV+ 25-58.5 kVA brochure independently names its standard controller as the **KG640C Controller** (same PDF, "State of the art Genset Controller" section, pictured with a photo captioned "KG640C Controller"). Two independently-obtained documents (a scraped OEM troubleshooting diagram, and a public marketing brochure) naming the identical, fairly specific controller part number is strong circumstantial confirmation that the wiring diagram belongs to this genset family. **Confirmed** (both documents state the KG640C designation directly; the *inference* is that they describe the same unit).

Step 4: Emissions tier

CPCB IV+ is confirmed as the applicable tier for this class of genset:

- Regulatory basis: **GSR 804(E), dated 3 November 2022**, Ministry of Environment, Forest and Climate Change — "Emission limits for new internal combustion engine up to 800 kW gross power used for power generating set (Genset) application," commonly referred to as the CPCB IV+ regulation. *(Confirmed via independent secondary source: <https://www.ricago.com/blog/revised-emission-standards-for-gensets-a-step-towards-cleaner-air>, which states the GSR number and date; corroborated by a second secondary source in the same search. Primary source page <https://cpcb.gov.in/genset-notifications/> (redirected from cpcb.nic.in) confirms the base notification chain "GSR 771(E) dated 11 Dec 2013" and its amendments exist at CPCB, but its listing text as fetched did not expose the GSR 804(E) reference by name — flagged as a fetch limitation, not a contradiction.)*
- Effective/compliance date: 1 July 2023 (secondary sources; not independently verified against gazette text).
- Power-band limits (19–56 kW band, which contains this engine's 26.5 kW prime rating): $PM \leq 0.03$ g/kWh, combined $NOx+HC \leq 4.7$ g/kWh, $CO \leq 3.5$ g/kWh, smoke ≤ 0.7 — per dieselnet.com's published summary table of the 2022/2023 Indian genset standard (<https://dieselnet.com/standards/in/genset.php>). **Inferred/secondary-sourced** — dieselnet is a respected industry standards clearinghouse but this is not the gazette text itself.
- The genset manufacturer's own literature states this engine/genset combination is CPCB IV+ certified and uses cooled EGR + DOC (no SCR, no DPF mentioned) to meet it — consistent with the 19–56 kW band not requiring the separate, stricter NOx-only limit (0.40 g/kWh) that appears at the 56–560 kW band, where SCR is more commonly needed to hit the tighter NOx number. This is **inferred**: no source states "this band never needs SCR," but the absence of SCR/DEF hardware, plus the combined NOx+HC limit form used below 56 kW, is internally consistent with why KOEL doesn't fit SCR to this genset size.

Implication for aftertreatment scope (informs U8): design should assume a **DOC-only aftertreatment path with cooled EGR** — no SCR, no DEF/urea system, no DPF regeneration logic. This matches the hardware evidence exactly (single catalyst-temperature-sensor connector, no urea pump/quality sensor anywhere in the harness).

Step 5: Summary — engine identity record

Field	Value	Status	Source
Engine make	Kirloskar (Kirloskar Oil Engines Ltd, KOEL)	Confirmed	KG4-25WS1 brochure (below)
Engine model	3R550ETA 4G1 (OCR'd in one source as "3R55OETA 4G1" — "O"/"0" artifact)	Inferred, high confidence (see Step 3)	Kirloskar Oil Engines CPCB IV+ 25-58.5 kVA brochure, https://www.kirloskaroilengines.com/documents/541738/2079942/2_A4_CPCB+IV+25-58.5+kVA.pdf ; corroborated by https://www.indiamart.com/proddetail/kirloskar-kg4-25ws1-diesel-generator-2854043860488.html
Engine platform/family	"R550" (Kirloskar's own naming)	Confirmed	Kirloskar Americas R550 G-Drive brochure, https://www.kirloskaramerica.com/documents/5928996/5929154/Kirloskar+R550+G+Drive+Brochure+25052022+Rev0.pdf
Genset model	KG4-25WS1	Confirmed	Kirloskar Oil Engines CPCB IV+ 25-58.5 kVA brochure (above); indiamart listing (above)
Displacement	1.65 L (1,650 cc, 3-cylinder)	Confirmed	Kirloskar Oil Engines CPCB IV+ 25-58.5 kVA brochure (above); cross-checked arithmetically: 86 mm bore × 94 mm stroke × 3 cyl ≈ 1,638 cc, consistent with rounded 1.65 L spec
Bore × stroke	86 mm × 94 mm	Confirmed	Kirloskar Americas R550 G-Drive brochure (above)
Cylinder count	3 (inline)	Confirmed	Kirloskar Oil Engines CPCB IV+ 25-58.5 kVA brochure (above); matches wiring-diagram injector wiring (cyl 1/2/3)
Compression ratio	19.7:1	Confirmed	Kirloskar Americas R550 G-Drive brochure (above)
Aspiration	Turbocharged, aftercooled (TA)	Confirmed	Both Kirloskar brochures (above)
Rated speed	1500 rpm	Confirmed	Kirloskar Oil Engines CPCB IV+ 25-58.5 kVA brochure (above) — matches 50 Hz expectation from the brief
Rated output	26.5 kW / 36 HP prime, per ISO 8528-1	Confirmed	Kirloskar Oil Engines CPCB IV+ 25-58.5 kVA brochure (above)

Field	Value	Status	Source
Genset rated output	25 kVA prime, 20 kW, 0.8 PF, 50 Hz, 230 V(1φ)/415 V(3φ)	Confirmed	Same brochure
Fuel injection	Common rail direct injection (CRDi), electronic governor	Confirmed	Same brochure; hardware corroboration via rail pressure sensor + PWM fuel metering unit in pinout doc
EGR	Cooled EGR fitted	Confirmed	Same brochure; hardware corroboration via EGR position sensor + EGR actuator pins in pinout doc
Aftertreatment	DOC only; no SCR, no DEF, no DPF mentioned	Confirmed (absence)	Same brochure narrative; hardware corroboration via single catalyst-temp-sensor connector and complete absence of urea/SCR hardware in pinout doc
Emissions tier	CPCB IV+ (GSR 804(E), 3-Nov-2022; effective 1-Jul-2023)	Confirmed regulation exists; inferred that this specific unit is certified to it (brochure says so but I have not seen a type-approval certificate)	Brochure states CPCB IV+ compliance; GSR number/date per https://www.ricago.com/blog/revised-emission-standards-for-gensets-a-step-towards-cleaner-air ; band limits per https://dieselnet.com/standards/in/genset.php
Genset controller	Kirloskar KG640C	Confirmed	Both the wiring-diagram pinout extraction and the CPCB IV+ brochure name this controller independently
Battery	12 V DC starting	Confirmed	Brochure ("Electrical Battery Start in R Voltage: 12 Volts-DC"); matches user-reported 12 V system
Battery Ah	User reports 75 Ah; generic Kirloskar Americas R550 brochure lists 65 Ah for the bare-engine starter spec; a KOEL-Green-branded 12 V/75 Ah battery (part 02.709.24.0.Pr, model 75D31R-A) is sold as a genuine Kirloskar spare	Inferred (discrepancy noted, not resolved)	Kirloskar Americas R550 G-Drive brochure (65 Ah, generic/global engine-only spec); https://www.safesparesonline.com/product/kirloskar-genuine-02-709-24-0-pr-12v-dc-75ah-lead-acid-koel-green-battery-model-75d31r-a/ (75 Ah, India-market genuine part)
OEM ECU part number	Not found	Not established	See Step 2 — no source ties a Bosch/Continental/etc. ECU part number to this engine or to manual 08-6P3-60-001
ECU manual/family code	08-6P3-60-001 / "GP3" — meaning of "GP3" not established	Not established	See Step 2 (negative result)

What would close this — the one photograph that settles it

The engine's own **rating plate** (usually riveted to the rocker cover, flywheel housing, or side of the cylinder block) would resolve every remaining "inferred" item above in one shot. Specifically, photograph it so the following are legible:

- Engine model string** — should read something matching `3R550ETA` (or a close variant) if this identification is correct. If it instead reads e.g. `3R1040...` or a different family entirely, this whole identification is wrong and should be discarded rather than patched.
- Serial number** — allows a direct KOEL/dealer parts-catalogue or warranty lookup, independent of any web search.
- Rated output, rated speed, and frequency** stamped on the plate — cross-check against the 26.5 kW / 1500 rpm / 50 Hz figures above.

4. **Emission certification marking** if present (CPCB IV+ or equivalent BEE/ARAI type-approval sticker) — would directly confirm Step 4 rather than relying on brochure-level marketing claims.
5. **The genset's own nameplate** (usually on the canopy or control panel, separate from the engine plate) — should read `KG4-25WS1` or a close variant if this identification is correct, and will separately carry the genset serial number, which is the cleanest path to an authoritative KOEL parts/spec lookup via an authorized dealer or KOEL Care.

Absent that photograph, treat every "inferred" row above as provisional. The hardware cross-check in Step 3 is strong (five independent constraints all satisfied by exactly one candidate in the only public document that lists 25 kVA CPCB IV+ engines), but it is still an inference from convergence, not a direct statement of identity from a primary source that names manual `08-GP3-60-001` and engine `3R550ETA 4G1` in the same document.

Sources consulted (full list)

- `/home/mohan/workws/ecu25kva/refs/GP3.8703.C4.pdf` — OEM wiring diagram, manual `08-GP3-60-001`, read in full via `pdftotext -layout`.
- `/home/mohan/workws/ecu25kva/refs/ecu-pinout-extracted.md` — pin-level extraction of the above, read in full.
- Kirloskar Oil Engines, "CPCBIV+ COMPLIANT — INDIA'S LARGEST FLEET OF GENSETS 25-58.5 kVA":
https://www.kirloskaroilengines.com/documents/541738/2079942/2_A4_CPCB+IV+25-58.5+kVA.pdf — downloaded directly and text-extracted in full (283 lines).
- Kirloskar Americas, "R550 G-Drive" brochure:
<https://www.kirloskaramerica.com/documents/5928996/5929154/Kirloskar+R550+G+Drive+Brochure+25052022+Rev0.pdf> — downloaded directly and text-extracted in full (blocked to the WebFetch tool with HTTP 403; retrieved successfully via direct HTTP GET).
- <https://www.kirloskaroilengines.com/products/power-gen/genset/lhp/diesel-generators/25-kva-diesel-genset> — fetched, gives KG4-25WS1 and engine model 3R550ETA 4G1.
- <https://www.indiamart.com/proddetail/kirloskar-kg4-25ws1-diesel-generator-2854043860488.html> — fetched, corroborates engine model, rated speed 1500 rpm, CRDi, EGR+DOC/CPCB IV+.
- <https://www.indiamart.com/proddetail/kirloskar-3r550-engine-2850959578848.html> — fetched, US-market bare-engine listing corroborating 3-cylinder, ~1.64 L (100 cu in), NA/TC/TA variants.
- <https://www.safesparesonline.com/product/kirloskar-genuine-02-709-24-0-pr-12v-dc-75ah-lead-acid-koel-green-battery-model-75d31r-a/> — found via search, corroborates a 12 V/75 Ah Kirloskar Green genuine battery part exists in the India market.
- <https://www.manualslib.com/manual/1698613/Kirloskar-Gk-Series.html> — found via search, used to rule out "GK Series" as a Kirloskar Brothers pump line unrelated to this engine (page itself returned HTTP 403 on direct fetch; ruled out based on search-result description only — flagged as weaker evidence).
- <https://ikonnnect.kirloskar.com/KOEL/Services/TroubleShootingWiringDiagram> (no query) and <https://ikonnnect.kirloskar.com/KOEL> — fetched directly; no manual-family index exposed without authentication.
- <https://dieselnet.com/standards/in/genset.php> — fetched, CPCB IV+ power-band emission limits table.
- <https://cpcb.gov.in/genset-notifications/> (redirected from <https://cpcb.nic.in/genset-notifications/>) — fetched, confirms existence of the GSR 771(E)/2013 notification chain and its amendments; did not by itself surface the GSR 804(E)/2022 CPCB IV+ reference.
- <https://www.ricago.com/blog/revised-emission-standards-for-gensets-a-step-towards-cleaner-air> — fetched, states GSR 804(E) dated 3 November 2022 as the CPCB IV+ notification.
- General web searches (not individually cited above where they returned only negative results or were superseded by direct fetches):
`Kirloskar GK550`, `Kirloskar Green GK 550 25 kVA genset`, `KOEL`

GK550 25 kVA diesel genset specification, Kirloskar "GP3" engine family diesel, ikonnnect.kirloskar.com TroubleshootingWiringDiagram manualName, Kirloskar "3R550ETA" engine specifications, Kirloskar "R550" OR "3R550" displacement bore stroke cc, site:kirloskaramerica.com R550, Kirloskar R550 engine "1500 rpm" OR "1800 rpm" kW genset industrial, Kirloskar "GP3" ECU OR "engine control unit" wiring diagram manual, "08-GP3-60-001" Kirloskar, CPCB IV+ norms diesel genset 19-56 kW NOx PM DOC EGR SCR requirement notification date, CPCB IV+ GSR notification number "3rd November 2022" gensets gazette.

Research Memo 02: KG640C Genset Controller (GCU) — J1939 Contract and the MPU Safety Gate

Task: U3 (safety gate), U4. Task 2, phase-1 research — highest priority in the phase. Date: 2026-09-13 Author: Phase-1 research (task 2)

SUPERSEDED, 18 Sep 2026 — the fitted controller is NOT a KG640C

The controller has been photographed. It is a Deep Sea Electronics DSE4522 MKII AMF (India SP), part 4522-001-01, serial 11021794, made in the UK. Not a KG640C. A Kirloskar-issued configuration for it was supplied alongside: GP3.314.01.0.PR — DSE 4522 CONTROLLER PROGRAM FOR 3R550.25KVA, 3PHASE (GK PROJECT), naming this engine and this rating.

This memo analyses a KG640C. The OEM wiring diagram in this repo shows a KG640C, and both documents carry GP3. numbering, so which controller is actually fitted is now an open question rather than a settled one. Spec section 3's safety position is built on this memo.

Two specific contradictions, neither yet resolved:

- **Output mapping.** This memo assigns output A to the start relay and output B to the ignition relay. The DSE4522 configuration assigns A = Fuel Relay, B = Start Relay. A separate reading of the wiring diagram gives -13RB1 = ignition on B and -13RB2 = start on A. Three sources, three mappings.
- **Speed source.** This memo's central finding is that the controller senses speed from alternator frequency independently of the ECU. The DSE4522 configuration carries Module To Use Engine Speed = No, whose meaning in DSE's own semantics must be established from their documentation before anything is concluded from it.

What the new document adds rather than contradicts: an Emergency Stop input wired fail-safe, ECU Data Fail → Shutdown, CAN source addresses 234 and 44, and the protection thresholds the controller applies to values our ECU supplies. Those partly answer U4.

Both contradictions are now resolved against this memo. DSE's own manual (057-260) states that with a CAN engine the ECU transmits engine speed to the controller — so the independent alternator-frequency *speed* path this memo describes does not exist on the fitted device. Spec §3 has been rewritten on the DSE4522, and the standalone overspeed trip module reverts to **mandatory**.

What survives and is still worth reading here: the reasoning method, the J1939 background, and the costing of the two trip-module candidates (GAC SSW675, Murphy HD9063), which is device-independent. The memo's conclusion that no magnetic pickup exists also turned out to be right — the DSE4522 has no MPU terminal either — but it was right by coincidence, about a different controller.

Treat everything below as **historical**: an analysis of a device that is not on this machine.

Evidence: refs/field-evidence-2026-09-18.md.

Verdict (U3 — read this first)

The KG640C genset controller has no dedicated magnetic pickup (MPU) input terminal. Verdict: No.

Status: inferred, high confidence (not "confirmed" outright — see caveat below) — from two Kirloskar Oil Engines Limited (KOEL) primary documents, not from the wiring diagram's silence:

1. The OEM wiring diagram itself

(/home/mohan/workws/ecu25kva/refs/GP3.8703.C4.pdf , page 1, "GENSET CONTROL UNIT KG640C" block) shows the controller's terminal groups **J1 through J8 in full** — every terminal on the block is labelled: J1 (terminals 1–2, 12/24 battery-sense), J2 (terminals 3–6, digital outputs A–D: Start Relay, Ignition Relay, MC Connect Relay, Common Fault), J3 (terminals 7–10: D+CHG ALT , digital outputs/inputs), J4 (terminals 11–16: digital input, CAN L/H, RS485 A/B), J5 (terminals 17–26: reserved, digital inputs D/E, analog inputs), J6 (terminals 27–30: genset alternator voltage/frequency sense, fed from the DG alternator's 415 VAC 3-phase output), J7 (mains voltage sense N/B/Y/R), J8 (terminals 35–42: CT current-sense inputs, SCP). **No terminal in J1–J8 is labelled MPU, speed pickup, or magnetic pickup.** Every terminal on the block is accounted for, so this is a complete negative, not silence about an unused pin. (*Confirmed — read directly from the diagram at 400 DPI.*)

2. A genuine KOEL controller manual for the same product family,

"KG640 CONTROLLER MANUAL" (KOEL letterhead, Laxmanrao Kirloskar Road, Khadki, Pune; internal document number SED-MAN-KG640-002), found hosted on Scribd (uploaded by an account named rajinder.kirloskar) at <https://www.scribd.com/document/878092942/SED-MAN-KG640-002-Koel-Manual-for-KG640-Genset-Controller> and retrieved in full (92 pages). This is the **base "KG640" variant** (no CAN bus at all — see §2 below), but its terminal architecture matches the KG640C block on the wiring diagram almost exactly: terminals 1–2 battery, 3–6 digital outputs, 7 D+ CHARGE ALT , 27–30 GEN_V IN (N,B,Y,R), 35–40 GEN_CT_IN , 41 SCP — the same terminal numbers doing the same jobs as the KG640C block in the wiring diagram. The manual's **complete 42-terminal table** (its Table 15, "Terminal Description") enumerates every terminal from 1 to 42 with a name; there is no MPU/speed-pickup terminal anywhere in it. Its glossary lists "MPU" as an abbreviation (page iii) but the term is **never used again anywhere in the 92-page body** — consistent with a shared glossary template across KOEL controller documents rather than evidence this unit has one. **Its own configuration parameter for engine speed, "Engine Speed Sense Source," offers only one value: Alternator frequency** (§16.8.2, "Speed monitoring engine screen": *"Allows to select electrical signal to be use for measurement of engine speed. Alternator frequency: Engine speed will be sensed from alternator frequency."*). No magnetic-pickup or CAN option is listed for that parameter anywhere in the document. (*Inferred, high confidence, from a primary KOEL manufacturer document — see caveat below on why this isn't "confirmed" outright.*)

Caveat, stated plainly: this manual documents "KG640," not "KG640C" by name, and it predates or excludes the CAN bus that the wiring diagram's KG640C block clearly has (J4 terminals 13/14, CAN L/H — repurposing what the base manual lists as "Reserved" terminals 13/14). The "C" suffix is therefore read as "CAN-added" variant of the same terminal architecture, not a different product — the terminal-for-terminal match on every other function (genset voltage sense, CT sense, digital I/O, D+ charge alt, RS485) is too exact to be coincidental.

But no manual titled specifically "KG640C" was located, so this is an inference from a closely related sibling document, not a direct statement about the exact SKU. This is why the verdict is **"No, inferred with high confidence"** rather than **"No, confirmed."** Per the task's own safety framing, concluding "no MPU" when the true answer might be "yes" is the safe-direction error (it triggers an unnecessary but harmless redundant hardware purchase); concluding "yes" when the answer is "no" is the dangerous-direction error. This memo is deliberately conservative.

What would upgrade this to "confirmed": a manual or datasheet with "KG640C" in the title/header (not just the base KG640), a photograph of a physical KG640C rear terminal block, or a call to KOEL technical support quoting manual number 08-6P3-60-001 (the engine wiring diagram's own manual number, from Task 1) and asking directly.

A materially important side-finding: the GCU already has a speed-sensing path independent of the engine ECU and CAN bus

The base KG640 manual's Table 5 ("Genset Voltages and Frequency") specifies terminals 27–30 sample the **genset's own AC output at 5 kHz**, deriving frequency over a **3–75 Hz range at 0.1 Hz resolution, 0.25% accuracy**. Because this 4-pole alternator's electrical frequency is rigidly tied to shaft speed (1500 rpm ↔ 50 Hz, no slip), and because the GCU's "Engine Speed Sense Source" defaults to reading this alternator frequency rather than a CAN message from the engine ECU, **the GCU's own over-speed and gross-over-speed shutdown logic (manual §16.8.2, alarms #27/#28 in the base manual's alarm table) does not depend on the engine ECU or the CAN bus between them** — it depends only on the GCU's own AC front-end and its own microcontroller. This is **not** a substitute for a hardware MPU trip (it still shares fate with the GCU's firmware/microcontroller, and a CAN-bus failure alone would not defeat it, but a GCU firmware hang would), and it does **not** change the verdict above. But it is directly relevant to the design decision that follows from the verdict, and should inform how the team weighs the residual risk before deciding how much redundancy the standalone module below needs to add. *(Inferred from the base KG640 manual's specification tables and alarm list; not confirmed for the KG640C variant specifically, since KG640C-specific documentation was not located.)*

Recommended standalone overspeed trip module candidates (since the answer is "No")

Two candidates, both electronic speed switches driven from a magnetic pickup at the flywheel ring gear, independent of any ECU or GCU microcontroller:

#	Part	Function	Key specs (manufacturer-stated)	India availability / price found
1	Governors America Corp (GAC) SSW675 "2-Element Speed Switch — Overspeed, Crank"	Dedicated dual-setpoint electromechanical-relay trip switch: low setpoint = crank disconnect, high setpoint = overspeed shutdown. Magnetic-pickup input, 5 A relay output, on-board overspeed test circuit.	Manufacturer product page: https://www.governors-america.com/product/ssw675/ (fetch blocked by site, confirmed via search-result excerpt quoting the manufacturer's own description)	India price not confirmed this session — GAC/Governors America Corp parts are stocked by Indian industrial resellers (e.g., Spares Bazaar Pvt Ltd, New Delhi, and HT Industrial Automation Ltd, which list other GAC part numbers), but no INR figure for SSW675 specifically was found. A related GAC part family (ESD5100-series analog governors, a different product — NOT a dedicated overspeed relay , and GAC's own ESD5500E documentation explicitly states a separate overspeed shutdown device is still required) is listed at ₹9,500/piece by Spares Bazaar Pvt Ltd, New Delhi (https://www.indiamart.com/proddetail/esd5500e-esd5111-esd5550-esd5221-gac-speed-control-unit-governor-america-corp-21293246791.html) — cited only as a rough order-of-magnitude anchor for this brand/class of device in the Indian market, not as pricing for the correct part.
2	Murphy by Enovation Controls HD9063 dual-setpoint electronic speed switch	Crank disconnect (250–6,000 Hz) + overspeed (1,100–10,000 Hz) on one module. Magnetic-pickup input, 0.35–60 Vrms signal range, 8–30 VDC supply, SPDT relay output 5 A/30 VDC, built-in overspeed test circuit and trip-point LEDs. Explicitly marketed for "generator sets, pumps, and compressors."	Manufacturer product page: https://www.enovationcontrols.com/products/hd9063-os77d-and-ss300/ (fetched directly, specs quoted above verbatim)	Indian distributor confirmed: " Murphy By Enovation Controls ," Pune, Maharashtra, IndiaMART storefront lists HD9063/OS77D/SS300 (https://www.indiamart.com/proddetail/hd9063-os77d-and-ss300-electronic-speed-switches-15889251691.html) but the listing is quote-only ("Get Latest Price") — no published INR figure .

Negative result, recorded honestly: this session did not obtain a firm INR list price for either correctly-matched dedicated overspeed module (SSW675 or HD9063). Both have confirmed Indian distributors who would need to be contacted directly for a quote. A single-function analog GAC governor (a different, not-directly-comparable product) was found at ₹9,500 in India as the only concrete INR figure this search turned up for the GAC brand; it is reported here only for order-of-magnitude context, flagged explicitly as the wrong product class, and should not be used as a quote basis.

1. Controller identity and sourcing basis

- **Controller:** Kirloskar KG640C, named independently by the OEM wiring diagram ([6P3.8703.C4.pdf](#) , page 1) and by KOEL's public CPCB IV+ 25–58.5 kVA brochure (Task 1, [01-engine-identity.md](#)). (*Confirmed.*)
- **Manual used:** [SED-MAN-KG640-002](#) , "KG640 CONTROLLER MANUAL," KOEL, found via Scribd (<https://www.scribd.com/document/878092942/SED-MAN-KG640-002-Koel-Manual-for-KG640-Genset-Controller>), retrieved and read in full (92 pages). This is a genuine KOEL-authored document (KOEL letterhead, internal doc number, copyright notice), located via a document-sharing site rather than kirloskar.com directly — the **content is primary** (manufacturer-authored), but its **hosting is secondary** (Scribd, not KOEL's own domain). Flagged for transparency; the manual's content is treated as manufacturer documentation for the purposes of the "no safety-relevant conclusion from a secondary source" rule, because the document itself is KOEL's own, not a reseller's or forum's description of KOEL's product.

- **Not investigated further as a "rebadge" lead:** the brief flagged Deep Sea Electronics, ComAp, Datakom, SmartGen and Lovato as common Indian-market OEM sources. A SmartGen HGM6100N manual was pulled during this investigation as a comparison candidate, but the KOEL-letterhead KG640 manual's near-exact terminal-for-terminal match to the KG640C block in the OEM wiring diagram is strong enough evidence that **KG640C is Kirloskar's own documented product**, not a rebadge requiring third-party manual substitution. This conclusion is **inferred**, not from a statement anywhere that KG640C is Kirloskar's original design, but from the convergence of two independent KOEL-branded documents describing the identical terminal architecture.
- **No manual titled specifically "KG640C" was located** despite searching Kirloskar KG640C manual , "KG640C" Kirloskar controller manual pdf , scribd "KG640C" OR "SED-MAN-KG640C" , and the ikonnect.kirloskar.com portal (which exposes no manual index to an unauthenticated fetch, consistent with Task 1's finding). A sibling document, "KG645 Genset Controller User Manual" (<https://www.scribd.com/document/407079798/Kirloskar-KG645-User-manual-for-genset-controller-pdf>, 123 pages), was found and may be a closer CAN-capable relative, but its full text could not be retrieved this session (Scribd's preview blocked the fetch tool; only metadata was obtained). **Flagged as an unresolved lead**, not pursued further under this task's budget constraint.

2. J1939 PGN expectations (Step 3) — mostly unconfirmed; documented honestly

This section could not be closed with primary-source confirmation. The only KOEL controller manual obtained in full (the base "KG640," §1 above) has **no CAN bus and no J1939 content whatsoever** — no mention of PGN, SPN, EEC1, ET1, EFL/P1, VEP1, DM1, or TSC1 anywhere in its 92 pages (verified by full-text search of the extracted manual). Its only digital communications are RS485/Modbus (for panel-to-PC configuration, terminals 15/16) and USB (configuration tool). This confirms the base KG640 does not need this section at all — but it also means **the base manual cannot answer Step 3 for the CAN-equipped KG640C variant actually in the field.**

What is confirmed directly from the two primary hardware documents:

Fact	Source	Status
KG640C has exactly one CAN interface on its terminal block: J4, terminal 13 = CAN L, terminal 14 = CAN H	OEM wiring diagram, page 1 (kg_bot.png crop)	Confirmed
The engine ECU has two CAN channels: channel 1 on pins 55 (CAN H)/77 (CAN L), channel 2 on pins 87 (CAN H)/86 (CAN L)	ecu-pinout-extracted.md , "CAN — two channels"	Confirmed
Which of the ECU's two CAN channels physically connects to the GCU's single CAN port is not stated in either document — the wiring diagram's harness connectors (XC1/XC11) were not traced pin-for-pin to this specific link in the pinout extraction	Both documents	Not established — open item
Baud rate is not stated on either the wiring diagram or in the base KG640 manual	—	Not established — open item
Engine ECU source address (whether 0x00 per SAE J1939 convention) is not stated anywhere in the available documents	—	Not established — open item

What is *not* confirmed, and is presented here only as an informed, explicitly-labelled expectation, not a finding: a J1939 link between a proprietary genset controller and a Tier-4/CPCB-IV+ common-rail ECU of this type would, by nearly universal industry convention (SAE J1939-71), be expected to carry at minimum: **EEC1 (PGN 61444)** for engine speed, **ET1 (PGN 65262)** for coolant temperature, **EFL/P1 (PGN 65263)**

for oil pressure, VEP1 (PGN 65271) for battery/system voltage, DM1 (PGN 65226) for active diagnostic trouble codes, and TSC1 (PGN 0) if the GCU (rather than the ECU alone) is meant to command engine speed/torque. Standard J1939 baud rate for this device class is 250 kbit/s, and the engine ECU node would conventionally sit at source address 0x00. **None of this is sourced to a KG640C document — it is a generic industry baseline offered so the design isn't starting from nothing, and it must be treated as inferred/unconfirmed until either a real KG640C CAN specification is found or the bus is logged directly off the existing panel.**

What would close this: (a) locating and fully retrieving the KG645 manual noted above, on the chance it documents the CAN link in detail; (b) a KG640C-specific manual or CAN database (DBC) from KOEL; (c) the most direct route — connecting a CAN logger to the existing genset's harness at the XC1/XC11 connectors and recording live traffic, which would settle baud rate, source addresses, and the actual PGN set in one measurement, superseding any manual-based inference.

3. Digital I/O contract (Step 4)

Confirmed terminal assignments on the KG640C block (wiring diagram page 1, `kg_top.png` / `kg_bot.png` crops), cross-referenced against the base KG640 manual's terminal table and against `ecu-pinout-extracted.md`'s ECU-side digital I/O list:

KG640C terminal	Group	Function	Direction relative to ECU
J2-3 (terminal 3, "A")	Digital output	Drives START RELAY coil	GCU issues the start <i>request</i> ; see reasoning below on what actually cranks the engine
J2-4 (terminal 4, "B")	Digital output	Drives IGNITION RELAY coil (-13RB1 , 70 A)	GCU-issued; corrected 14 Sep 2026 — this memo originally named this relay -13RB2 , which is the START relay on output A; the ignition relay is -13RB1 . The binding spec always had it right, so the error did not propagate. This relay's contact almost certainly switches battery+ onto ECU pin 71 (<code>IGNITION</code> , confirmed active-high/switched-battery per <code>ecu-pinout-extracted.md</code>), i.e. the GCU powers the ECU on
J2-5 (terminal 5, "C")	Digital output	Drives MC CONNECT RELAY (mains contactor connect)	Not ECU-related — genset/mains transfer logic, panel-side only
J2-6 / J3 area	Digital output	COMMON FAULT relay, GC CONTACT RELAY (genset contactor)	Not ECU-related — panel/contactor logic
J3-7	Combined I/O	D+ CHARGE ALT — GCU provides alternator field excitation and reads back charge-alternator voltage for charge-fail detection	Entirely GCU-and-alternator; the ECU is not involved. <code>ecu-pinout-extracted.md</code> 's connector appendix lists a 2-pin "Charging Alternator" connector present in the harness but never traced to an ECU pin, consistent with this.
J4-13/14	CAN L/H	Single CAN port	Almost certainly the GCU's link to one of the ECU's two CAN channels (see §2) — exact channel not traced
J5-21/22 ("D")/"E")	Digital inputs	Wired to OVERRIDE SS SW - S7	Matches <code>ecu-pinout-extracted.md</code> 's note that ECU pin 24 ("OVERSIDE SWITCH") traces via harness connectors XC1/XC11 to the same -S7 Override SS SW component — this is a shared physical switch, not two different switches , and per the pinout doc it is explicitly not a dedicated overspeed switch: <i>"OVERSIDE SWITCH (pin 24) and Override SS SW -S7 are the same component. There is no separate overspeed switch."</i>

What actually executes cranking — inferred, not fully traceable: `ecu-pinout-extracted.md` lists ECU pin 50 (`MAIN RELAY`) as a **low-side output from the ECU itself**, and separately names a starter (`8STR1`) with its own solenoid (`-S0L`) among the panel components. The most consistent reading of the two documents together is: the GCU's Ignition Relay output energizes the ECU (powers it up); the ECU then runs its own internal start sequence (preheat, fuel priming, crank-permission

logic) and asserts its own `MAIN_RELAY` output to engage the starter solenoid — i.e., **the GCU requests a start (by powering the ECU on) and the ECU executes it** (drives the actual crank relay), matching the brief's framing exactly. However, **the GCU's own "START RELAY" terminal (J2-3) has no confirmed downstream connection to any ECU pin in either document** — it may feed a separate/legacy manual-crank path, a crank-permissive input the pinout extraction didn't capture, or something not traceable without the full XC1/XC11 harness cross-reference (which page 1 of the wiring diagram, the page examined in most detail for this memo, does not itself contain — that harness detail lives on later pages not fully re-examined under this task's time budget). **Flagged as an open item**, not asserted as resolved.

4. What remains unknown

1. **No document titled specifically "KG640C" was found** — the MPU verdict and terminal architecture rest on a sibling "KG640" (non-CAN) manual plus the OEM wiring diagram's own KG640C block, not on a KG640C-specific manual. High confidence, not full confirmation.
 2. **J1939 baud rate, engine-node source address, and the exact PGN set are not confirmed** — no KG640C CAN specification or DBC was located. Direct CAN bus logging off the existing panel is the fastest way to close this and should be prioritized over further document search.
 3. **Which of the ECU's two CAN channels (55/77 or 87/86) physically connects to the GCU's single CAN port (J4, 13/14) is not established.** If the second channel is used for something else (a dealer diagnostic tool, telematics/remote-monitoring per KOEL's own marketing material mentioning a "remote monitoring system"), that has implications for whether diagnostic tooling can coexist with the GCU link without contention.
 4. **The GCU's own "START RELAY" output (terminal J2-3) has no confirmed destination** — whether it reaches the ECU, the starter solenoid directly, or a legacy manual-start path is unresolved from the documents in hand.
 5. **INR pricing for the two correctly-matched standalone overspeed modules (GAC SSW675, Murphy/Enovation HD9063) was not obtained** — both have confirmed Indian distributors but require a direct quote request.
 6. **The "KG645" manual** (a possible CAN-capable sibling found via search, <https://www.scribd.com/document/407079798/>) was not retrieved in full this session and may resolve some or all of items 1–3 above if obtained.
 7. **As with Task 1, a photograph of the physical KG640C rear terminal block would be the single most valuable piece of evidence** this design phase could still obtain — it would upgrade the MPU verdict from "inferred, high confidence" to "confirmed" in one image, and would likely also show CAN port labelling that clarifies item 3.
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Sources consulted (full list)

- `/home/mohan/workws/ecu25kva/refs/GP3.8703.C4.pdf` — OEM wiring diagram, page 1, rendered at 400 DPI and read directly (crops examined: KG640C block terminals J1–J8, in this session's scratchpad).
- `/home/mohan/workws/ecu25kva/refs/ecu-pinout-extracted.md` — ECU-side pin-level extraction of the same diagram, read in full.
- `/home/mohan/workws/ecu25kva/docs/research/01-engine-identity.md` — Task 1 memo, engine and controller identity.

- KOEL, "KG640 CONTROLLER MANUAL" / SED-MAN-KG640-002 , 92 pages, retrieved via <https://www.scribd.com/document/878092942/SED-MAN-KG640-002-Koel-Manual-for-KG640-Genset-Controller> — read in full.
- KOEL, "KG645 Genset Controller User Manual," <https://www.scribd.com/document/407079798/Kirloskar-KG645-User-manual-for-genset-controller-pdf> — located, **not** retrieved in full (fetch returned only page metadata); flagged as an open lead.
- KOEL, "KG934V1 Genset Controller User Manual" (2010), <https://5.imimg.com/data5/SELLER/Doc/2021/7/JH/DB/YP/28202182/kirloskar-green-kg934v1-genset-controller.pdf> — located as an older/smaller sibling product during search, not used as evidence in this memo (different, smaller controller family).
- SmartGen (Zhengzhou) Technology Co., "HGM6100N Series Genset Controller User Manual" — pulled as a rebadge-comparison candidate; ruled out as the KG640C's underlying product per §1.
- Governors America Corp (GAC), SSW675 product page, <https://www.governors-america.com/product/ssw675/> (direct fetch blocked by site; description obtained via search-result excerpt of the manufacturer's own page).
- Governors America Corp (GAC), ESD5500E documentation (via search snippet) — cited only to show GAC's own explicit statement that a separate overspeed shutdown device is required in addition to its analog governors.
- Enovation Controls (Murphy), HD9063/OS77D/SS300 product page, <https://www.enovationcontrols.com/products/hd9063-os77d-and-ss300/> — fetched directly, specs quoted verbatim.
- IndiaMART listing, Spares Bazaar Pvt Ltd (New Delhi), GAC ESD5500E/ESD5111/ESD5550/ESD5221 at ₹9,500/piece, <https://www.indiamart.com/proddetail/esd5500e-esd5111-esd5550-esd5221-gac-speed-control-unit-governor-america-corp-21293246791.html> — cited only as order-of-magnitude brand/market context, not as pricing for the correct part (see caveat in the recommendation table).
- IndiaMART listing, Murphy By Enovation Controls (Pune), HD9063/OS77D/SS300, <https://www.indiamart.com/proddetail/hd9063-os77d-and-ss300-electronic-speed-switches-15889251691.html> — quote-only, no published price.
- ikonnect.kirloskar.com portal — checked for a KG640C-specific manual index; none exposed without authentication (consistent with Task 1's finding).
- General web searches (recorded for completeness, mostly negative or superseded by the direct fetches above): Kirloskar KG640C manual , KG640C genset controller datasheet , KOEL KG640C wiring , Kirloskar KG640C manual J1939 CAN engine PGN , "KG640C" Kirloskar controller manual pdf , scribd "KG640C" OR "SED-MAN-KG640C" Kirloskar CAN J1939 , GAC ESD5500 electronic overspeed shutdown device price India magnetic pickup , GAC ESD5111 electronic overspeed shutdown device specification magnetic pickup frequency , "Governors America" OR GAC overspeed shutdown device dedicated relay model number , GAC SSW675 speed switch price India , "SSW675" OR "SSW676" OR "speed switch" GAC indiamart price generator overspeed India , Murphy overspeed shutdown module magnetic pickup speed switch price India , Murphy OS77D speed switch India IndiaMART price , Murphy Enovation OS77D specification frequency input voltage amplitude magnetic pickup , Electra Brandt SSW675 price .

Research Memo 03: ECU Connector Identification

U1 CLOSED, 18 Sep 2026 — this memo's leading candidate is confirmed

The connector has been photographed with its markings legible. It carries the **Bosch** name and armature logo, part numbers **1 928 405 192** and **1 928 405 194**, code **C**, and **2.7**. The face-on photograph shows 94 cavities in two chambers: 8 large power contacts (1–8) plus four fine rows numbered 9–28, 29–50, 51–72, 73–94.

This memo ranked "Bosch EDC17-style 94-pin" as the leading candidate on inference. Confirmed, with part numbers read off the physical part.

One detail this memo did not anticipate: code **C** is a **mechanical coding variant**. Bosch supplies these housings in coded versions that will not mate across codes, so any replacement or adapter housing must be code C.

A search for the two part numbers returned no direct catalogue hit, so sourcing should go through a Bosch distributor quoting the numbers rather than a search engine. The shortlist and sourcing analysis below remain useful for that conversation.

Evidence: `refs/field-evidence-2026-09-18.md`.

Task: U1 (ECU connector part number / mating-half sourcing), partial closure — full closure needs a photograph. Consumes Task 1 (engine identity). Feeds the adapter-harness deliverable (spec §7.1). **Date:** 2026-09-13 **Author:** Phase-1 research (task 3) **Engine/genset:** Kirloskar 3R550ETA 4G1 / KG4-25WS1, per `docs/research/01-engine-identity.md` (inferred, high confidence).

Verdict (read this first)

The OEM ECU part number **could not be established** — same negative result already recorded in `01-engine-identity.md`, Step 2, and not re-openable with the sources available here. What this memo adds is a **physical characterisation of the 94-pin ECU connector** from the wiring diagram, and a **ranked shortlist of connector families** that plausibly match it, each with a buyability verdict for India. No part number is

asserted with confidence — per the task's own framing, a line drawing cannot support that, and a confident wrong connector is worse than an honest shortlist.

Rank	Family	Visual match	Buyable in India	Verdict
1	Bosch EDC17/EDC15/EDC16-style 94-pin ECU connector (TE-Connectivity-built, Bosch-proprietary tooling)	Strong — pin count exact, mixed large/small contact pattern plausible, widely used on Bosch common-rail diesel ECUs of this class	Yes, but only through the automotive ECU-repair/chiptuning aftermarket, not mainstream component distributors — order via specialty ECU-connector shops (international, ships to India)	Leading candidate — photograph required to confirm
2	Aptiv/Delphi GT-series (150-position family) rectangular multi-row connector	Moderate — rectangular, multi-row, mixed contact sizes are a documented feature of this family; exact 94-position variant not confirmed	Yes — stocked by Mouser (mainstream distributor with an India storefront)	Plausible alternate
3	TE Connectivity AMPSEAL (16/23/35/direct-descendant 90-series)	Weak — AMPSEAL housings are characteristically oval/rounded single-lever shells; the PDF drawing shows a rectangular, sharp-cornered, multi-row grid that fits this family less well	Yes — mainstream Mouser/DigiKey stock	Weakest visual fit of the three buyable candidates
—	Molex CMC / high-density cylindrical automotive headers	Ruled out — CMC-family connectors are round/cylindrical pin-and-socket bundles, not a flat rectangular multi-row grid	N/A	Ruled out on shape alone

None of these is asserted as "the" connector. All three buyable candidates are plausible; the photograph brief (final section) is designed to distinguish them in a single site visit.

Step 1: What the wiring diagram actually shows

Source: `refs/GP3.8703.C4.pdf` , page 4, "Engine Control Unit Connector" panel (one of six connector-detail panels on that page), re-rendered at 400 dpi with `pdftoppm` and cropped with PIL for close reading.

Confirmed directly from the drawing: - The connector is drawn as **one single housing outline** (one continuous molded shell in the 2D projection) — not two or three physically separate ganged blocks side by side. Pins 1–94 are called out on this one outline. - The pin field is arranged as **four rows** of small, evenly pitched square contacts, split top/bottom: an upper pair of rows (running from a low pin number

near "7" up to "28" at the top-right corner) and a lower pair of rows (running from "73" at bottom-left to "94" at bottom-right corner). This matches `refs/ecu-pinout-extracted.md`'s statement that the connector is 94-pin. - At the **top-left and bottom-left corners**, outside the dense small-pin grid, sit two separate 2-contact clusters, individually labelled "PIN NO.1" (top) and "PIN NO.5" (bottom). These two clusters are drawn with the **identical stepped-rectangle pin symbol** used elsewhere on the *same PDF page* for the standalone 2-pin **Fuel Pump Connector** and **Charging Alternator Connector** panels — both of which are physically large, single-current-path power/current connectors in this harness. **Inferred** (by direct comparison of drawing conventions within one source document, not an explicit label): pins 1, 2, 5, 6 are drawn as large/keyed contacts, distinct in kind from the ~88 small pins in the main grid. This is consistent with the brief's expectation that a 94-pin automotive ECU header of this class mixes large power/injector contacts with small signal contacts in one shell. - Between the upper and lower row-pairs, **four elongated rectangular features** run across the housing's mid-band. None of them carries a pin number anywhere in the source (every actual contact in this diagram set is numbered). **Inferred:** these are molded polarization/alignment ribs in the plastic shroud (a standard feature on wide, high-pin-count automotive headers to prevent pin damage on blind mating), not additional large contacts. - On the **right-hand (mating/cable) side** of the housing, the outline shows a curved, bowed wall plus two small rectangular tabs protruding further right than the main shell, near the top-right and bottom-right corners. **Inferred, not confirmed:** this is most consistent with pivot bosses and body of a **lever-actuated latch** (used on high pin-count automotive connectors to provide the large mating force ~94 contacts require) — but a cable-entry strain-relief boot is a plausible alternative reading of the same 2D outline, and the drawing does not disambiguate. This is the single largest open question the photograph needs to resolve. - Small rectangular tabs are also present at even intervals along the top and bottom edges of the housing (visible in the full-page crop) — consistent with either mounting/latch-catch features or additional polarization ribs; not resolvable further from a line drawing. - **Not a housing marking:** the "Kirloskar / Oil Engines" watermark visible across the page image is the ikonnect web portal's page watermark (confirmed present on every page of this browser-printed PDF, per `01-engine-identity.md` Step 2), not a molded marking on the physical connector. This is flagged explicitly so it is not mistaken for a part marking later.

What the drawing does not show and cannot be inferred: exact housing dimensions, contact pitch, wire-seal type (individual seals vs. a single mat seal), material/color, or any molded part number/logo.

`refs/ecu-pinout-extracted.md` already records that only 44 of the 94

pins are traceable to a signal in the wiring diagram; the connector-detail panel on page 4 is a generic "how to count pins" reference (it exists to let a technician find pin 1/5/28/73/94 as anchor points), not a manufacturer datasheet.

Step 2: Candidate families, ranked, with reasoning

1. Bosch EDC15/EDC16/EDC17-style 94-pin ECU connector (leading candidate)

- **Why it's plausible:** The pin count (94) is an exact match, not just "in the ballpark" — multiple independent listings in the automotive ECU-repair/chiptuning aftermarket specifically describe a "Bosch EDC17 94-pin" (and a companion 60-pin, for two-connector EDC17 variants) ECU connector, sold as bench/pigtail cable kits: e.g. [u-obd.com](https://www.u-obd.com/product/edc17-connector/) ("BOSCH EDC17 Connector 94 PIN 60 PIN plug cable set full PIN wired," <https://www.u-obd.com/product/edc17-connector/>, US\$99), [ecudepot.com](https://ecudepot.com/product/bosch-edc17-connector/) (same product, <https://ecudepot.com/product/bosch-edc17-connector/>), [autoecupart.net](https://www.autoecupart.net/products/original-full-94-pin-connector-for-bosch-edc17-electronic-control-unit-ecu) (multiple listings, e.g. <https://www.autoecupart.net/products/original-full-94-pin-connector-for-bosch-edc17-electronic-control-unit-ecu>), and eBay listings (e.g. <https://www.ebay.com/itm/265128037857> "OEM Full 94 Pin Connector for Bosch ME17.9.11 EDC17"). **Confirmed:** a documented, commercially available "Bosch 94-pin" ECU connector family exists and is actively sold as a discrete product. **Not confirmed:** that this specific family is the one on the Kirloskar genset — no source ties a Bosch part number to manual 08-GP3-60-001 or to this engine.
- **Corroborating context from this project's own prior research (not new evidence, but relevant):** <docs/research/04-injector-drive.md> independently concludes the injector architecture is most likely a Bosch solenoid common-rail family (CRIN2/CRIN3-class), based on wiring-diagram evidence (rail-pressure sensor + PWM metering-unit pattern) and Kirloskar's confirmed practice of re-badging genuine Bosch injectors under Kirloskar part numbers for at least one other engine family (Bosch 0445120646 = Kirloskar F6.248.08.0.pr). If the injectors are Bosch, an ECU built around a Bosch-family connector (rather than a mixed-vendor harness) is the more parsimonious assumption — **inferred, not a separate confirmed fact.**
- **Retention/shape match:** the EDC15/16/17 94-pin housing family is documented in the tuning trade as using a mixed grid of small signal pins with a smaller number of larger power/injector pins in one shell, closely matching the page-4 drawing's large-corner-pair-plus-dense-grid layout described in Step 1. This comparison is based on general familiarity with these connectors in the aftermarket record, not a side-by-side dimensioned drawing — **inferred, moderate confidence.**

- **Buyability (India): Yes, but not through a normal electronic-component distributor.** No listing was found on Mouser India, DigiKey India, or Element14 for a "Bosch EDC17 94-pin" connector under any manufacturer part number — searches for a TE Connectivity part number tied to this housing returned no result (**negative result**). It is however readily orderable from the international automotive ECU-repair trade (u-obd.com, ecudepot.com, autoecupart.net, eBay/AliExpress sellers), all of which ship internationally including to India, typically **US\$65–99 per full connector-pair kit** (94-pin + companion 60-pin, wired pigtail) plus international shipping and India customs duty/GST on import (landed cost not independently verified — treat the \$65–99 base figure as confirmed-at-source, the India-landed total as **unverified, order-of-magnitude estimate only**). This satisfies "a connector we cannot buy is not a candidate" — it is buyable — but the sourcing channel is the aftermarket repair trade, not a component distributor, which has implications for lead time, quality/counterfeit risk, and lack of a formal datasheet.

2. Aptiv/Delphi GT-series rectangular multi-row connector (150-position family)

- **Why it's plausible:** The GT150 family is a genuine, rectangular, multi-row automotive ECU/PCM connector platform distributed through mainstream channels (Mouser lists it directly: <https://www.mouser.com/Delphi-Connection-Systems/Connectors/Automotive-Connectors/GT-150-Series/>, and specialist motorsport/ECU-wiring shops such as Corsa Technic and Waytek Wire stock GT150 housings, terminals, and seals across a range of position counts). Its general silhouette — rectangular shell, dense rows of small pins with provision for larger power contacts in the same housing family — is structurally similar to what the page-4 drawing shows. **Confirmed:** this connector family exists, is rectangular/multi-row, and is mainstream-distributor-stocked.
- **What is not confirmed:** whether a 94-position GT-series variant exists and matches the specific layout (four small-pin rows plus two 2-large-pin corner blocks) seen on page 4. This was not verified in this pass — **negative result, flagged as an open item**, not a ruled-out candidate.
- **Buyability (India): Yes** — Mouser operates an India storefront with INR pricing and ships GT-series connectors; DigiKey/Element14 India stock was not separately checked in this pass (**not verified**, time-boxed). Exact INR unit pricing for a 94-position variant was not obtained in this pass — recommend confirming against Mouser India's live catalog once/if this candidate is pursued further.

3. TE Connectivity AMPSEAL (90-series descendants)

- **Why it's a weaker fit:** AMPSEAL is a genuinely mainstream, India-buyable automotive sealed-connector family (Mouser lists the series directly: <https://www.mouser.com/en/c/connectors/automotive-connectors/?m=TE+Connectivity&series=AMPSEAL>) and is used on some diesel ECUs. But the AMPSEAL housing silhouette is characteristically **oval/rounded with a single center or side lever**, single or double staggered pin rows — not the sharp-rectangular, four-row, dense micro-pin grid with two separate large-pin corner clusters that page 4 shows. **Inferred:** shape mismatch downgrades this candidate relative to #1 and #2, even though it is the easiest of the three to actually buy through normal distribution.
- **Buyability (India): Yes, straightforwardly** — standard Mouser/DigiKey stock, no aftermarket-only caveat. This is the safest fallback if the photograph rules out #1 and #2.

Ruled out: Molex CMC and similar high-density cylindrical automotive headers

Molex's CMC (and comparable high-pin-density automotive families built around round/cylindrical pin-and-socket bundles rather than a flat rectangular grid) do not match the page-4 drawing's flat, rectangular, multi-row silhouette at all. **Ruled out on shape alone** — no further sourcing work done on this family. (*Negative result, recorded per the global instruction that negative results are results.*)

Step 3: Sourcing summary table

Candidate family	Housing/mating-half source	Contacts/seals source	Price (as found)	India availability	Buy verdict
Bosch EDC15/16/17 94-pin (+60-pin companion)	u-obd.com, ecudepot.com, autoecupart.net, eBay sellers (aftermarket ECU-repair trade)	Bundled with the wired pigtail kits above; loose terminals also listed separately on some of the same sites	US\$65–99 per kit (source-confirmed); India-landed cost incl. shipping/duty not verified, estimate only	Yes, via international aftermarket shipping — not stocked by Mouser/DigiKey/Element14 India (checked, negative result)	Buyable — proceed if photograph confirms Bosch-style housing
Aptiv/Delphi GT150-series	Mouser (mainstream distributor); Corsa Technic, Waytek Wire (motorsport specialty)	Same distributors, per-position contact/seal kits	Not obtained in this pass for a 94-position variant — unverified	Yes — Mouser has an India storefront	Buyable — needs live-catalog price/position check before ordering
TE Connectivity AMPSEAL (90-series)	Mouser, DigiKey (mainstream, confirmed stocked)	Same distributors	Not itemised in this pass	Yes — mainstream stock	Buyable — safest fallback, weakest visual match

Candidate family	Housing/mating-half source	Contacts/seals source	Price (as found)	India availability	Buy verdict
Molex CMC / cylindrical high-density	—	—	—	—	Not a candidate — shape mismatch

Step 4: The photograph brief that closes this

One site visit, done right, resolves which of the three buyable candidates is correct. Photograph the ECU connector **on the engine harness side (the plug, not the ECU-mounted socket)** and the ECU case itself, as follows:

1. **Full-face shot of the connector's mating face**, camera square-on (not at an angle) to the pin field, filling the frame. This is the single most important shot: it will immediately show whether the housing is one rectangular shell with a flat multi-row grid (Bosch/Delphi-style — candidates #1/#2) or an oval/rounded single-lever shell (AMPSEAL-style — candidate #3), settling the ranking's top-level question in one image.
2. **A shot of the retention mechanism actuated/engaged and again disengaged** (if it can be safely operated with the engine off and battery disconnected) — specifically to determine whether it is a pivoting lever, a slide latch, a center bolt, or simple friction/CPA-clip retention. This resolves the Step-1 ambiguity about the bowed feature on the connector's right side.
3. **A raking (low-angle, side-lit) shot of every flat surface of the housing** — top, both sides, and the cable-entry boot — specifically hunting for a **molded (raised or recessed) part number, date code, or manufacturer logo**. These are usually small (2–4 mm character height) and shallow, so: use a raking light held at a low angle roughly parallel to the surface (a phone flashlight held to the side works well) rather than the on-axis camera flash, which washes out shallow embossing. Take multiple shots at different light angles if the first pass doesn't reveal legible text. Manufacturer logos to watch for specifically: the Bosch roundel, a Delphi/Aptiv wordmark or triangle logo, or a TE Connectivity ("te" stylized) mark — presence of any one of these directly resolves the family question.
4. **The ECU's own case label/nameplate**, photographed flat-on and well-lit — looking for a Bosch part number of the pattern `0 281 0xx xxx` (EDC-family) or `F 01R 0xx xxx` (Bosch reseller code, sometimes used by Kirloskar per the injector cross-reference in `04-injector-drive.md`), or any Continental/Delphi/Denso equivalent. Finding this would independently resolve the still-open "OEM ECU part number" question from `01-engine-identity.md` Step 2, which is a bigger win than the connector question alone.

5. A wide shot showing the connector still mated to the harness, to capture the harness-side (socket) connector's backshell and cable exit angle/strain relief — useful for the adapter-harness design even if it adds nothing to family identification.
6. A **pin-count check shot**: with the connector unplugged (engine off, battery disconnected), count and photograph the pin rows directly, confirming the "four rows, two 2-pin corner blocks" structure this memo infers from the line drawing, and specifically checking whether the two corner-pin contacts are visibly larger in diameter/blade-width than the grid pins — this is the single check that would upgrade Step 1's "inferred" finding about mixed contact sizes to "confirmed."

Lighting/angle summary: natural or diffuse white light for the full-face and pin-count shots (avoid on-axis flash — it flattens the pin field and hides depth); raking side-light specifically for hunting molded text. A macro mode or phone "portrait"/close-focus mode is recommended for the part-number hunt, since the expected character height is small.

Sources consulted

- `/home/mohan/workws/ecu25kva/refs/GP3.8703.C4.pdf` — page 4, "Engine Control Unit Connector" panel, re-rendered at 400 dpi (`pdftoppm -r 400 -f 4 -l 4`) and examined via cropped PIL renders.
- `/home/mohan/workws/ecu25kva/refs/ecu-pinout-extracted.md` — confirms 94-pin count and that only 44/94 pins are traceable in the wiring diagram.
- `/home/mohan/workws/ecu25kva/docs/research/01-engine-identity.md` — engine/genset identity; Step 2 negative result on OEM ECU part number, carried forward unchanged (not re-openable with sources available here).
- `/home/mohan/workws/ecu25kva/docs/research/04-injector-drive.md` — Bosch-injector-family inference used as corroborating (not standalone) context for candidate #1.
- <https://www.u-obd.com/product/edc17-connector/> — Bosch EDC17 94-pin+60-pin connector kit, US\$99, fetched directly.
- <https://ecudepot.com/product/bosch-edc17-connector/> — same product, found via search.
- <https://www.autoecupart.net/products/original-new-full-94-pin-connector-for-bosch-edc17-electronic-control-unit-ecu> and related autoecupart.net listings — found via search, corroborate the "Bosch 94-pin EDC17" connector as a distinct, repeatedly-listed aftermarket product.

- <https://www.ebay.com/itm/265128037857> and <https://www.ebay.com/itm/254545773762> — eBay listings, found via search, further corroboration.
- <https://www.mouser.com/Delphi-Connection-Systems/Connectors/Automotive-Connectors/GT-150-Series/> — Aptiv/Delphi GT150 series distributor listing, found via search.
- https://www.corsa-technic.com/category.php?category_id=178 and <https://www.waytekwire.com/catalog/connectors/aptiv-gt150-connectors> — GT150 specialty-distributor listings, found via search.
- <https://www.mouser.com/en/c/connectors/automotive-connectors/?m=TE+Connectivity&series=AMPSEAL> — TE Connectivity AMPSEAL series distributor listing, found via search.
- <https://bosch-connectors.com/> (PS Connectors) — fetched directly; page returned only navigation/legal boilerplate, no usable technical content (**negative result**, recorded rather than omitted).
- Searches returning negative or inconclusive results (recorded per the global instruction): Bosch EDC17 94 pin connector India price Delhi distributor OR Amazon.in OR IndiaMART (no India-specific listing found); Bosch MD1 MDG1 154 pin ECU connector ampseal delphi GT94 comparison (no direct technical comparison found); TE Connectivity AMPSEAL 90 way OR Delphi GT150 150 pin connector Mouser India price (distribution channel confirmed, exact India pricing not obtained).
- Prior-session scratchpad files reviewed for reusable material (/tmp/claude-1000/ ... /scratchpad/): `drivven.txt` (Drivven DI Driver Module manual — relevant to Task 4, not this task), `bpc960.txt` (Bliss Powercom BPC960 genset-controller manual — a different controller, not the KG640C or the engine ECU, not used in this memo), `apalrd.txt` (generic common-rail-diesel-control-system slide deck — not connector-specific, not used). None of these contained ECU-connector-family evidence directly usable here; reviewed to avoid duplicate fetching, not cited as sources for any claim above.

Research Memo 04: Injector Identification and Drive Profile

Task: U5 (injector drive stage — boost rail voltage, peak/hold current, pulse timing). Consumes Task 1 (engine identity). Feeds the injector driver sheet and Phase-2 boost converter design. **Date:** 2026-09-13

Author: Phase-1 research (task 4) **Engine:** Kirloskar 3GK550ETA 4SR1 (this memo was written as 3R550ETA 4G1 ; corrected 18 Sep 2026 from the rating plate — the family envelope this memo derives is unaffected, since it was never tied to a part number). 3-cyl inline, 3-cyl inline, 1.65 L, common-rail, turbocharged-aftercooled, 1500 rpm, 26.5 kW prime (per docs/research/01-engine-identity.md , inferred with high confidence). **Genset:** Kirloskar KG4-25WS1.

Verdict (read this first)

Item	Value	Status
Injection system type	Solenoid common-rail, Bosch-family architecture (not piezo)	Inferred
Injector supplier/part number	Not established for this specific engine (3R550ETA)	Negative result — see Step 1
Boost rail voltage envelope	65–115 V DC, most solenoid CR designs cluster 90–115 V; this engine most likely sits near ~100 V ± 15 V	Inferred envelope
Peak current	12–24 A, most designs 18–20 A	Inferred envelope
Peak current duration	~0.3–1.0 ms (boost phase, until current threshold reached)	Inferred envelope
Hold current	~8–13 A, most designs ~10–12 A	Inferred envelope
Minimum controllable pulse width	~0.1–0.3 ms	Inferred envelope
Required driver slew rate	40–100 A/ms covers the family	Inferred envelope (matches brief's design target)
Bank-sharing timing check (Step 3)	PASSES, with large margin (≥240 crank degrees / 26.67 ms minimum separation vs. ≤~3 ms max injection duration)	Confirmed by calculation — see Step 3
Calibration boundary (Step 5)	Hardware can be built and bench-verified; cannot run this engine correctly without flow-vs-pulsewidth maps not publicly available	Stated plainly — see Step 5

The headline finding for Step 3 is good news, stated loudly because the brief asked for it either way: **the bank-shared wiring (cyl 1+3 on one high side, pin 03) is inherently safe for any plausible 3-cylinder firing order**, because a 3-cylinder 4-stroke engine with even firing intervals has exactly three, evenly-spaced injection events per 720° cycle — so *any* two of the three cylinders are always separated by one full firing interval (240°) in the closer direction, regardless of which firing order (1-2-3 or 1-3-2) is used. This is a property of "3 evenly spaced cylinders," not of the specific order label. See Step 3 for the full derivation and the one assumption that could break it.

Step 1: Injector family identification

Search performed: Kirloskar R550 common rail injector Bosch part number , Kirloskar 3R550 injector Denso , cross-reference of [crdiparts.com](#) and [turbotech.in](#) Kirloskar-branded injector listings.

What was found: - Kirloskar packages Bosch common-rail injectors under its own part-number scheme for at least one other engine family: Bosch injector 0445120646 is sold as Kirloskar part F6.248.08.0.pr, "suitable for Kirloskar engine in genset and Liugong loader" (confirmed listing, <https://www.crdiparts.com/product/common-rail-fuel-injector-0445120646-f6-248-08-0-pr-suitable-for-kirloskar-engine-in-genset-and-liugong-loader/>, mirrored at <https://turbotech.in/product/common-rail-fuel-injector-0445120646-f6-248-08-0-pr-suitable-for-kirloskar-engine-in-genset-and-liugong-loader/>). The "F6.248" prefix is consistent with Kirloskar's engine-family/displacement coding (analogous to the "3R550" pattern already established in Task 1) and most plausibly denotes a different, larger (6-cylinder-class) Kirloskar engine, **not** the 3-cylinder R550 family used in the KG4-25WS1. **Confirmed:** Kirloskar sources genuine Bosch solenoid CR injectors and re-badges them with Kirloskar part numbers of the form `<code>.<code>.<code>.<code>.pr` . **Not confirmed:** which Bosch part number corresponds to the 3R550ETA

specifically. - No source found ties a specific Bosch (or Denso) injector part number to "3R550" or "3R550ETA" by name. Searches tried: Kirloskar 3R550 common rail injector Bosch part number, Kirloskar R550 CRDI injector Denso. **Negative result**, recorded per the global instruction that negative results are results. - The wiring evidence itself (rail-pressure sensor on pins 32/35/08, PWM-driven metering unit on pin 88 per refs/ecu-pinout-extracted.md) confirms a solenoid-type high-pressure common-rail architecture (a metering-unit-controlled rail, not a unit-injector or unit-pump system), which is consistent with the Bosch CRSN2/CRIN family design pattern used broadly in this displacement/power class. **Inferred** from architecture, not a part-number match.

Working assumption for this memo (inferred, not sourced): the injector is a **Bosch CRIN-family solenoid injector**, most likely in the CRIN2/CRIN3 generation used across Bosch's small-to-mid displacement common-rail industrial/genset engines circa the CPCB IV+ generation (2022–2023 introduction per Task 1). This is a reasonable design assumption but should not be treated as a confirmed part number. **A rating-plate or injector-body photograph (stamped Bosch part number, e.g. 0445 1xx xxx) would close this** — the same recommendation Task 1 made for engine identity.

Step 2: Drive profile envelope

Exact peak/hold/boost figures are proprietary to the injector manufacturer and are not published for a specific part number without an NDA or a physical sample. Per the brief, the correct engineering response is **an envelope that covers the family**, not a guess at one number.

The envelope, and why these bounds

Solenoid common-rail injectors (Bosch CRIN, Denso G3/G4, Delphi/Continental DFI) all use the same two-stage drive strategy for the same physical reason: the injector solenoid must open against rail pressure (which can exceed 1600–2000 bar in this class) within a few hundred microseconds, which requires driving far more current into the coil, far faster, than a 12 V battery rail can deliver into a ~0.3–1 mH / <1 Ω coil. The universal solution:

1. **Boost phase:** a boost converter (12 V $\rightarrow$ tens to ~100+ V) is switched across the coil to slew current up rapidly to a peak threshold.
2. **Peak hold:** current is regulated (chopped) at the peak level for a short duration to fully open the pin/needle against flow forces.
3. **Hold phase:** current is dropped to a lower "hold" level (higher-voltage boost removed, battery rail or PWM-chopped boost used) for the remainder of the injection, to reduce power dissipation and heating.
4. **Fast turn-off:** the low side is opened and freewheel/clamp circuitry rapidly collapses the field to close the injector precisely.

Sourced figures found for this class of driver: - A peak-and-hold strategy using **~100 V boost to reach 20 A peak, then 24 V (battery-derived) to hold ~10 A** is described directly in an automotive-oscilloscope diagnostic reference for Bosch common-rail solenoid injectors (confirmed, <https://www.picoauto.com/library/automotive-guided-tests/bosch-current-at-idle>). - NXP's own reference design for its dedicated injector-driver IC (MC33816, see Step 4) documents a three-stage waveform: **48 V boost to 19 A, regulated hold at 19 A for the peak duration, then regulated hold at 11 A** for the remainder of injection (confirmed, drawn from NXP application material referenced in general search results for the MC33816 common-rail reference design). - A peer-reviewed overview of common-rail injector driving circuitry states boost voltages in the 65–115 V range and peak currents around 20 A, with coil resistance $\leq 1 \Omega$ and inductance ≤ 1 mH typical for this injector class (confirmed, "Circuit for driving common rail diesel injectors," ResearchGate, https://www.researchgate.net/publication/350426572_Circuit_for_driving_common_rail_diesel_injectors). - NXP's MC33816 product page states its pre-driver stage operates up to 72 V (confirmed, <https://www.nxp.com/products/analog-and-mixed>).

signal/gate-drivers/high-precision-differential-injector-driver:MC33816)
— lower than the ~100–115 V boost used by some Bosch-native designs, which means a driver built around this IC's pre-driver alone would need external high-voltage MOSFETs/gate stages if the target boost rail is pushed toward the top of the family envelope.

Envelope adopted for this project (matches the brief's design target and is corroborated by the above):

Parameter	Envelope	Where this engine likely sits, and why
Boost rail voltage	65–115 V	Likely ~90–105 V. This is a mid-displacement (0.55 L/cyl) industrial/genset engine, not a high-speed passenger-car CR engine chasing sub-300 µs injector response; it does not need the most aggressive (115 V-class) boost used by some high-speed automotive applications, but a genset running at a fixed 1500 rpm still benefits from fast, repeatable injector opening for combustion-timing accuracy, so the low end of the range (65 V, associated with older/smaller injectors) is also unlikely. Inferred , no part-number-specific source.
Peak current	12–24 A	Likely ~18–20 A, matching both cited reference designs (Picoscope: 20 A; NXP MC33816 reference: 19 A) which cluster tightly despite being independent sources. Inferred .
Peak duration	0.3–1.0 ms	Set by however long it takes the boost stage to reach threshold current against coil L/R and back-EMF; no engine-specific source. Inferred envelope only .
Hold current	8–13 A	Likely ~10–12 A, again bracketed by the two cited reference designs (10 A, 11 A). Inferred .
Min. pulse width	0.1–0.3 ms	Standard for pilot-injection-capable CR solenoid injectors; this engine's CPCB IV+ emissions calibration likely uses pilot injection for NOx/noise control, so the low end of this range should be treated as a hard requirement, not headroom. Inferred .
Driver slew rate	40–100 A/ms	Per the brief's own design guidance; consistent with reaching ~20 A peak within under 0.5 ms at a ~90–100 V boost rail against a sub-1 mH coil (a first-order L/di/dt estimate: 100 V / 1 mH ≈ 100 A/ms at turn-on, tapering as back-EMF rises — internally consistent with the cited figures). Derived, not directly sourced .

Design conclusion: a driver spanning 65–115 V boost, 12–24 A peak, 8–13 A hold, 40–100 A/ms slew will drive any solenoid CR injector in this displacement/power class, this engine's specific injector included, without knowing its exact part number. This is the number set to carry into the Phase-2 boost converter and driver-IC selection.

Step 3: Bank arrangement vs. firing order — the safety check

The wiring fact being checked

Per `refs/ecu-pinout-extracted.md`: pin 03 is a single high-side driver output shared, via harness splice V5, between cylinder 1 and cylinder 3. Pin 05 is a dedicated high side for cylinder 2. Low sides are independent per cylinder (73 → cyl1, 07 → cyl3, 29 → cyl2). **The failure mode to rule out:** if cylinder 1 and cylinder 3 ever needed to inject at overlapping times, sharing one high-side node would prevent independent control — both injectors would be forced onto the same boosted supply simultaneously, which the low-side switches alone cannot arbitrate into two independently-timed pulses.

Establishing the firing order

Three-cylinder inline 4-stroke engines fire every $720^\circ/3 = 240^\circ$ of crank rotation when cylinders are evenly spaced (the standard configuration — inline-3 crankshafts use 120° crank-throw spacing, which combined with the 720° four-stroke cycle produces exactly this even 240° firing interval). The two firing orders seen across the industry, 1-3-2 and 1-2-3, both satisfy this even spacing; they differ only in which physical cylinder fires second, not in the interval between any two firing events. **Confirmed general mechanics** (basic four-stroke engine theory; see e.g. https://en.wikipedia.org/wiki/Firing_order for the general formula). No source was found stating the 3R550ETA's specific firing order by name — searches for `Kirloskar 3R550 firing order`, `Kirloskar R550 firing order 1-2-3` OR `1-3-2` returned no Kirloskar-specific document (**negative result**). This does not block the safety check, for the reason below.

The check

At 1500 rpm rated speed (per Task 1, confirmed): one crankshaft revolution = 60,000 ms / 1500 = 40 ms. A four-stroke cycle is $720^\circ = 2$ revolutions = **80 ms**, matching the brief exactly.

Three evenly-spaced injection events per 80 ms cycle → one event every $80/3 \approx 26.67$ ms, i.e. every 240° of crank rotation.

Label the three firing instants within one 720° cycle as 0°, 240°, 480° (order 1-2-3: cyl1@0°, cyl2@240°, cyl3@480°; order 1-3-2: cyl1@0°, cyl3@240°, cyl2@480° — the derivation below is order-independent, done for both explicitly):

- **Order 1-2-3:** cyl1 at 0°, cyl3 at 480°. Forward gap cyl1 → cyl3 = 480° (53.3 ms). Backward gap (cyl3 in the *previous* cycle, at 480–720 = –240°, to cyl1 at 0°) = 240° (26.67 ms). **Minimum separation = 240° = 26.67 ms.**
- **Order 1-3-2:** cyl1 at 0°, cyl3 at 240°. Forward gap cyl1 → cyl3 = 240° (26.67 ms). **Minimum separation = 240° = 26.67 ms.**

Both orders produce the **identical minimum separation: 240 crank degrees, 26.67 ms**. This is not a coincidence specific to either order — it is a general property of any 3 evenly-spaced points on a 720° cycle: with exactly three points, every pair is separated by exactly one interval (240°) in one rotational direction and two intervals (480°) in the other, so the *minimum* pairwise gap is always one interval, **regardless of firing-order labeling**. This is why Step 3 does not actually hinge on resolving the unconfirmed firing order — the bank-sharing safety property holds for both candidate orders identically. **This derivation is original calculation for this memo (basic four-stroke kinematics), not drawn from an external source; it is presented as confirmed by calculation, not inferred.**

Margin check: injection duration for common-rail solenoid injectors in this engine class runs from under 0.3 ms (pilot injections, light load) up to roughly 1.5–2.5 ms for a single main injection at higher load, per the cited peak/hold reference designs' typical operating envelope (Step 2 sources); even a generous upper bound of 3 ms for a long, high-load main injection leaves:

26.67 ms available – 3 ms max injection duration ≈ 23.67 ms of margin, an ~8.9× safety factor.

Even stacking a pilot + main injection (e.g., 0.3 ms + 2 ms + a few hundred μs dwell between them, well under 5 ms total) on one cylinder leaves over 20 ms of clearance before the shared high side would need to serve the other cylinder on that bank.

Verdict

The bank-sharing arrangement (cyl 1+3 on pin 03) is safe. Cylinders 1 and 3 are never required to inject within less than 240 crank degrees / 26.67 ms of each other under either plausible firing order, and typical common-rail injection durations (even generously bounded) are roughly an order of magnitude shorter than that window. There is no finding to raise loudly here in the negative sense the brief warned about — the check **passes**, and passes with enough margin that even significant transient-response slop in the driver electronics (boost recharge time between pulses, current decay tails) will not threaten it. The one assumption underlying this result — **even 240° firing spacing**, i.e., a conventional 120°-throw inline-3 crankshaft — is standard for this engine class and considered safe to rely on absent contrary evidence; it would be worth a one-line confirmation against a Kirloskar service manual or a firing-order marking on the timing cover if one becomes available, but it is not a blocking risk for the driver design.

Step 4: Candidate driver ICs

Given budget constraints, this section is intentionally thinner, per the brief's own prioritization (Steps 3 and 5 are load-bearing; this one may be lighter).

Candidate	Channels	Boost handling	Current regulation	High-side bank switching	Package	India availability
NXP MC33816 (SMARTMOS precision differential injector/solenoid driver)	5 high-side + 7 low-side external MOSFET pre-drivers (up to ~4 full injector channels typical in reference designs)	Pre-driver operates up to 72 V; does not itself generate boost — needs an external boost converter stage; current regulation and PWM channel sequencing are on-chip (confirmed, https://www.nxp.com/products/analog-and-mixed-signal/gate-drivers/high-precision-differential-injector-driver:MC33816)	On-chip programmable current-profile sequencer (the part is explicitly documented and used for common-rail diesel injector waveforms: boost-to-peak, peak-hold, hold, per NXP's own AN4849 application note referenced from the product page)	Yes — external high/low-side FETs are user-selected and the pre-driver sequences which pair is active, so a shared-high-side bank topology like this ECU's can be implemented in firmware/config, not hardware-forced	Not confirmed from the fetched page (typical NXP power packages; needs datasheet confirmation)	Confirmed general-market part — listed ' with NXP's Product Longevity Program (1 availability commitment), stocked at Dig: (https://www.digikey.com/en/products/detail/mouser-usa-inc/MC33816AE/4693868) and Mouser (https://www.mouser.com/ProductDetail/Semiconductors/MC33816AE) with normal now, ships today" retail terms — not automotive-allocation-only , orderable in small quantities
Discrete architecture (boost converter IC + external high-side/low-side power MOSFETs or IGBTs + current-sense resistor/comparator, sequenced by an MCU or FPGA)	N/A — built from discrete blocks	Fully custom; boost voltage set by converter design (any point in the 65–115 V envelope)	Closed-loop current regulation implemented in firmware/analog comparator, not integrated	Fully under designer control — the natural way to implement any custom bank topology, including this one	N/A	All constituent parts (boost controllers, automotive MOSFETs, current-sense amp) broadly available in small quantities from standard distributors

Note on automotive-allocation risk: several dedicated common-rail injector driver ASICs used by Tier-1s (proprietary Bosch/Denso/Continental in-house silicon, and some Infineon/Elmos/ST parts specifically qualified and allocated for OEM automotive programs) are **not available through normal small-quantity distribution** — searches for Infineon and Elmos part numbers plausibly in this space (`TL8090` , `E526`) did not surface a confirmed, orderable, small-quantity datasheet within this memo's search budget (**negative result** — this does not prove such parts don't exist or aren't orderable, only that this search did not locate and confirm one; it should be treated as an open item, not a closed "allocation-only" finding, for any part not named above). The **MC33816 is the one confirmed, small-quantity-orderable candidate found in this pass** and is recommended as the primary driver-IC direction for Phase 2, paired with an externally designed boost stage sized to the Step 2 envelope.

Step 5: The calibration boundary — stated plainly

This is the most important paragraph in this memo, and it is written to be read on its own.

What this hardware can do without any injector-specific calibration data: the drive stage described above — a boost converter in the 65–115 V range, a peak/hold current-regulated driver spanning 12–24 A peak and 8–13 A hold, sequenced across the bank-shared topology this ECU actually uses — **can be fully designed, built, and verified on the bench**.

It can be proven correct against a **known inductive load** (a resistor/inductor dummy load, or a bare injector solenoid off the engine, driven on a bench rail) — confirming boost rise time, peak/hold current accuracy, turn-off decay, minimum pulse width, and the bank-sequencing logic from Step 3, all without ever touching the running engine. This is real, verifiable engineering progress and should proceed.

What it cannot do without more data: run this specific engine correctly. A correctly *built* driver that outputs the right current profile is not the same as a driver that delivers the right *fuel quantity* at the right *crank angle* at the right *rail pressure* — and that mapping (fuel mass injected as a function of pulse width, at each of several rail pressures, separately for pilot and main injections, including the "ballistic region" where pulse width is too short for the injector to reach full lift) is exactly the calibration data that **Bosch and Denso do not**

publish. It is proprietary flow-bench data tied to the specific injector part number's hydraulic geometry (nozzle hole count/size, needle lift, spring preload), and normally reaches an OEM only bundled with the supplier's own ECU and calibration tools. Without it, this hardware would open and close the injectors with electrically correct pulses but with **no reliable way to know how much fuel each pulse actually delivered** — which means no reliable air-fuel ratio control, no reliable torque control, and for this specific engine, no reliable path to the CPCB IV+ NOx/PM limits its EGR-and-DOC aftertreatment strategy was designed around (Task 1). This is not a minor gap to be tuned out empirically on a running genset; incorrect fueling on a compression-ignition engine at 19.7:1 compression risks knock-like combustion damage, turbocharger overspeed/overtemperature, and DOC/EGR fouling well before a trial-and-error tuning process could converge.

The three routes to flow-vs-pulsewidth data, named plainly:

1. **Supplier licence (Bosch or Denso).** Approach the injector's OEM supplier (once the exact part number is identified, per Step 1's open item) for calibration data or a development/licensing agreement. **Cost:** realistically out of reach for a project at this scale — these programs are structured around OEM production volumes (tens of thousands of units), not a single genset retrofit; even where a "development kit" path nominally exists, expect NDA gates, minimum engineering-service fees, and multi-month lead times with no guarantee of access for a non-OEM customer. **Legality:** fully legal — this is the intended, sanctioned route. It is listed first for correctness and last for realism.
2. **Dyno characterization with rate-of-injection measurement.** Buy or borrow injectors of the identified part number, mount them on an injector-rate test bench, and directly measure fuel delivery vs. pulse width vs. rail pressure using an established method — the **Bosch tube method** (pressure-wave measurement in a long fuel-filled tube) or the **Zeuch method** (constant-volume chamber), both well-documented in the injection-research literature (confirmed: e.g. "Experimental Research on the Injection Rate of DME and Diesel Fuel in Common Rail Injection System by Using Bosch and Zeuch Methods," <https://doi.org/10.3390/en11020273>; "Benchmark between Bosch and Zeuch method-based flowmeters," https://www.researchgate.net/publication/331523136_Benchmark_between_Bosch_and_Zeuch_method-based_flowmeters_for_the_measurement_of_the_fuel_injection_rate). **Cost:** substantial but bounded and controllable — either access to an existing university/industry fuel-injection lab (a rate-of-injection rig is specialized, not off-the-shelf test equipment, and few Indian institutions maintain one) or building a simplified rate tube in-house, plus the engineering time to sweep pulse width × rail pressure × injector unit-to-unit variation into a usable map. **Legality:** entirely legal — this is independent, first-party measurement of purchased hardware, no reverse engineering of anyone's IP.
3. **OEM calibration extraction** (reading the flash/calibration data out of a genuine Kirloskar/Bosch ECU for this engine, by bench-side flash dump or bootloader access, and decoding the injector maps embedded in it). **Cost:** potentially the cheapest in engineering time if a donor ECU and the right tools are available, but requires reverse-engineering an undocumented, likely-encrypted or checksum-protected calibration binary — a nontrivial and uncertain effort in itself. **Legality: do not soften this.** Extracting and using a manufacturer's proprietary calibration data, without a license, sits in genuinely contested legal territory — it implicates the OEM's copyright in the calibration software/data, and depending on jurisdiction and exact method, protections analogous to anti-circumvention rules for access-controlled software. This is squarely the kind of activity that has drawn civil action against aftermarket tuning firms in other jurisdictions (well-documented in the passenger-vehicle ECU-tuning industry). It is **not recommended** as this project's route absent explicit legal clearance, and should not be treated as a routine engineering shortcut.

This is the single largest risk in the project, larger than any hardware selection question in this memo. The drive electronics are a solved, bench-verifiable problem. Getting *this engine* to run correctly, safely, and within its designed emissions strategy is gated on data this project

does not currently have a confirmed, legal, affordable path to. The owners have been told this before; this memo is the durable written record: **build and bench-verify the driver now; do not plan to commission it on the running genset without first closing Route 1 or Route 2 above (or accepting the documented legal exposure of Route 3).**

Sources consulted

- `/home/mohan/workws/ecu25kva/.superpowers/sdd/2026-09-13-phase1-research/task-4-brief.md` — task requirements (read in full).
- `/home/mohan/workws/ecu25kva/docs/research/01-engine-identity.md` — engine identity memo (read in full).
- `/home/mohan/workws/ecu25kva/refs/ecu-pinout-extracted.md` — ECU pinout extraction, injector bank wiring (read in full).
- <https://www.crdiparts.com/product/common-rail-fuel-injector-0445120646-f6-248-08-0-pr-suitable-for-kirloskar-engine-in-genset-and-liugong-loader/> — fetched; Kirloskar-branded Bosch injector cross-reference (different engine family than 3R550).
- <https://turbotech.in/product/common-rail-fuel-injector-0445120646-f6-248-08-0-pr-suitable-for-kirloskar-engine-in-genset-and-liugong-loader/> — mirror listing, found via search.
- <https://www.picoauto.com/library/automotive-guided-tests/bosch-current-at-idle> — found via search; Bosch CR solenoid injector current profile (~100 V boost to 20 A peak, 24 V to 10 A hold).
- https://www.researchgate.net/publication/350426572_Circuit_for_driving_common_rail_diesel_injectors — found via search; boost voltage 65–115 V, peak ~20 A, coil R/L typical values.
- <https://www.nxp.com/products/analog-and-mixed-signal/gate-drivers/high-precision-differential-injector-driver:MC33816> — fetched; MC33816 channel count, pre-driver voltage (up to 72 V), Product Longevity status.
- <https://www.digikey.com/en/products/detail/nxp-usa-inc/MC33816AE/4693868> and <https://www.mouser.com/ProductDetail/NXP-Semiconductors/MC33816AE> — found via search; small-quantity distributor stocking and pricing (~US\$5.19–5.61).
- https://en.wikipedia.org/wiki/Firing_order — found via search; general four-stroke firing-interval formula ($720^\circ/n$ cylinders).
- <https://doi.org/10.3390/en11020273> ("Experimental Research on the Injection Rate of DME and Diesel Fuel... Bosch and Zeuch Methods") — found via search; rate-of-injection measurement methods.
- https://www.researchgate.net/publication/331523136_Benchmark_between_Bosch_and_Zeuch_method-based_flowmeters_for_the_measurement_of_the_fuel_injection_rate — found via search; corroborates Bosch/Zeuch method equivalence.
- Searches returning negative results (recorded, not omitted): `Kirloskar R550 common rail injector Bosch part number`, `Kirloskar 3R550 3 cylinder diesel firing order 1-2-3`, `Infineon TLE8090 OR ELMOS E526 OR STMicroelectronics common rail injector driver IC datasheet` (no confirmed small-quantity-orderable alternative IC located in this search pass).

Research Memo 05: MCU Selection — S32K148 Confirmation

Verdict: CONFIRMED. NXP S32K148, Arm Cortex-M4F, recommended in the 144-pin LQFP package (orderable P/N FS32K148HAT0MLQT — 80 MHz clock grade, -40...+125 °C, CAN-FD feature set, tape-and-reel), is retained as the MCU for this design. It clears the channel budget in spec §5 with large margin, is in stock at an India-serving distributor today (not a 40-week part), has a workable — if not ideal — KiCad footprint path, and a free toolchain. **One part of the spec's reasoning is weaker than it looks and is replaced below (§6): the "constant speed relaxes timing" argument checks crank-edge rate, which was never the binding constraint. The binding constraint is injector pulse-width/output-compare resolution, and that also clears with large margin — so the conclusion holds, but for a different reason than §5 states.**

1. Channel budget vs. the S32K148 datasheet

Source: NXP, *S32K1XX — S32K1xx Data Sheet*, Rev. 15, 5 March 2026, <https://www.nxp.com/docs/en/data-sheet/S32K1xx.pdf> (cached copy inspected locally; page numbers below refer to this PDF's own footer pagination, "N / 108").

Budget item (spec §5)	Need	S32K148 datasheet	Cite	Status
CAN	2×	3× FlexCAN, all 3 with CAN-FD (ISO/CD 11898-1)	p.6, Fig. 3	Confirmed — 1 spare instance
ADC	~14 ch (8 sensor + 2 batt-rail + 4 current-sense)	2× 12-bit SAR ADC, 1 Msps each, up to 32 ch/module = up to 64 analog-capable pins at full-chip level	p.2 (Key Features), p.6 Fig. 3	Confirmed at instance/full-chip level. Inferred, not yet confirmed: how many of those 64 channels are actually broken out on the 144-pin package specifically requires the pin-mux table in the <i>S32K1xx Reference Manual</i> — the datasheet itself says pinouts/signal descriptions live there, not in this document (p.10, §10.1). Flagged for Phase 2.
Timer	~10 ch (crank, cam, 3 inj-lo, 2 inj-hi, metering PWM, 2 EGR PWM)	8× independent 16-bit FlexTimer (FTM) modules, up to 64 standard IC/OC/PWM channels total (8 ch/instance)	p.3, p.6 Fig. 3	Confirmed — >6× the required channel count

Budget item (spec §5)	Need	S32K148 datasheet	Cite	Status
I2C (baro sensor)	1×	2× LPI2C	p.6 Fig. 3	Confirmed
Flash	—	up to 2 MB program flash (up to 1,984 KB when EEPROM emulation/data flash in use)	p.2, p.6 Fig. 3 footnote 2	Confirmed, generous for this application's code size
RAM	—	up to 256 KB SRAM (incl. FlexRAM)	p.2, p.6 Fig. 3	Confirmed, generous
Core / clock	—	Arm Cortex-M4F, 80 MHz (RUN) or 112 MHz (HSRUN)	p.2, p.6 Fig. 3	Confirmed

The spec's own summary of the datasheet (§5: "3× FlexCAN... dual 12-bit ADC... FlexTimer instances well beyond the ~10 timing channels required") is accurate and is now backed by a primary-source page reference rather than recollection.

2. Package selection

Recommendation: 144-pin LQFP (NXP case outline SOT486-2, drawing 98ASS23177W — confirmed document numbers, per datasheet §9.1 "Obtaining package dimensions," p.108 of the PDF; the dimensional drawing itself was **not** independently fetched in this session — **inferred, not confirmed**: SOT486 is the standard 20×20 mm / 0.5 mm-pitch LQFP144 outline used across NXP's automotive MCU lines, but this specific drawing should be pulled from nxp.com and checked pad-by-pad before footprint lock, per the PCB Roadmap's warning that a hand-sourced footprint is the most likely place to introduce a silent error).

Why 144-pin over the smaller options, and why LQFP over BGA:

- Package options that exist for S32K148: 100-pin LQFP (QuadSPI *not* supported in this package per datasheet footnote 6, p.6), 100-pin MAPBGA, 144-pin LQFP, 176-pin LQFP.
- The confirmed pin map (spec §4) resolves only 44 of 94 OEM ECU pins; U1 in the unknowns register carries the other 50. The MCU-side signal budget from §5 alone (2 CAN ×2 pins + 14 ADC + 10 timer + 1 I2C ×2 pins ≈ 30 dedicated pins) plus SWD debug, oscillator, and power/ground pins for a part this size (typically another 30–40 pins on a 144-pin LQFP) is comfortably inside even the 100-pin package. The 144-pin package is chosen as headroom against U1, not because the confirmed budget requires it.

- LQFP over MAPBGA: a 100-pin MAPBGA buys nothing the 144-pin LQFP doesn't already have (same instance counts — CAN/ADC/FTM/I2C counts are silicon-level, not package-level, per Fig. 3), while a BGA package requires X-ray or destructive inspection to verify solder joints — a real cost for a small team doing early hand/low-volume assembly. 144-pin LQFP is 0.5 mm pitch, which is reflow-soldering-standard and inspectable by eye or AOI.

Recommended orderable part number: FS32K148HAT0MLQT — decoded against the datasheet's own ordering-information key (p.8, Fig. 5): - **H** = 80 MHz speed grade (RUN mode) - **A / A1** = CAN-FD + FlexIO (+ Security) feature option - **M** = -40 °C to +125 °C temperature grade - **LQ** = 144-pin LQFP package code - **T** = trays/tubes (an **R** suffix variant, e.g. FS32K148HFT0VLQR, exists for tape-and-reel production quantities)

Confirmed the **U** (112 MHz HSRUN) speed-grade parts are explicitly **not valid at the 125 °C (M) temperature grade** (datasheet footnote 1 under the ordering-option table, p.8: "Not valid with M temperature/125C"). This matters directly for the recommendation: an engine-bay-adjacent ECU should be specified to the 125 °C grade, and \$6 below shows the extra 32 MHz of HSRUN clock isn't needed to close the timing budget — so there is no reason to trade thermal margin for clock headroom the design doesn't use. **This is a concrete correction to the spec's framing in §5**, which cites "a Cortex-M4F at 112 MHz" as the relevant number; the part that should actually go on the BOM runs at 80 MHz and gets the better temperature rating in exchange.

3. Availability and pricing — India

Distributor pages for the specific chip part numbers (as opposed to the eval board) are behind bot-detection on live fetch (Mouser: DataDome 403; DigiKey: 403) in this session, so the figures below come from each distributor's own indexed page content returned via search, dated 2026-09-13. This is flagged so a live re-check before BOM lock is understood as still worthwhile — component pricing/stock is volatile — but the figures are not fabricated; they are quoted from the cited distributor URL's own content as indexed.

S32K148, 144-pin LQFP family (FS32K148UJT0VLQT / FS32K148HFT0VLQR / FS32K148HAT0MLQT):

Distributor	Part	Stock / lead time	Price	Cite
DigiKey (India storefront, digikey.in)	FS32K148UJT0VLQT	In stock, ships to India in ~4 days	₹1,900.22 (qty 1)	DigiKey
DigiKey	FS32K148UJT0VMHT (100-BGA variant, for comparison)	In stock	₹2,016.80 (qty 1)	DigiKey

Distributor	Part	Stock / lead time	Price	Cite
DigiKey (global .com listing)	FS32K148HAT0MLQT	"Ships today" per indexed listing	not captured in INR	DigiKey
Third-party stocking distributor (components-store.com)	FS32K148UJT0VLQT	7,948 units	\$4.28/pc	components-store.com
Third-party stocking distributor	FS32K148HFT0VLQR	10,530 units	\$3.59/pc	(same search)
Element14 India	S32K148EVB-Q176 (eval board, not bare chip)	listed, "pricing unavailable, contact customer service" for chip SKUs	₹17,922.07 (eval board)	in.element14.com

Verdict on S32K148 availability: confirmed pass. Real stock at DigiKey's India-facing storefront in single-digit quantities, at low-hundreds-of-rupees-to-few-thousand-rupee unit pricing, with ordinary shipping — not a lead-time problem. No need to invoke the S32K146 fallback.

S32K146 fallback (checked per brief step 2, not needed but recorded):

Part	Stock	Cite
FS32K146HFT0VLQT (144-LQFP)	Out of stock, backorder at DigiKey	DigiKey
FS32K146HAT0VLQT (144-LQFP)	Out of stock, backorder	DigiKey
FB32K146HAT0VLLT (related SKU, 100-LQFP)	Manufacturer standard lead time 52 weeks	found via search index
FS32K146HAT0MLLT (related SKU, 100-LQFP)	Manufacturer standard lead time 12 weeks	found via search index

Inferred: the 144-pin LQFP S32K146 SKUs specifically are on backorder with no confirmed lead time in the indexed snippets; related SKUs in the same family show a wide 12–52 week spread, which would have made S32K146 a materially worse availability bet than S32K148 if it had been needed. This reinforces, rather than undercuts, the S32K148 recommendation — the fallback is in worse shape right now than the primary choice.

Element14 India does not surface public per-unit pricing for the bare MCU chip in either family (only the eval board); it would need a quote request. Not a blocker given DigiKey's confirmed stock, but worth noting Element14 is the weakest of the three distributors for this specific part.

4. KiCad symbol and footprint provenance

- KiCad's own stock symbol library does not carry the S32K1xx family. A direct search of the official [KiCad/kicad-symbols](#) GitHub repository for NXP MCU symbols returns only legacy Freescale parts

(`MCU_NXP_HC11.lib` , `MCU_NXP_HC12.lib` , `MCU_NXP_NTAG.lib`) — confirmed, no `S32K` entry exists. Cite: <https://github.com/KiCad/kicad-symbols>

- **NXP's own EVB design files exist but are schematic-level, not a reusable KiCad/parametric MCU symbol.** The S32K148EVB-Q176 has a published schematic (document SCH-29642, dated 2017-07-12, mirrored via Octopart) but this is a board-level PDF schematic, not a native KiCad library part — using it as a footprint source would mean hand-deriving pin assignments, exactly the "hand-sourced footprint" risk the PCB Roadmap warns about. Cite: <https://datasheet.octopart.com/S32K148EVB-Q144-NXP-Semiconductors-datasheet-103093927.pdf>
- **SnapEDA (now SnapMagic Search) is the practical source.** It lists symbol/footprint/ 3D-model downloads, exportable to KiCad, for S32K148 parts including 144-LQFP variants (e.g. `FS32K148HAT0MLQT`) — confirmed listing exists. Cite: <https://www.snapeda.com/parts/FS32K148HAT0MLLT/NXP%20Semiconductors/view-part/>, <https://octopart.com/part/nxp-semiconductors/FS32K148HAT0MLQT> (Octopart surfaces the same "trusted partner" symbol/footprint/3D-model set)

Pad pitch verification: not done in this session — flagged as an open Phase 2 item. The datasheet points to case outline SOT486-2 / drawing 98ASS23177W for the exact mechanical dimensions (§9.1, p.108 of the PDF) but does not inline the pitch value, and the drawing PDF itself was not fetched here. Per the PCB Roadmap's explicit warning, do **not** carry a SnapEDA footprint into layout without diffing its pad pitch/pad size against document 98ASS23177W pulled fresh from nxp.com — this is exactly the class of silent error the Roadmap calls out, and it has not been closed by this memo.

5. Toolchain and debug

- **Compiler/IDE: NXP S32 Design Studio for ARM**, free, Eclipse-based, bundles the GNU Arm Embedded Toolchain (GCC/GDB) — confirmed, S32DS for ARM 2.2 explicitly lists S32K148 support. NXP is migrating S32K1 support toward "S32 Design Studio for S32 Platform" going forward; both remain usable today. Cite: <https://www.nxp.com/design/design-center/software/automotive-software-and-tools/s32-design-studio-ide/s32-design-studio-for-arm:S32DS-ARM>, release notes PDF via NXP Community. Third-party toolchains (IAR, GHS, Arm/Keil, Lauterbach, iSystems) are also listed as supported ecosystem options in the datasheet itself (p.6, Fig. 3, "Ecosystem" row).

- **Evaluation board exists: S32K148EVB-Q176**, Arduino-Uno-pinout-compatible, confirmed available from DigiKey India (₹17,487.61) and Element14 India (₹17,922.07) — the two prices differ by ~2%, ordinary distributor variance, not a discrepancy worth chasing. Cite: <https://www.digikey.in/en/products/detail/nxp-usa-inc/S32K148EVB-Q176/7929314>, <https://in.element14.com/nxp/s32k148evb-q176/eval-board-32bit-arm-cortex-m4f/dp/2917581>. **Confirmed** this lets firmware bring-up (HAL, crank/cam capture, CAN/J1939 stack skeleton, injector timer logic) start well before the custom PCB is fabricated, per spec §10 Phase 4 sequencing.
- **Debug probe for the custom board.** The EVB has an onboard OpenSDA debug/serial adapter (confirmed via NXP's own EVB documentation, indexed) usable during EVB-based bring-up at no extra cost. For the custom PCB (bare SWD header, no onboard debug circuit), a standalone probe is needed. **Important gotcha, not called out in the brief:** SEGGER's cheapest J-Link variant, the **J-Link EDU Mini**, is **licensed for non-commercial/educational use only** — SEGGER's own product description states it is "sold to private persons, colleges, schools, universities and NPOs for educational purposes only and is not sold to companies." This project is a commercial product development effort, so J-Link EDU Mini is **not** a legally usable option here despite being the cheapest one that shows up in a casual search. Commercially-licensed options:
- **SEGGER J-Link BASE:** \$798 (8.08.00) or **J-Link BASE Compact:** \$598 (8.19.00) — confirmed pricing at DigiKey (USD; India-specific INR not separately captured in this session — inferred to land in the ₹50,000–₹67,000 band at typical GST/duty markup, needs a live India quote before budgeting). Cite: <https://www.digikey.com/en/products/detail/segger-microcontroller-systems/8-08-00-J-LINK-BASE/2175882>, <https://www.digikey.com/en/products/detail/segger-microcontroller-systems/8-19-00-J-LINK-BASE-COMPACT/7386652>
- **PE Micro U-MULTILINK-FX:** \$738.27 at DigiKey — comparable price to J-Link BASE, no commercial-use restriction, and PEmicro explicitly supports S32 Design Studio. Cite: <https://www.digikey.com/en/products/detail/nxp-usa-inc/U-MULTILINK-FX/3247905> Recommendation: budget for one J-Link BASE Compact or PEmicro U-MULTILINK-FX for lab/ production debug; use the EVB's free onboard OpenSDA for early firmware work.

6. Testing the constant-speed timing argument

The spec's §5 argument, restated: a genset governs at a fixed 1500 rpm; a crank degree is 111 μ s at that speed; a 60-tooth wheel gives a tooth edge every 667 μ s; the Cortex-M4F/ FlexTimer combination has "several

orders of magnitude of headroom" against that, so the eTPU2's angle-synchronous hardware isn't needed.

This is checking the wrong variable. Crank-edge *capture* rate was never the hard constraint that motivated eTPU2 use in automotive engines. A tooth arriving every several hundred microseconds to a few milliseconds is trivial for a Cortex-M4F interrupt or a FlexTimer input-capture channel regardless of whether the engine is a genset or a car — 667 μ s is $\sim$ 75,000 CPU cycles at 80 MHz, and NVIC latency on a Cortex-M4F is on the order of 12 cycles ($\sim$ 150 ns at 80 MHz), several thousand times smaller than the interval being measured. This was true for automotive engines too, even at 6000 rpm (crank edge every $\sim$ 28 μ s at 6000 rpm / 60 teeth — still $>$ 2000 CPU cycles). **So the comparison the spec makes was never actually close, in either the automotive or the genset case, and citing constant-speed operation as what makes it comfortable is not quite the right causal story.**

The eTPU2's real value in automotive engines is different: it is a co-processor that autonomously schedules and re-schedules many angle-domain output-compare events (start of injection, duration, sometimes multi-pulse rate-shaping) across a wide, *rapidly changing* speed range, without loading the main CPU's interrupt budget, and without the main core's scheduling jitter (cache effects, higher-priority ISRs, RTOS latency) touching the injection timing chain. That is a real constraint when rpm sweeps 800–6000 rpm with tight closed-loop transient response, many simultaneous channels, and multi-pulse strategies (pilot/main/post injection) that all need to land within microseconds of a moving target angle.

The constraint that should actually be checked is injector pulse-width/output-compare resolution — how finely the MCU can control injector on-time, independent of crank speed. This is what determines fuel-quantity metering accuracy for a given injector's flow characteristic, and it does not go away just because rpm is constant. Checking it:

- FTM output-compare channels run off the bus/peripheral clock, up to 112 MHz (HSRUN) or 80 MHz (RUN) per the datasheet (p.2). At 80 MHz, one timer tick = 12.5 ns; at 112 MHz, $\sim$ 8.9 ns.
- Typical diesel solenoid injector pulse-width control for quantity repeatability needs on the order of 1–4 μ s resolution in demanding automotive multi-pulse (pilot/main/post) strategies — this is a widely-cited engineering figure for common-rail injector drive, not a datasheet fact, so it is marked **inferred** rather than confirmed by a specific citation here; it is not sourced from memory of a specific document, and should be checked against the actual injector's datasheet once U5 (injector part number, in the spec's unknowns register) is closed.

- The pin map (spec §4) shows a single main injection pulse per cylinder per cycle (one high-side bank switch + one low-side select per cylinder, "boosted high-side switch" + "current-controlled low-side, peak-and-hold" — no evidence of multi-pulse rate-shaping in scope for this genset). A single-pulse-per-cycle strategy is materially less demanding on timing resolution than automotive pilot/main/post injection.
- Either way, an 8.9–12.5 ns timer tick against a ~1–4 μ s (or looser, for single-pulse metering) requirement leaves roughly **100–450× headroom** — comparable in kind to the margin the spec claimed for crank-edge capture, but now measured against the constraint that actually matters.

Conclusion: the spec's recommendation survives, but its stated reasoning should be corrected, not just accepted. Crank-edge capture rate was never the tight constraint, in either an automotive or a genset engine. The tight constraint — injector pulse-width resolution — is also comfortably cleared by the S32K148's FlexTimer output-compare granularity, with margin that would likely hold even if this design later grew a multi-pulse injection strategy (not currently in scope; U5 in the spec's unknowns register should be revisited if it changes). The two PDBs (Programmable Delay Blocks, 2× on K148, confirmed p.6 Fig. 3) are also relevant here and not mentioned in the spec: they allow triggering ADC conversions or timer events at a defined delay from a capture event in hardware, without CPU intervention — useful headroom for keeping the injection-timing chain jitter-free even if firmware complexity grows, and a partial answer to the "what does the eTPU2 buy that we're giving up" question the spec doesn't fully address.

One caveat this memo cannot close: the flow-rate/resolution figure used above is a general diesel-injector engineering figure, not a citation to this project's actual injector. U5 (injector part number and drive profile) is still open per the spec's unknowns register. If the injector's actual metering-quantity-vs-pulse-width sensitivity turns out to need sub-microsecond control (unusual for single-pulse metering, but not impossible for a very small idle-quantity injector), this section's margin claim should be re-run against the injector's actual datasheet before Phase 2 sign-off. This is the one place where "confirmed" in this memo rests on an inferred industry figure rather than a project-specific document, and it is flagged as such rather than smoothed over.

7. Residual items for Phase 2

1. Pull NXP package drawing 98ASS23177W (144-pin LQFP, SOT486-2) directly and diff its pad pitch/pad geometry against whatever KiCad footprint (SnapEDA-sourced or hand-built) is actually used in the schematic sheet — not done in this memo.

2. Pull the **S32K1xx Reference Manual** pinout/IO-signal-description tables to confirm which of the 64 ADC channels and 64 FTM channels are actually broken out on the 144-pin LQFP package specifically (the datasheet defers this to the Reference Manual, §10.1) — the instance-level counts are confirmed, the exact pin-level assignment for this package is not.
 3. Get a live India quote (not indexed-search pricing) for FS32K148HAT0MLQT and one commercial debug probe (J-Link BASE Compact or PEmicro U-MULTILINK-FX) before BOM lock — Mouser/DigiKey blocked live automated fetch in this session (bot protection); the figures above are real but were retrieved via search-index snippets of the distributors' own pages, not a live page load.
 4. Revisit \$6's injector timing margin once U5 (injector part number) is closed.
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Citation index

- NXP, *S32K1XX S32K1xx Data Sheet*, Rev. 15, 5 March 2026:
<https://www.nxp.com/docs/en/data-sheet/S32K1xx.pdf>
- KiCad stock symbols repo: <https://github.com/KiCad/kicad-symbols>
- SnapEDA / SnapMagic Search, S32K148 parts:
<https://www.snapeda.com/parts/FS32K148HAT0MLLT/NXP%20Semiconductors/view-part/>, <https://octopart.com/part/nxp-semiconductors/FS32K148HAT0MLQT>
- S32K148EVB-Q176 schematic (SCH-29642):
<https://datasheet.octopart.com/S32K148EVB-Q144-NXP-Semiconductors-datasheet-103093927.pdf>
- NXP S32 Design Studio for ARM: <https://www.nxp.com/design/design-center/software/automotive-software-and-tools/s32-design-studio-ide/s32-design-studio-for-arm:S32DS-ARM>
- DigiKey India: <https://www.digikey.in/en>; DigiKey FS32K148HAT0MLQT: <https://www.digikey.com/en/products/detail/nxp-usa-inc/FS32K148HAT0MLQT/8628226>; S32K148EVB-Q176: <https://www.digikey.in/en/products/detail/nxp-usa-inc/S32K148EVB-Q176/7929314>
- DigiKey S32K146 SKUs:
<https://www.digikey.com/en/products/detail/nxp-usa-inc/FS32K146HFT0VLQT/10815998>,
<https://www.digikey.co.il/en/products/detail/nxp-usa-inc/FS32K146HAT0VLQT/10816081>
- Element14 India, S32K148EVB-Q176:
<https://in.element14.com/nxp/s32k148evb-q176/eval-board-32bit-arm-cortex-m4f/dp/2917581>

- SEGGER J-Link EDU Mini (educational-use-only licensing):
<https://www.segger.com/news/segger-introduces-j-link-edu-mini-a-low-cost-j-link-for-education/>
- SEGGER J-Link BASE / BASE Compact pricing:
<https://www.digikey.com/en/products/detail/segger-microcontroller-systems/8-08-00-J-LINK-BASE/2175882>,
<https://www.digikey.com/en/products/detail/segger-microcontroller-systems/8-19-00-J-LINK-BASE-COMPACT/7386652>
- PEmicro U-MULTILINK-FX:
<https://www.digikey.com/en/products/detail/nxp-usa-inc/U-MULTILINK-FX/3247905>
- Project spec: `docs/superpowers/specs/2026-09-13-ecu-reconciled-spec-design.md` §4, §5

Research Memo 06: Standards, Test Requirements, and Certification Scope

UPDATED 18 Sep 2026 — the certificate is now a specific number

The engine's rating plate has been photographed. Two things change for this memo, both in the direction of making its conclusions **more** concrete, not less:

- The engine is 3GK550ETA 4SR1, application code GK3.8703 , not the 3R550ETA 4G1 this memo inherited from research memo 01. The genset model KG4-25WS1 in the header was inferred from the same chain and should be treated as **unconfirmed** — no genset nameplate has been seen.
- The plate carries the actual type approval:
ARAI/MoEF/DGTA/IGES4/KOEL-P25/2825/24 . IGES4 is India Genset Emission Standard IV, confirming the CPCB IV+ tier from the machine itself rather than from a brochure.

This memo's regulatory conclusions stand, and now attach to a numbered certificate against a specific serial number rather than to a product family in the abstract. That is worth stating plainly to whoever takes the decision.

One gap this memo did not examine, raised by a later review: the **state pollution board Consent-to-Operate** regime is a separate gate from central type approval and was never looked at. Recorded as unexamined.

The controller references below to the KG640C are also superseded — the fitted unit is a DSE4522 MKII AMF. See [refs/field-evidence-2026-09-18.md](#) .

Task: U6 (J1939 layers, ISO 7637-2 transient immunity, CISPR 25 conducted emissions, certification position). Consumes Task 1's emissions tier. Produces the test/design numbers Phase 3 layout must honour and an honest scope statement on certification. **Date:** 2026-09-13 **Author:** Phase-1 research (task 6) **Genset:** Kirloskar KG4-25WS1, 25 kVA, 415 V 3-phase / 230 V 1-phase, 50 Hz. **Engine:** Kirloskar 3R550ETA 4G1, 26.5 kW prime (19–56 kW CPCB IV+ band). **Emissions tier:** CPCB IV+, cooled EGR + DOC, no SCR/DEF/DPF (per [docs/research/01-engine-identity.md](#)). 12 V electrical system. In service, in India.

Verdict (read this first)

Item	Finding	Status
J1939 physical/transport/diagnostic layers	Standard 250 kbit/s twisted-pair CAN with 120 Ω termination, 1 m stub limit; BAM for broadcast multi-packet, RTS/CTS for destination-specific; DM1/DM2 for active/previously-active DTCs; address-claim on power-up	Secondary-sourced (SAE texts paywalled); PGN set itself carried over from Task 2, there marked unconfirmed for this specific controller
ISO 7637-2 transient levels (12 V ref.)	Pulse 1: -150 V (Level IV); Pulse 2a: +112 V; Pulse 2b: +10 V; Pulse 3a: -220 V; Pulse 3b: +150 V (2 ms-0.1 ms widths, 2-50 Ω source impedance per pulse); Pulse 5b (load dump, now properly an ISO 16750-2 pulse, still colloquially cited under 7637-2): 40-105 V suppressed, 0.5-8 Ω , 40-400 ms	Secondary source (industry technical note quoting the standard's table), clearly marked — standard text itself not accessed
CISPR 25 conducted emissions	Class 5 (most stringent), 150 kHz-108 MHz, voltage method via 5 μ H/50 Ω LISN; toughest bands are VHF/FM (68-108 MHz): 18 dB μ V avg / 38 dB μ V peak	Secondary source, clearly marked
Certification position	Confirmed, from the gazette text itself: CPCB IV+ type approval is granted to an engine/genset family and obtained by the manufacturer/importer/assembler at the point of manufacture or import — it is not a per-unit approval that travels indefinitely with the physical genset regardless of what is later done to it. A separate CPCB regulatory document (the RECD System & Procedure) states directly that ECU characteristics governing injection timing, air-mass metering, and emission-reduction strategy must not be altered even under a government-sanctioned retrofit programme, and that modifications limited to engine-control changes fall outside the one retrofit-approval pathway CPCB operates. No pathway was found by which a third-party ECU replacement on this genset could be approved, and General Condition 3 of GSR 804(E) states plainly that no person shall use a genset lacking a valid Type Approval + CoP certificate. This is a decision for the project owners to take to qualified counsel and/or CPCB/ARAI directly — not an engineering call, and not something this memo resolves.	Confirmed (primary gazette text and primary CPCB procedure text, both fetched and read in full) for what the rules say; not legal advice on what they mean for this specific project

Step 1: SAE J1939 layers that apply

SAE's J1939 series texts are paywalled (sae.org); nothing below was read from the standard itself. Every figure is from secondary industry sources, cited individually, and should be treated as **secondary/summarized**, not as a verbatim standard quote.

- **J1939-11 (physical layer, twisted-pair variant).** Shielded twisted-pair backbone, nominal 120 Ω characteristic impedance, 250 kbit/s data rate for this device class, 120 Ω ($\pm 1\%$, $\frac{1}{4}$ W typical) termination resistors at both physical ends of the backbone, and stub ("drop line") length limited to ≈ 1 m per node to control reflections. (*Secondary:* <https://simmasoftware.com/products/j1939/sae-j1939-11-physical-layer/>);

<https://copperhilltech.com/content/SAE%20J1939%20Wiring%20Requirements.pdf>
; <https://obd-cable.com/j1939-backbone-termination-stub-length-guide/>.) **Design relevance:** the genset's CAN segment (GCU ↔ ECU, per Task 2's KG640C memo) should be laid out to this backbone/stub/termination discipline; a single genset with two nodes is a short, simple topology relative to the standard's truck/trailer use case, so termination is the only piece likely to bite if omitted.

- **J1939-21 (transport protocol).** Multi-packet (>8 byte) messages use either **BAM** (Broadcast Announce Message — connectionless, one-to-all, no flow control, used for messages any node may want, e.g. some diagnostic broadcasts) or **RTS/CTS** (Request-to-Send/Clear-to-Send — connection-oriented, destination-specific, flow-controlled, used for point-to-point large transfers). (*Secondary:* <https://www.spcontroller.com/j1939-transport-protocol-bam-vs-rts-cts/>; <https://copperhilltech.com/blog/sae-j1939-programming-with-arduino-transport-protocol-rtscts-session-sae-j193921/>.) **Design relevance:** if the replacement ECU needs to send anything larger than a single CAN frame (e.g., a multi-DTC DM1 payload) to the GCU, the firmware needs both mechanisms implemented — most stacks default to BAM for DM1 and would only need RTS/CTS if a proprietary large-payload message is added.
- **J1939-71 (application layer / PGN definitions).** Task 2 (`docs/research/02-kg640c-gcu.md`, §2) already established the working PGN set for this specific genset, **explicitly flagged there as an inferred industry-baseline, not confirmed against a KG640C document:** EEC1 (PGN 61444, engine speed), ET1 (PGN 65262, coolant temp), EFL/P1 (PGN 65263, oil pressure), VEP1 (PGN 65271, battery/system voltage), DM1 (PGN 65226, active DTCs), and TSC1 (PGN 0, torque/speed control, if the GCU is to command the engine). This memo does not re-derive that list; it is carried forward as-is, with the same caveat Task 2 attached.
- **J1939-73 (diagnostics, DM1/DM2).** **DM1 — Active Diagnostic Trouble Codes** (PGN 65226/0xFECA): broadcast by default at 1 Hz and on state change, carries currently-active DTCs. **DM2 — Previously Active Diagnostic Trouble Codes** (PGN 65227): available on request only, carries DTCs that were active but have cleared. (*Secondary:* <https://www.csselectronics.com/pages/j1939-73-dm1-diagnostic-message-dtc>.) **Design relevance:** the replacement ECU should broadcast DM1 unsolicited at 1 Hz to be a drop-in match for what the KG640C GCU's fault-display logic almost certainly expects (per Task 2's finding that the GCU is a CAN consumer, not just a display).
- **J1939-81 (network management / address claiming).** On power-up, each node transmits an Address Claimed message; every other node records/verifies it; if two nodes claim the same address, priority is resolved by the NAME field (a 64-bit value encoding manufacturer, function, and an arbitrary/self-configurable-address capability bit). (*Secondary:* <https://copperhilltech.com/blog/sae-j1939-address-claim-procedure-sae-j193981-network-management/>.) **Design relevance:** the

engine ECU node conventionally claims source address 0x00 by industry convention (again per Task 2, unconfirmed for this controller specifically) — the replacement ECU should claim the same address the GCU expects, which direct bus logging (Task 2's recommended next step) would settle definitively.

Net for Phase 3 layout: none of this drives component values the way Steps 2–3 do; it drives firmware behaviour and connector/stub layout. The one open item carried from Task 2 — the real PGN set and source address, unconfirmed from any KG640C-specific document — should be closed by logging the existing bus before the replacement ECU's CAN stack is finalized.

Step 2: ISO 7637-2 transient immunity — the numbers that set the TVS and buck input rating

Source note: the ISO 7637-2 standard text itself is paywalled and was not accessed. The table below is drawn from an ON Semiconductor automotive applications technical note that reproduces the ISO 7637-2:2011(E) severity-level table directly (title "Supplemental Data for ISO7637 Pulses," <https://www.onsemi.com/download/design-notes/pdf/tnd6424-d.pdf>, fetched and text-extracted in full) — **marked here as secondary**, since it is an industry document quoting the standard, not the standard.

Pulse	Description	Amplitude, 13.5 V nominal system, Level I/II	Level III	Level IV	Source impedance / pulse width	Test duration
1	Supply disconnection (inductive load switch-off, e.g. relay coil)	−75 V	−112 V	−150 V	10 Ω, 2 ms	500 pulses, min. 0.5 s repeat
2a	Sudden interruption of current in a device sharing the harness with the DUT (inductive)	+37 V	+55 V	+112 V	2 Ω, 0.05 ms	500 pulses, min. 0.2 s repeat
2b	Same mechanism, lower-energy variant	+10 V	+10 V	+10 V	2 Ω, 0.05 ms	10 pulses, min. 0.5 s repeat
3a	Fast transient bursts (switching-type, EFT-like)	−112 V	−165 V	−220 V	50 Ω, 0.1 ms	1 hr test time, 90 ms min. repeat
3b	Same mechanism, opposite polarity	+75 V	+112 V	+150 V	50 Ω, 0.1 ms	1 hr test time, 90 ms min. repeat

(All figures per ON Semiconductor TND6424-D, secondary source.)

Load dump (Pulse 5b) — a nuance worth stating plainly: the 2011 edition of ISO 7637-2 removed the load-dump pulses (5a/5b); they now live in ISO 16750-2 (and are referenced in ISO 21848 for some classes), though industry practice — including many component datasheets and this project's own brief — still refers to "ISO 7637-2 pulse 5b" as shorthand. *(Secondary: search-result synthesis, not independently verified against either standard's text — flagged as an open item rather*

than a confirmed edition history. If the design team needs to cite the pulse in a test plan, ISO 16750-2 §4.6.4 is the more defensible current reference.) For a **suppressed** load dump (the realistic case on a genset with an alternator-integrated or centralized suppression diode/clamp, as is now common) — pulse 5b: peak voltage 40–105 V for a 12 V system, source impedance 0.5–8 Ω , pulse width 40–400 ms. (*Secondary, search-result synthesis — not read from either standard's own table.*)

What this sets for Phase 2/3 hardware: - **TVS clamp voltage:** must sit above the normal 12 V (13.5–14.4 V charging) rail with margin, but clamp well below the injector boost-converter's and any other sensitive node's absolute maximum — the working design window is a clamp that lets pulses 2a/2b (+37 to +112 V) and suppressed 5b (up to ~105 V) pass through to a controlled clamp voltage, while pulses 1/3a (down to –220 V) are clamped on the negative side. A single automotive-grade bidirectional TVS rated for these levels (many parts are explicitly binned "meets ISO 7637-2 pulse 1/2a/3a/3b," per component datasheets — not independently verified against this list here) is the usual answer; the –220 V pulse 3a figure (Level IV, 50 Ω /0.1 ms — low energy despite high voltage) is the most demanding negative excursion to clamp. - **Buck converter input rating:** the input stage (from battery rail to the point after transient protection) should be rated for a working input up to the clamped level of the worst transient the TVS lets through (commonly designed to ~40–45 V clamped for a suppressed load-dump system, consistent with the Vishay/EDN load-dump note found in this search pass but not independently verified here) rather than the full unclamped 220 V pulse, which the TVS/clamp stage is responsible for absorbing before it reaches the buck's input pins.

Step 3: CISPR 25 conducted emissions — band limits and what a bench setup can tell you

Source note: the CISPR 25 standard text itself is paywalled and was not accessed. Figures below are from two secondary sources: a Texas Instruments technical article (<https://www.ti.com/lit/pdf/sszt673>, fetched and read in full) and an EMC-limits compilation site (<https://emccalc.com/limits/cispr25-cev/>, fetched). **Both are marked secondary — neither is the standard itself.**

- **Method and frequency range:** conducted emissions are measured at the DUT's power terminals via a 5 μH /50 Ω LISN (line impedance stabilization network), voltage method, over 150 kHz to 108 MHz, using peak and average detectors. (*TI, sszt673, confirmed as stated by that source.*)
- **Class 5** is the most stringent of CISPR 25's five classes and the one automotive OEMs typically mandate for safety-/receiver-adjacent electronics. (*TI, sszt673.*) Selected Class 5 limits, voltage method (secondary, per emccalc.com's compilation, which states the 2016 (Ed.4) and 2021 (Ed.5) voltage-method limits are identical):

Band	Frequency	Peak limit	Average limit
LW (broadcast)	0.15–0.30 MHz	70 dB μ V	50 dB μ V
MW (broadcast)	0.53–1.8 MHz	54 dB μ V	34 dB μ V
SW (broadcast)	5.9–6.2 MHz	53 dB μ V	33 dB μ V
CB	26–28 MHz	44 dB μ V	24 dB μ V
VHF (low)	30–54 MHz	44 dB μ V	24 dB μ V
VHF (high)	68–87 MHz	38 dB μ V	18 dB μ V
FM broadcast	76–108 MHz (overlapping VHF-high per source)	38 dB μ V	18 dB μ V

The 68–108 MHz VHF/FM region carries the tightest limit (18 dB μ V average / 38 dB μ V peak) — explicitly called out by TI as "extremely challenging." (*TI, sszt673, confirmed as stated by that source.*) This is the band a switching converter's harmonics most often violate, and it is where PCB layout (input filter corner frequency, output ripple, switching-edge control) earns its keep.

What a LISN + spectrum analyser bench setup can and cannot tell you, short of a test house: - **Can:** give a reliable relative measurement of your own converter's conducted noise floor and harmonic content across the 150 kHz–108 MHz range, using a 5 μ H/50 Ω LISN (buildable or purchasable as a discrete unit) feeding a spectrum analyser with peak-hold — enough to catch a converter that is grossly non-compliant (wrong switching frequency choice, missing input filter, poor grounding) long before a certification test, and to A/B compare layout or filter revisions against each other. - **Cannot:** substitute for accredited peak/average/quasi-peak detector calibration, the specific bonded ground-plane/test-bench geometry CISPR 25 mandates, cable harness length and dress requirements, or the formal measurement uncertainty budget a test house applies — none of which a bench LISN setup replicates exactly. A bench pass is **necessary-but-not-sufficient** evidence; it de-risks the design before a paid test-house slot, it does not replace one. (*This paragraph is general EMC test-practice knowledge, not tied to a specific citation found in this search pass — flagged as such rather than presented as sourced.*)

Step 4: The certification position — stated plainly

This is the section the brief calls the one that matters most, and it is written to stand on its own.

4.1 What CPCB IV+ actually is, and where it comes from — confirmed from the gazette text itself

This memo obtained and read, in full, the actual Gazette of India notification: G.S.R. 804(E), Ministry of Environment, Forest and Climate Change, New Delhi, 3 November 2022 — the *Environment (Protection) Third Amendment Rules, 2022*, made under sections 6 and 25 of the Environment (Protection) Act, 1986, amending Schedule I, serial number 88 of the Environment (Protection) Rules, 1986. (*Confirmed — primary source, fetched directly at*

<https://cpcb.gov.in/openpdf.php?id=TGF0ZXN0RmlsZS8zNThfMTY2ODE2NjYzNI9tZWRpYXBob3RvMTkwNTgucGRm>

and text-extracted via `pdftotext -layout` from the saved PDF; this upgrades Task 1's "inferred/secondary" GSR citation to confirmed.) It comes into force **1 July 2023**, exactly as Task 1 recorded.

The band limits are now confirmed directly from the gazette's own Table 1, not from dieselnet's summary as Task 1 had it: for the $19 < P \leq 56$ kW power category (this engine's 26.5 kW prime rating falls here) — combined $\text{NO}_x + \text{HC} \leq 4.7$ g/kWh, $\text{CO} \leq 3.5$ g/kWh, $\text{PM} \leq 0.03$ g/kWh, $\text{smoke} \leq 0.7 \text{ m}^{-1}$ (light absorption coefficient), for CI engines, no separate NO_x -only limit in this band. This matches dieselnet's figures exactly and is now **confirmed**, not inferred.

4.2 What the certification covers, and who holds the obligation — confirmed from the gazette's General Conditions

- **General Condition 1 (Applicability):** the rules apply to "new engines for power generation application and products **manufactured, assembled or imported to India**" — this is a point-of-manufacture/import regime, not a running in-service inspection scheme (there is no equivalent here to a vehicle's periodic PUC check). (*Confirmed, direct gazette quote.*)
- **General Condition 2 (Requirement of certification):** "Domestic manufacturer, importer or assembler... shall obtain **Type Approval** from authorised certifying agency and also comply with **Conformity of Production (CoP)** test of their product(s)..., valid for the next Conformity of Production year." (*Confirmed, direct gazette quote.*) The obligation to *hold* a Type Approval and CoP certificate is placed on the **manufacturer/importer/assembler**, not on the end owner or operator.
- **General Condition 3 (Sale/import/use of non-compliant product):** "No person shall sell, import or use an engine and genset for power generation application which is not having a valid Type Approval certificate and certificate of Conformity of Production." (*Confirmed, direct gazette quote — this is the clause that reaches beyond the manufacturer to anyone who uses such equipment, which would include a genset owner/operator.*)
- **General Condition 5 (Nodal agency):** the Central Pollution Control Board (CPCB) is the nodal agency for implementation. (*Confirmed.*)
- **General Condition 6 (Authorised certifying agencies):** Automotive Research Association of India (ARAI, Pune), International Centre for Automotive Technology (ICAT, Manesar), and Indian Institute of Petroleum (IIP, Dehradun) are the agencies authorised to test and certify Type Approval and CoP for engines and gensets. (*Confirmed, direct gazette quote; a related CPCB procedure document for the separate retrofit-device scheme, §4.2.1, also lists Vehicle Research and Development Establishment (VRDE, Ahmednagar) and any NABL-accredited agency meeting that scheme's requirements — this fourth agency and the NABL route are confirmed only for the retrofit-device scheme, not shown in the GSR 804(E) text itself for new-engine Type Approval.*)

- **General Condition 11:** CPCB, the Commission for Air Quality Management, State Pollution Control Boards, or Pollution Control Committees may issue **more stringent** local norms. *(Confirmed — a reminder that state-level rules could add obligations beyond the national floor; not researched further here, flagged as an adjacent open item.)*

What "Type Approval" covers, **structurally**: it is granted to an engine/genset **family or model** — the regulation's own "engine family" concept (defined in the companion CPCB retrofit-device procedure, §1.1.18: "a manufacturer's grouping of engine systems, which, through their design, have similar exhaust emission characteristics; all members of the family must comply with the applicable emission limit values," citing ISO 8178-7) is confirmed only in that companion document but is standard practice across this same regulatory architecture (same nodal agency, same three certifying agencies, same GSR-804(E)-referenced testing regime). Individual units are not independently re-tested against Table 1 one-by-one; the manufacturer's ongoing conformance is checked by periodic Conformity of Production (CoP) sampling of production units, not a certificate that attaches permanently, in isolation, to one physical serial number regardless of what happens to it afterward.

4.3 The direct hit: CPCB's own text on ECU/calibration modification

This memo located and read, in full, CPCB's "System and Procedure for Emission Compliance Testing of Retro-fit Emission Control Devices (RECD) for Diesel Power Generating Set Engines up to Gross Mechanical Power 800 kW" (fetched from https://cpcb.gov.in/uploads/RECD_Procedure_Revised.pdf, text-extracted via `pdftotext -layout` from the saved PDF in full, 3258 lines). This is CPCB's own regulatory document governing the *one* pathway CPCB operates for modifying an already-type-approved genset's emission-control configuration after manufacture. Two passages from it are decisive for this project, quoted verbatim:

§4.1.2.2(e): "The characteristics of the electronic engine control unit (as regards, for example, injection timing, air-mass flow metering, or exhaust emissions reduction strategies) **shall not be altered** by the retrofitting. Any modification of the test engine that changes the original emission behaviour (for example alternation of the injection timing) **is not permitted**."

§1.1.35 (definition of "Retrofit emission control device"): "...Systems that **only modify the existing engine system controls** are not considered to be RECDs."

(Both confirmed — direct quotes, primary CPCB source, fetched and read in full.)

Read together, these two passages mean: (1) CPCB explicitly identifies **injection timing, air-mass flow metering, and exhaust-emission-reduction strategy** — precisely the calibration parameters this engine's EGR-rate-scheduling-and-injection-timing NOx/PM strategy runs on (per Task 1's finding of EGR+DOC-only aftertreatment, no SCR) — as characteristics that must not be touched, even under a CPCB-sanctioned emissions-improvement retrofit programme; and (2) a modification that is *purely* a change to the engine's own control system, with no added physical after-treatment hardware — which is exactly what an ECU replacement is — **does not qualify as an RECD at all** and therefore has **no approval pathway under this scheme, full stop**. It is neither covered by the new-manufacture Type Approval/CoP scheme (GSR 804(E), which governs manufacturers at point of production/import, §4.2 above), nor by the one modification-approval scheme CPCB operates for in-service units (RECD, which by its own definition excludes pure control-system changes).

Negative result, recorded per the brief's instruction: no CPCB-administered, or any other, approval pathway was found in this search by which replacing this genset's OEM ECU with a third-party unit of different calibration could be formally re-certified or otherwise sanctioned as compliant. This does not prove no such pathway exists anywhere in Indian regulatory practice — only that this search, within its budget, did not locate one. This is exactly the kind of fact pattern that should go to CPCB, ARAI, or qualified counsel directly, not be inferred further by this memo.

4.4 The calibration angle the brief asked to check — confirmed as directly relevant, not merely plausible

The brief's suspicion is correct, and the gazette text itself confirms the mechanism: GSR 804(E) Table 1 sets limits in **g/kWh over a defined test cycle** (a mass-of-pollutant-per-unit-of-work measure), not as a hardware bill-of-materials requirement. Compliance is therefore inherently a function of *how the engine is run* — injection timing, EGR rate scheduling, fuelling — as much as *what parts are bolted to it*. Combined with §4.3's finding that CPCB itself names injection timing, air-mass metering, and emission-reduction strategy as the calibration parameters that determine emission behaviour, and with Task 1's finding that this specific engine meets its 19–56 kW-band NOx+HC limit (≤ 4.7 g/kWh) through cooled EGR and DOC alone, with **no SCR** as a chemical backstop: **a replacement ECU with different EGR-scheduling or injection-timing calibration can move this engine out of compliance with its certified emission limits without any hardware changing at all**. This is not a remote or theoretical risk — it is the same mechanism by which the engine meets its limits in the first place, run in the wrong direction. It is directly relevant to, and reinforces, the calibration-data gap already flagged as the project's largest single risk in `docs/research/04-injector-drive.md` (Step 5): that memo's finding was about getting the engine to run *correctly and safely* without OEM fuelling maps; this memo's finding is that even a "correctly running"

replacement ECU, if its EGR/injection-timing calibration differs from the certified original, has no assurance of remaining within the CPCB IV+ limits the genset was certified against — an emissions-compliance risk layered on top of the combustion/mechanical risk Task 4 already documented.

4.5 Statutory exposure — flagged, not advised on

Separately from the certification-scheme mechanics above, the **Environment (Protection) Act, 1986, Section 15** provides a general penalty for contravention of the Act or rules made under it. *(Secondary — this memo relied on a legal-database rendering of the statute, <https://indiankanoon.org/doc/24023684/>, rather than a freshly fetched Gazette copy of the Act itself, and the search result returned text that appears to blend the original 1986 penalty (imprisonment up to 5 years and/or fine up to ₹1 lakh, escalating for continuing contravention) with a later monetary-penalty band (₹10,000–₹15 lakh) that reads as a post-2023 Jan Vishwas (Amendment of Provisions) Act amendment. Which version is currently in force, and whether it would apply to this specific fact pattern, is exactly the kind of question that needs a lawyer, not this memo — flagged, not resolved.)*

4.6 The decision, framed as the brief asked

This memo does not tell the project owners whether to proceed. What it establishes, with the citations above:

1. CPCB IV+ Type Approval is a family/model-level certificate obtained by the manufacturer, not a permanent per-unit status immune to what happens to the physical genset afterward.
2. The obligation to *hold* Type Approval + CoP sits with the manufacturer/importer/assembler; the obligation not to *use* a non-compliant engine/genset (General Condition 3) reaches the owner/operator directly, by the gazette's own words — "no person shall... use."
3. CPCB's own regulatory text, in the one scheme it operates for modifying in-service gensets, states plainly that altering ECU characteristics governing injection timing and emission-reduction strategy is not permitted even under a sanctioned retrofit programme, and that pure engine-control modifications fall outside that scheme's approval process entirely.
4. No approval pathway for a third-party ECU replacement on this genset was found in this search.
5. The mechanism by which this specific engine meets its CPCB IV+ limits (EGR scheduling + injection timing, no SCR backstop) is exactly the mechanism a replacement ECU's calibration would change.

Whether this means the genset's certification is void, whether it creates a compliance or disclosure obligation, and what exposure attaches to the owner/operator versus this project team, is a legal and regulatory question — not an engineering one. It should go to CPCB,

one of the three authorised certifying agencies (ARAI/ICAT/IIP), or qualified counsel, before this project proceeds past the bench-verification stage Task 4 already scoped as safe to continue.

Sources consulted

- `/home/mohan/workws/ecu25kva/.superpowers/sdd/2026-09-13-phase1-research/task-6-brief.md` — task requirements, read in full.
- `/home/mohan/workws/ecu25kva/docs/research/01-engine-identity.md` — engine identity and emissions-tier memo, read in full (Step 4 specifically).
- `/home/mohan/workws/ecu25kva/docs/research/02-kg640c-gcu.md` — KG640C GCU/J1939 memo, grepped for PGN references (Step 2 of that memo).
- `/home/mohan/workws/ecu25kva/docs/research/04-injector-drive.md` — injector drive/calibration-boundary memo, read in full (Step 5 specifically, for the calibration-risk cross-reference in §4.4 above).
- **G.S.R. 804(E), 3 November 2022**, Ministry of Environment, Forest and Climate Change — fetched directly (<https://cpcb.gov.in/openpdffile.php?id=TGF0ZXN0RmlsZS8zNThfMTY2ODE2NjYzNl9tZWRpYXBob3RvMTkwNTgucGRm>), saved and text-extracted in full via `pdftotext -layout`. **Primary source, read in full.** This is the actual Gazette of India notification text — confirms GSR number, date, ministry, Table 1/Table 2 emission limits, and all General Conditions cited in Step 4.
- **CPCB, "System and Procedure for Emission Compliance Testing of Retro-fit Emission Control Devices (RECD) for Diesel Power Generating Set Engines up to Gross Mechanical Power 800 kW"** — fetched directly (https://cpcb.gov.in/uploads/RECD_Procedure_Revised.pdf, redirected from cpcb.nic.in), saved and text-extracted in full via `pdftotext -layout` (3258 lines). **Primary source, read in full** for the definitions and §4.1.2.2(e) quoted in Step 4.
- <https://cpcb.gov.in/genset-notifications/> — fetched, confirms the GSR 804(E)/GSR 436(E) amendment chain exists on CPCB's own notification index.
- <https://indiankanoon.org/doc/24023684/> (via search) — Environment (Protection) Act 1986, Section 15 text, secondary rendering of the statute, not independently fetched from a gazette copy of the Act.
- <https://www.onsemi.com/download/design-notes/pdf/tnd6424-d.pdf> — ON Semiconductor technical note reproducing the ISO 7637-2:2011(E) pulse severity table; fetched, saved, and text-extracted in full via `pdftotext -layout`. Secondary source.
- <https://www.ti.com/lit/pdf/sszt673> — Texas Instruments, "A Review of EMI Standards, Part 1 – Conducted Emissions"; fetched, saved, and text-extracted in full via `pdftotext -layout`. Secondary source, for CISPR 25 method and Class 5 VHF/FM limits.

- <https://emccalc.com/limits/cispr25-cev/> — CISPR 25 Class 5 voltage-method limit table compilation; fetched. Secondary source.
- <https://www.araiindia.com/certification/certification/genset> ;
<https://cpcb.nic.in/type-approval-cop-data/> ;
<https://cpcb.nic.in/displaypdf.php?id=Tm9pc2VfYW5kX0dlbnNldHMvU3lzX1Byb2RfZm9yX0VtaXNzX0NvbXBfb2ZfREdTZXRzLnBkZg%3D%3D>
— found via search, corroborate the general Type Approval/CoP scheme description in §4.2 without being individually quoted.
- Searches returning only partial or negative results (recorded, not omitted): `ISO 7637-2 pulse 1 2a 2b 3a 3b 5b amplitude 12V test levels table` (general web search did not surface a full table directly, resolved via the onsemi PDF instead); `CISPR 25 class 5 limits table dBuV band I II III IV V` (resolved via emccalc.com); direct WebFetch of <https://keystonecompliance.com/iso-7637-2/> and <https://www.electronicdesign.com/.../cispr-25-class-5...> returned only qualitative descriptions, no numeric tables — superseded by the sources above.

Research Memo 07: Consolidated BOM and Sourcing

Task: Consolidated, costed bill of materials covering every functional block in spec §4 and §6. Consumes Tasks 3 (connector), 4 (injector driver), 5 (MCU) — the three hard sourcing problems — plus the simulation-fixed component values supplied directly for this task. **Date:** 2026-09-13 **Author:** Phase-1 research (task 7) **Scope note, stated once and binding throughout:** this ECU has no DEF/urea sensor, no LIN bus, no SCR, no magnetic pickup, and one catalyst temperature sensor. Fuel level and water-in-fuel are panel signals, not ECU signals. None of these appear below. Earlier design artifacts that included them were wrong and are not carried forward.

Read this first — the headline

	Qty 1	Qty 10
BOM total (excl. PCB fab, excl. assembly)	≈ ₹14,200	≈ ₹1,34,500 (≈ ₹13,450/unit)

Two findings drive that number and its shape:

1. The 94-pin OEM-mating connector is ~45–50% of the per-unit BOM cost, and it does not get cheaper with volume — it is bought as an aftermarket wired-pigtail kit at a fixed per-kit price, not from a distributor with quantity breaks. This is the single biggest lever on total cost and the single biggest sourcing risk in the design.
2. Real quantity discounts on the catalog parts (MCU, injector driver, CAN transceiver, buck, supervisor) only partly show up at qty 10 — DigiKey's own price breaks for these lines mostly land at qty 100, not qty 10 — so the qty-10 total is not proportionally cheaper than 10× the qty-1 total.

Single-source / long-lead lines flagged: 3 confirmed, 1 partially mitigated, 1 open Phase-2 item. See §3.

1. Method and honesty note

Per the task's own budget instruction, this memo prices the expensive and risky lines from live or near-live distributor data — **MCU, injector driver, connector, CAN transceivers, the buck** — and gives the remaining functional blocks a **reasonable aggregate estimate, marked as such**, built from a named representative automotive-grade part rather than an invented number with no anchor. Every "estimate" line is a candidate part in a real, in-production automotive family, priced by class-typical India distributor pricing, not fabricated from nothing — but it has **not** been individually fetched from a live distributor page this session, and Phase 2 must confirm it before BOM lock.

Distributor pricing fetches in this session hit DigiKey India live pages successfully; Mouser India pages timed out repeatedly (likely bot-mitigation, consistent with what Memo 05 reported for Mouser/DigiKey chip pages). Where a Mouser-only line is cited, the price is from search-indexed content, not a live page load, and is marked accordingly.

2. BOM by functional block

Legend: AEC = automotive qualification status. C = confirmed this session (live fetch or direct citation). I = inferred / estimated, not independently priced this session. Currency conversion where only USD was found: ₹88/US\$1, noted per line.

#	Functional block (spec §4/§6)	Part	Distributor	Stock	Qty 1 (₹)	Qty 100 (₹)	Lead time	AEC-Q	Status	Cite
1	EMI input common-mode choke	Würth WE-CMB automotive CMC (representative, e.g. 744232801-class)	Mouser/DigiKey (class-typical)	—	~90	~55	—	AEC-Q200 (passive)	I — aggregate estimate	class prici not fetch

#	Functional block (spec §4/§6)	Part	Distributor	Stock	Qty 1 (₹)	Qty 100 (₹)	Lead time	AEC-Q	Status	Cite
2a	Reverse-battery ideal-diode controller	TI LM74610-Q1	DigiKey / Mouser	listed	~320 (est. from \$1.25@1k)	~140	standard	AEC-Q100	I — price scaled from confirmed \$1.25@1000 unit	TI
2b	Reverse-battery P-FET	Infineon IPB80P03P4L-01 (representative AEC-Q101 P-FET)	Mouser/DigiKey (class-typical)	—	~150	~90	—	AEC-Q101	I — aggregate estimate	class price not fetch
3a	Transient clamp — TVS	SMBJ33CA bidirectional TVS (fixed by simulation)	Mouser/DigiKey (widely stocked jellybean)	—	~18	~8	—	Commercial-grade common; automotive-grade (-13-F type suffixes) available from Littelfuse/Vishay/onsemi	I — aggregate estimate	class price not fetch
3b	Transient clamp — 10 µF bulk cap	Automotive X7R/electrolytic, 10 µF (fixed by simulation)	Mouser/DigiKey (class-typical)	—	~15	~7	—	AEC-Q200 (ceramic)	I — aggregate estimate	class price not fetch
4	Buck pre-regulator, 6–40 V → 5 V	TI TPS54360B-Q1	DigiKey (global; India page not confirmed live)	listed	~458 (\$5.20 × 88)	~280 (est.)	standard	AEC-Q100	C (price), 60 V-rated device chosen over 42 V LM5175-Q1 for load-dump margin — I (India-specific stock/lead time not confirmed)	DigiKey search \$5.20 found via search index
5a	5V_SENSOR LDO	TI TLV70233-Q1-class automotive LDO (representative)	Mouser/DigiKey (class-typical)	—	~120	~70	—	AEC-Q100	I — aggregate estimate	class price not fetch
5b	5V_SENSOR PTC resettable fuses ×4 (per sensor group)	Bourns MF-USMF-class automotive PTC	Mouser/DigiKey (class-typical)	—	~100 (4×~25)	~60	—	AEC-Q200	I — aggregate estimate	class price not fetch
6	3V3_MCU LDO	TI TLV1117-33QDCYRQ1-class (representative)	Mouser/DigiKey (class-typical)	—	~70	~35	—	AEC-Q100	I — aggregate estimate	class price not fetch
7	Supervisor (windowed WD + BOR)	TI TPS3823-33QDBVRQ1	DigiKey India	4,605 units in stock	162.44	95.60	Mfr. standard 16 wk (moot — stock covers near-term demand)	AEC-Q100	C — live DigiKey India fetch	DigiKey in
8	MCU	NXP FS32K148HAT0MLQT, 144-LQFP	DigiKey India	In stock, ships ~4 days (sibling SKU confirmed; exact SKU "ships today" per global listing)	~1,950 (₹1,900–2,000 band)	~1,400 (est., no qty100 break found)	Not a lead-time problem	AEC-Q100	C — per Memo 05	Mem 05, DigiKey
9	Crystal (MCU osc.)	NDK/Abracon AEC-Q200 automotive crystal, 8 MHz-class (representative)	Mouser/DigiKey (class-typical)	—	~35	~18	—	AEC-Q200	I — aggregate estimate	class price not fetch
10	CAN transceiver ×2 + protection	TI TCAN1042HGVDRQ1 (×2) + ESD array (representative NUP2105L-class, ×2)	DigiKey India	Out of stock; 2,500 units expected 21-Sep-2026	426.16 (2× 213.08) + ~40 (2× ~20 ESD)	257.50 (2× 128.75) + ~30	Mfr. standard 16 weeks	AEC-Q100	C (transceiver price/stock) — flagged, see §3	DigiKey in
11	VR conditioner (crank sensor)	Melexis MLX92242-class or NXP MC33199-class differential VR interface (representative — not yet selected)	Mouser/DigiKey (class-typical)	—	~180	~100	—	AEC-Q100	I — aggregate estimate, part not yet firmly selected (Phase 2)	class price not fetch

#	Functional block (spec §4/§6)	Part	Distributor	Stock	Qty 1 (₹)	Qty 100 (₹)	Lead time	AEC-Q	Status	Cite
12	EGR H-bridge	Infineon TLE9201SG-class automotive H-bridge (representative)	Mouser/DigiKey (class-typical)	—	~220	~140	—	AEC-Q100	I — aggregate estimate	class price not fetch
13	Metering-unit low-side driver + current sense	Infineon BTS443P-class automotive smart low-side switch (representative)	Mouser/DigiKey (class-typical)	—	~250	~160	—	AEC-Q100	I — aggregate estimate	class price not fetch
14a	Injector driver IC	NXP MC33816AE	DigiKey India	868 units	1,040.54	693.12	Std.; note: page shows "Last Time Buy Date 06/08/2027" — flagged, see §3	Industrial/automotive SMARTMOS, not itself AEC-Q100-badged in fetched listing — needs Phase-2 confirmation	C — live DigiKey India fetch	DigiKey India
14b	Injector boost stage (65–115 V, 12–24 A pk, 8–13 A hold)	Discrete: boost controller (e.g. LM5155-Q1-class) + HV automotive MOSFETs + magnetics + HV caps	Mouser/DigiKey (class-typical, multi-part)	—	~800	~500	—	Mixed AEC-Q100/Q101	I — aggregate estimate for the whole discrete stage, no specific BOM chosen yet (Phase 2)	Memorandum
15	Relay FETs ×2 (main relay, buzzer relay) — low-side N-FET + flyback diode each	Automotive N-FET (e.g. IPD50N04S4L-13-class) + automotive Schottky (e.g. SS34-class), ×2 pairs	Mouser/DigiKey (class-typical)	—	~80 (2×~40)	~40	—	AEC-Q101	I — aggregate estimate	class price not fetch
16	Barometric sensor (onboard, per spec §7 deviation #2)	NXP MPXH6115A6U-class automotive absolute pressure sensor (representative)	Mouser/DigiKey (class-typical)	—	~450	~300	—	Automotive-grade family (NXP MPXH line is automotive-qualified)	I — aggregate estimate, part not yet firmly selected (Phase 2)	class price not fetch
17a	94-pin ECU-mating connector (adapter harness side)	Bosch EDC17-style 94-pin + 60-pin companion, wired pigtail kit — leading candidate, unconfirmed part number	u-obd.com / ecudepot.com / autoocupart.net (aftermarket ECU-repair trade)	Listed, multiple sellers	~6,600 (US\$65–99 × 88, band)	~6,600 (no bulk break — aftermarket kit pricing)	Order-dependent, international shipping + India customs, not independently verified — Phase 2 item	N/A — not a distributor part number	C (price band at source) / I (India-landed total) — flagged, see §3	Memorandum 03, u-obd.com
17b	New ECU-side split connectors (spec §7 deviation #1)	TE Connectivity sealed automotive connector family (e.g. MCON 1.2/Superseal-class, representative)	Mouser/DigiKey (mainstream stock)	—	~300 (aggregate, multiple housings)	~180	—	Automotive sealed connector family	I — aggregate estimate, exact housing/pin-count split not yet designed	class price not fetch
18	Fixed-value passives (battery-sense divider ×2, discrete-input front-end ×3 sets, ratiometric front-end ×4 sets, plus generic bypass/pull-up passives)	1% metal-film resistors, MLCC caps, zener diodes per the simulation-fixed values below	Mouser/DigiKey (class-typical)	—	~300	~150	—	Mixed AEC-Q200 (passive) availability	I — aggregate estimate, see breakdown below	class price not fetch

Passives aggregate breakdown (line 18), values fixed by simulation, not re-derived: - Battery-sense divider, ×2 (pins 04, 06): 120 kΩ / 10 kΩ, 1% metal film — 4 resistors. - Discrete-input front-end, ×3 (pins 20, 24, 71): 47 kΩ / 68 kΩ, 220 nF, 3.0 V zener — 12 components. - Ratiometric sensor front-end, ×4 (boost pressure, EGR position, oil pressure, rail pressure): 10 kΩ / 16 kΩ, 22 nF, 3.3 V zener — 16 components. - Plus an allowance for generic bypass caps, gate resistors, and pull-ups not individually itemized in this memo. - ~32 fixed-value components + ~30 generic ones ≈ 60–70 small passives total, priced at India-typical qty-1

rates of ₹2–4/resistor, ₹3–6/MLCC cap, ₹8–15/zener; qty-100 rates roughly a third lower. This is the class of line the task explicitly authorizes as an honest aggregate rather than 30 invented individual prices.

3. Single-source and long-lead flags

Per the task's global instruction: any line with one supplier, or a lead time beyond eight weeks, gets a named second source or an explicit "no alternative identified." These are the lines that stall a build.

3.1 94-pin ECU-mating connector — FLAGGED, no distributor alternative identified

Single-source in the distribution sense that matters: the leading candidate (Bosch EDC17-style 94-pin housing) is sold **only through the international ECU-repair aftermarket** (u-obd.com, ecudepot.com, autoecupart.net, eBay sellers) — confirmed absent from Mouser India, DigiKey India, and Element14 in Memo 03's search. That channel carries real risks a normal distributor doesn't: no formal datasheet, counterfeit/quality risk, order-dependent lead time, and customs/duty exposure not yet quantified.

Named second sources, both weaker matches, both mainstream-distributor-buyable: - Aptiv/Delphi GT-150 series — Mouser-stocked, moderate visual match, exact 94-position variant not yet confirmed to exist (Memo 03, negative result on that specific check). - TE Connectivity AMPSEAL (90-series) — Mouser/DigiKey-stocked, weakest visual match (oval/rounded shell vs. the drawing's rectangular multi-row grid), but the safest fallback if the Phase-2 photograph rules out the other two.

This line cannot be closed further without the photograph brief in Memo 03 §4. It is the top risk item in this BOM — both for build-stalling potential and for the ~45–50% of per-unit cost it represents.

3.2 Injector driver IC (MC33816AE) — FLAGGED, no alternative identified

Memo 04 searched specifically for allocation-free alternatives (Infineon TLE8090, Elmos E526) and returned a **negative result** — no confirmed, small-quantity-orderable alternative dedicated common-rail injector-driver IC was located. The MC33816AE is **confirmed in stock (868 units) at DigiKey India** today, which is good news, but it is a single-supplier (NXP), single-product-line dependency with no named second source.

New finding this session, not yet resolved: the live DigiKey India page for this part carries a "Last Time Buy Date: 06/08/2027" field. Memo 04 separately reports the part as "Active" under NXP's Product Longevity Program (a 10-year-availability commitment), which is not the same thing as an EOL notice — a longevity-program part can still show a programmatic LTB-style date without meaning the product is being discontinued. **This memo cannot resolve which reading is correct with the sources available this session** — it is flagged, not smoothed over, as a Phase-2 item: confirm directly with NXP or DigiKey sales whether MC33816AE has an active EOL notice before committing the injector driver architecture to this part.

No alternative identified. If the LTB field turns out to mean what it says, this becomes the most urgent open item in the entire BOM, ahead of the connector.

3.3 CAN transceiver (TCAN1042HGVDRQ1) — FLAGGED, second source named

Live DigiKey India data this session: **out of stock**, with 2,500 units expected 2026-09-21, and a **manufacturer standard lead time of 16 weeks** — over the task's 8-week threshold on both counts (current stockout, and the manufacturer-quoted lead time for a fresh order).

Named second source: NXP TJA1051 / TJA1042 high-speed CAN transceiver family — AEC-Q100, functionally equivalent (ISO 11898-2 high-speed CAN transceiver), confirmed listed at Mouser (Memo-05-style search-index confirmation; India-specific live price not obtained this session, flagged as a Phase-2 confirmation item). Given the restock

date is roughly one week out, this may resolve itself before it matters, but the flag stands per the letter of the instruction — it was out of stock and over 8 weeks lead time at time of writing.

3.4 Supervisor IC (TPS3823-33QDBVRQ1) — flagged on paper, low real risk

DigiKey's manufacturer standard lead time field shows 16 weeks (>8-week threshold), but **4,605 units are in stock right now** at DigiKey India, which resolves the practical risk for any order size this project plausibly needs. Named second source anyway, for completeness: Microchip MCP130-class or ON Semi automotive supervisor family, both mainstream-distributor-stocked.

3.5 Injector boost stage, VR conditioner, EGR H-bridge driver, barometric sensor — open Phase-2 selections, not yet single-source risks

These four lines carry a representative candidate part, not a firmly selected one. They are not flagged as single-source because no sourcing decision has been locked in yet — but Phase 2 must close each of them with the same rigor Tasks 3/4/5 applied to the connector, injector driver, and MCU, before this memo's estimate lines convert to real BOM lines.

3.6 MCU (FS32K148HAT0MLQT) — mitigated, not flagged

Per Memo 05: real DigiKey India stock today, ordinary shipping, multiple orderable SKUs within the same S32K148 family (UJT0VLQT / HFT0VLQR / HAT0MLQT) across DigiKey and at least one third-party stocking distributor (components-store.com, thousands of units). The checked fallback family (S32K146) is in *worse* supply shape (12–52 week spread, backordered on the 144-LQFP SKUs), so it is not offered as the named second source — the in-family SKU spread already provides one. Not flagged.

4. Headline cost, worked

Qty 1 (sum of the "Qty 1" column above): $\approx ₹14,175$, rounded to $\approx ₹14,200$.

Of that, the connector kit (line 17a) alone is $\approx ₹6,600$ — about 46% of the total. Remove it and the rest of the BOM (electronics + new-side connectors + passives) is $\approx ₹7,575$ at qty 1.

Qty 10, worked with an explicit, stated assumption rather than silently guessed: catalog IC lines are assumed to fall to roughly 90% of their qty-1 unit price at qty 10 (a conservative read — DigiKey's own price breaks for the lines fetched live this session mostly land at qty 100, e.g. TPS3823 falls 41% qty1 $\rightarrow$ 100 and TCAN1042 falls 40%, so a qty-10 point partway there is a reasonable, stated interpolation, not a measured figure). The connector kit (line 17a) is assumed to hold its per-unit price flat at qty 10 — it is bought per-kit from an aftermarket seller with no confirmed bulk-order mechanism.

- Non-connector BOM at qty 1: $₹14,175 - ₹6,600 \approx ₹7,575 \rightarrow \times 0.90 \approx ₹6,820/\text{unit}$ at qty 10 $\rightarrow \times 10 \approx ₹68,200$
- Connector at qty 10: $₹6,600 \times 10 = ₹66,000$ (flat, per above)
- **Total, 10 units:** $\approx ₹1,34,200$, i.e. $\approx ₹13,420/\text{unit}$ — barely below the qty-1 unit price, because the largest line item does not get cheaper in volume through this channel.

This is the headline finding worth repeating in plain language: a conventional component BOM would show a strong per-unit cost curve from qty 1 to qty 10. This one barely moves, because its largest cost driver is not a catalog part.

5. PCB fabrication estimate (separate from the BOM total above)

No layout exists yet — this is a Phase-2-pending, explicitly inferred board-area assumption, built from the parts on this BOM: a 144-pin LQFP MCU, a 94-pin connector-shadow footprint plus new-side connectors, a buck pre-regulator, an injector boost stage with HV magnetics and bulk caps, an H-bridge, and assorted discrete front-ends.

A board in the 100–150 mm × 120–160 mm class (≈150–240 cm²) is a reasonable planning assumption for this component count; this memo adopts 120 mm × 150 mm (180 cm²) as the working figure.

- **China-based low-volume fab, shipped to India** (JLCPCB/PCBWay-class service, the common route for small Indian teams): a 4-layer, standard-FR4, no-ENIG prototype run of 5–10 boards at this area typically lands in the US\$40–70 total band for the panel, i.e. ≈ ₹3,500–6,200 for the run (≈ ₹600–1,200/board), plus international shipping (days to ~2 weeks) and possible customs handling. **Inferred**, not a live quote — Phase 2 should submit real Gerbers once layout is complete.
- **India-based fab** (faster turnaround, no import wait, typically higher per-board cost at prototype quantities for 4-layer with plated vias at this class of complexity): ≈ ₹1,000–2,000/board for a small prototype run. **Inferred**, same caveat.

6. What this memo did not do, and why

- Did not individually fetch live pricing for ~14 of the 22 BOM lines. The task's own budget instruction says this explicitly is the right trade: price the expensive and risky lines properly, mark the rest as an aggregate estimate. Ten well-sourced lines (MCU, injector driver, connector, both CAN transceiver components, buck, supervisor) plus one clearly-marked aggregate covers everything the task asked for without chasing forty invented numbers.
- Did not resolve the MC33816AE "Last Time Buy Date" field. This is flagged loudly in §3.2 rather than either alarmed over or ignored — it needs a direct NXP/distributor sales conversation, not another web search.
- Did not select final parts for the VR conditioner, EGR H-bridge driver, metering-unit driver, barometric sensor, or injector boost-stage BOM. These carry representative, real, in-production automotive-grade candidates so the cost estimate has an anchor, but none of the five is a locked sourcing decision. Flagged in §3.5, not smoothed over.

Citation index

- Memo 03 — ECU connector
- Memo 04 — Injector drive
- Memo 05 — MCU selection
- Project spec: `docs/superpowers/specs/2026-09-13-ecu-reconciled-spec-design.md` §4, §6
- DigiKey India, TPS3823-33QDBVRQ1:
<https://www.digikey.in/en/products/detail/texas-instruments/TPS3823-33QDBVRQ1/1894636> — fetched live 2026-09-13
- DigiKey India, MC33816AE:
<https://www.digikey.in/en/products/detail/nxp-usa-inc/MC33816AE/4693868> — fetched live 2026-09-13
- DigiKey India, TCAN1042HGVDRQ1:
<https://www.digikey.in/en/products/detail/texas-instruments/TCAN1042HGVDRQ1/5967638> — fetched live 2026-09-13
- TI LM74610-Q1 product page and \$1.25@1k pricing:
<https://www.ti.com/product/LM74610-Q1>,
https://www.how2power.com/newsletters/1512/products/H2PToday1512_products_TexasInstruments.pdf
- TI TPS54360B-Q1 / TPS54360BQDDARQ1 pricing (search-indexed, \$5.20–5.24): <https://www.ti.com/product/TPS54360B-Q1>, DigiKey listing found via search
- DigiKey FS32K148HAT0MLQT:
<https://www.digikey.com/en/products/detail/nxp-usa-inc/FS32K148HAT0MLQT/8628226>
- u-obd.com, Bosch EDC17 94-pin connector kit: <https://www.u-obd.com/product/edc17-connector/>

- Mouser, Delphi GT-150 series: <https://www.mouser.com/Delphi-Connection-Systems/Connectors/Automotive-Connectors/GT-150-Series/>
- Mouser, TE Connectivity AMPSEAL:
<https://www.mouser.com/en/c/connectors/automotive-connectors/?m=TE+Connectivity&series=AMPSEAL>

Engine Inspection Brief — Field Checklist

STATUS 18 Sep 2026 — part of this brief is done

A first visit has happened and photographs came back. **Four register items closed**, and the tasks that closed them are ticked below. What remains is mostly the multimeter work, which is the part that actually gates the design.

Closed: U11 engine identity (task 3.1 — the plate reads 3GK550ETA 4SR1 , *not* the model memo 01 inferred) · U1 the connector (tasks 3.4–3.8 — Bosch 1 928 405 192 / 194 , **code C**) · U13 which controller is fitted (DSE4522 MKII AMF, not a KG640C — this was not a task in this brief, it was raised by the evidence itself) · U3 no speed pickup (from DSE's manual).

UPDATE, later on 18 Sep: U12 is CLOSED, and it closed badly. Reported from the machine: **the fuel relay cuts a signal, it does not remove power.** That is the outcome task 2.1's table told the reader to flag immediately. §3's second line of defence terminates in the failure it was meant to catch, and the standalone trip module is now the only shutdown path on this machine that does not require the cooperation of the ECU being protected against. Task 1.4's contact-type check is still wanted as confirmation — see [refs/field-evidence-2026-09-18.md §4](#) for why nothing depends on it.

Also reported: the intake throttle and the pre/post-heaters are **battery-fed**. That answers who supplies them, not who commands them — tasks 3.11 and the U14 trace both still stand.

Also still open and cheap: task 1.3 (U2, one resistance reading) and task 3.11 (U8, follow one cable by hand).

New since this brief was written, worth adding to the next visit: trace the **pre-heat / post-heat** supply — the DSE4522's configuration enables both at 50 °C and this design has no heater output at all (U14).

Purpose: close every open unknown that no amount of desk research can close — only a person with a multimeter and a camera, at the machine, can. Closes or advances **U1, U2, U4, U6, U7, U8, U9, U10, U11, U12**, plus the pin 01/02 ground question raised in the reconciled spec.

Print this. Carry it. Tick boxes as you go — a half-finished visit is still useful if the ticks say which half.

Sources: reconciled spec `docs/superpowers/specs/2026-09-13-ecu-reconciled-spec-design.md` (§3 safety, §8 unknowns register), `refs/ecu-pinout-extracted.md`, research memos `docs/research/01-engine-identity.md` and `docs/research/03-ecu-connector.md`. Every expected reading below cites one of these. Anything not sourced is marked **inferred**.

SAFETY — read this before touching anything

This is a 25 kVA genset: a diesel engine that can start, and a 415 V three-phase alternator. This checklist is bench/inspection work only.

- **Battery disconnected before any continuity or resistance measurement.** Negative terminal off first, isolated, out of reach of the positive post.
- **Nothing inside the alternator output enclosure.** No task here requires it. If you find yourself reaching toward the 415 V output terminals, stop — that is not this checklist.
- **The engine does not run during any of this.** Every task below is done with the engine off. Ignition-on tasks mean the key/master switch position only, never cranking or running.
- **If a measurement needs the set running** (this includes the full CAN log in Group 4, which may need it), **that is a separate supervised activity**, done by someone qualified to run the set, not a solo checklist item. Note it as such and move on.
- Reconnect the battery only after Group 1 is fully complete and you've checked no tools or leads are left on the terminals.

Materials: digital multimeter (V, Ω , continuity beep), clamp meter if available (safer than breaking a fused circuit for current), phone/camera with flashlight, notebook or voice memo, 10 mm socket for the battery terminal, tape/marker for labelling. **Group 4 additionally needs a USB-CAN adapter and a laptop** — it may warrant its own trip.

Group 1 — Battery disconnected (continuity / resistance)

Do these first, in this order, with the battery negative terminal removed and isolated.

- ☐ **1.1 — Pin 01/02 ground hypothesis.** Continuity from ECU connector pins 01 and 02 to battery negative / chassis ground.
- *Why:* the connector drawing places 01/02/05/06 as large power contacts, but 01/02 are labelled `G_G_B AT1 / AT2` — switch-return legs, which wouldn't need a large contact. The alternate reading: `G_G_B` = battery

ground, and these are the ECU's main power grounds (also the only explanation for why no power-ground pin appears anywhere in the 44-pin extraction). (*spec §4 note; pinout "Digital inputs"*)

- **Continuity (near 0 Ω)** → **hypothesis confirmed**: 01/02 are power grounds, not switch returns. Update the pin map and power/ground plan.
- **Open circuit** → **hypothesis wrong**. They're isolated switch-return legs as originally labelled; the power-ground pin(s) remain unidentified — keep looking during Group 3 tracing.
- [] **1.2 — Pin 09 "synchronization ground."** Continuity from pin 09 to (a) battery negative and (b) a known sensor ground pin (e.g. 34 or 08).
- *Why*: pin 09 is named as a ground but drawn in **red** (switched-battery colour class), which is contradictory on its face. (*spec §4, "Power"; pinout, U6 entry*)
- **Continuous to battery negative** → it likely is a ground point, oddly coloured in the diagram for harness-routing reasons, not a live signal. Note this for §4 but treat as resolved.
- **Open to both** → it is *not* a simple ground. Carry to Group 2 (2.3) for a voltage measurement — the red wire colour then means what it usually means.
- [] **1.3 — Boost sensor thermistor topology (closes U2 outright)**. Boost P&T sensor unplugged, engine **cold**. Resistance across sensor pins 2 and 1 (ECU pins 79 / 34 per the pinout — temperature signal / sensor ground). (*brief; spec §8 U2; pinout "Analog sensors"*)
- A few k Ω , falling if you re-check once the engine's warmed a little → NTC thermistor. Design a pull-up divider front-end. U2 closed.
- **Open circuit, or a fixed high reading that doesn't move with temperature** → not a bare NTC — a conditioned/ratiometric output. Design the ratiometric front-end (same as the pressure channel). U2 closed.
- This is a single measurement. Either outcome closes U2 — there is no third case.

▣ **1.4 — Ignition relay contact type.** (*Still wanted 18 Sep as confirmation; task 2.1 has since been answered the other way round — see the status banner.*) **Do this BEFORE task 2.1; it decides how 2.1's result is read.** Locate ignition relay -13RB1 on the GCU relay board. With the battery disconnected and the relay coil unpowered, measure continuity across its **load contacts** (not the coil).

Why this matters and why 2.1 is ambiguous without it: task 2.1 reads voltage with the ignition switch off, assuming that is the relay's de-energised state. If the relay turns out to be normally-closed, "de-energised" means contacts **made** and power still present — so a battery-voltage reading in 2.1 would mean the test state was wrong, not that the relay only signals. Those two have opposite conclusions and the same reading. This measurement separates them. **Inferred concern, not sourced — raised because the relay's drive logic is not documented anywhere we have.**

Continuity across load contacts, coil unpowered	Meaning for task 2.1
Open (normally-open, energise to run)	Expected case. 2.1's table reads as written.
Closed (normally-closed, energise to cut)	2.1's table is inverted . Battery voltage with the switch off is then normal, and U12 must instead be tested by energising the relay and re-measuring. Note it and flag it.

Group 2 — Ignition switch tests (battery connected, engine OFF, not cranking)

Reconnect the battery for this group only. Key/master switch is the only thing that moves. **Do not crank or start.**

- 2.1 — U12, the priority item. Complete task 1.4 first. Ignition/master switch **OFF**. This is assumed to be ignition relay -13RB1 's de-energised state — **task 1.4 confirms that assumption, and if it fails, the table below inverts**. You do not need to pull or force the relay. If you can see or hear the relay (GCU relay board), confirm it looks/sounds unenergised before measuring; photograph its label to confirm 70 A rating.

With the switch OFF, measure voltage (relative to chassis/battery negative) at: - ECU connector pin 21 (battery+ feed) - ECU connector pin 04 (V_V_BAT_1R) - ECU connector pin 06 (V_V_BAT_2R) - the **fuel metering unit's own pin 2** (its battery-fed pin, via 20 A fuse 8F1 — not ECU pin 88, which is the ECU's PWM output to metering unit pin 1)

Why this is the whole point of the trip: spec §3's safety argument — that a hung ECU can't keep injecting because the GCU physically removes fuel-system power — depends entirely on this. (*spec §3, §8 U12*)

Reading at all four points	Meaning
0 V (no battery voltage present)	Relay removes power . §3's second line of defence is real. U12 closed in the safe direction.
Battery voltage present at any of them	Relay signals only — power stays up through a hung ECU. §3's relaxation is void: reverts to "standalone MPU trip module mandatory." Flag this immediately, do not wait for the writeup.

- ▣ 2.2 — U7: are pins 04/06 sense-only or load feeds? Switch ON (engine still off). Re-measure voltage at pins 04 and 06 (expect battery voltage present now, per 2.1's contrast — that alone doesn't answer U7).
- **Then measure current**, preferably with a clamp meter on the wire (safer — no circuit break). If only a multimeter is available, break the circuit at the fuse holder for 04/06 and measure DC current in series, briefly.
- *Why*: sense lines draw milliamps; something that's actually feeding a load draws amps, and each fuse is rated 10 A. (*brief; spec §8 U7; spec §4 "Power" — both pins are 10 A fused*)

Current	Meaning
mA range	Sense-only — these just feed the battery-voltage ADC divider. Power sheet needs no extra current rating here.
Amps, approaching the 10 A fuse rating under some load state	Load feed — something downstream draws real current through the ECU. Power sheet must size for it.

- ▣ 2.3 — Pin 09, if 1.2 left it open. Switch ON. Voltage at pin 09 vs. chassis.
- ~0 V → confirms a ground reference despite the odd wire colour.
- **Switched battery voltage** → it's a real signal wearing a "ground" name — treat as an undocumented discrete/analog input, not a ground. Note whatever else in the harness moves when this changes (if observable) as a clue to its function.

Group 3 — Photography (engine off; battery may be either state, disconnect

preferred while working around the engine bay)

For each: square-on unless noted, fill the frame, flash off / raking light for part numbers (on-axis flash washes out shallow embossing).

- ▣ ~~3.1 DONE 18 Sep~~ — **Engine rating plate** (rocker cover / flywheel housing / block side). Get legible: model string (expect something matching **3R550ETA** — anything else means the engine-identity inference in memo 01 is wrong and should be discarded, not patched), serial number, rated output/speed/frequency, any emissions certification mark. **Closes U11.** (*memo 01, "What would close this"*)
- ▣ **3.2 Genset nameplate** (canopy/control panel). **Do not go looking for a string you expect.** This brief originally said to expect **KG4-25WS1**, inferred from memo 01's engine identification — and that identification turned out to be wrong when the engine plate was read. The genset model inference came from the same chain and is equally unsupported. Photograph whatever is actually on the plate, including the serial number. (*memo 01, and its correction*)

- ▣ **3.3 ECU case label.** Flat, well-lit. Hunting a vendor part number — Bosch EDC pattern `0 281 0xx xxx`, Bosch-reseller pattern `F 01R 0xx xxx`, or any Continental/Delphi/Denso mark. **Now more likely to pay off than when this was written:** the connector is confirmed Bosch (`1 928 405 192 / 194`), so a Bosch part number on the case would be consistent and would identify the OEM ECU outright. Would resolve the still-open OEM-ECU-vendor question. (*memo 03 §4 item 4*)
- ▣ ~~3.4 DONE 18 Sep~~ — **ECU connector — full-face, mating side, connector unplugged.** Determines at a glance: flat rectangular multi-row grid (Bosch/Delphi-style) vs. oval/rounded single-lever shell (AMPSEAL-style) — the top-level branch of the connector shortlist. **Advances U1.** (*memo 03 §4 item 1, ranked candidate table*)
- ▣ ~~3.5 DONE 18 Sep~~ — **Connector retention mechanism**, actuated and disengaged, if it can be operated safely with the connector unplugged. Lever vs. slide latch vs. bolt vs. friction/CPA clip. (*memo 03 §4 item 2*)
- ▣ ~~3.6 DONE 18 Sep~~ — **Raking-light pass over every flat face of the connector housing** — top, both sides, cable boot. Hunting molded/raised part number, date code, or logo: Bosch roundel, Delphi/Aptiv wordmark, TE Connectivity mark. Take several shots at different light angles if the first doesn't show text. **Would settle U1's connector-family ranking outright if a logo turns up.** (*memo 03 §4 item 3*)
- ▣ ~~3.7 DONE 18 Sep~~ — **Connector still mated, wide shot** — backshell, cable exit angle, strain relief. Feeds the adapter-harness design even without a family ID. (*memo 03 §4 item 5*)
- ▣ ~~3.8 DONE 18 Sep~~ — **Pin-count/contact-size check, connector unplugged.** Confirm four rows of small pins plus two 2-pin corner blocks (near pins 1 and 5). **Check specifically whether the corner-block contacts (pins 1/2/5/6) are visibly larger/wider than the small-pin grid.** This is the photographic half of the 1.1 pin 01/02 question — larger contacts support "these are power," normal-sized contacts support "these are ordinary switch returns after all." (*memo 03 §4 item 6; spec §4 note*)
- ▣ **3.9 Boost P&T sensor body** — part number, if legible. Backs up 1.3's electrical answer with a datasheet-checkable part.
- ▣ **3.10 Injector body** — part number, if accessible without disturbing the fuel system. Corroborates the Bosch-family injector inference already used in memo 03/04.
- ▣ **3.11 Intake throttle — 6-pin connector and cable run (closes U8).** Photograph the connector, then follow the cable by hand through the loom.
- **Terminates at the ECU's 94-pin connector** → in scope. Photograph which physical position on the ECU housing it plugs into (row/corner), so it can be matched to a pin number once U1 is resolved.

- Joins the XC11/XC1 panel loom instead (same path as fuel level/WIF, per spec §2.3) → out of scope for this ECU; a throttle driver sheet is not needed.
- ▣ 3.12 Catalyst temperature sensor — 2-pin connector and harness run (closes U9). Same method as 3.11: photograph, then trace.
- Terminates at ECU → in scope, an EGT-style front-end is needed.
- Joins the panel/GCU loom → out of scope, same treatment as fuel level/WIF.
- ▣ 3.13 KG640C terminal strip (J1–J8), photographed with terminal labels legible. Reference material for the MPU-input question (U3) and future GCU interface work; not expected to close anything by itself this visit.
- ▣ 3.14 Wire-colour spot check at each analog sensor connector (boost, coolant, EGR, oil pressure, rail pressure) — flag toward U10. SENT and CAN H share the same yellow colour code in this harness's legend (*pinout*, "*Wire colour classes*"), so this can't be resolved by colour alone. Photograph any single yellow wire that is not paired with a green wire (a CAN pair) and is not part of a 3-wire AC bundle — that's the SENT candidate. This will likely only narrow it down, not close it outright — say so in the notes rather than guessing a pin.

Group 4 — Separate trip: CAN bus log (closes U4 in practice, not from a manual)

Needs a USB-CAN adapter and a laptop. Do this as its own visit unless you're already bringing the hardware.

- ▣ 4.1 With the OEM ECU still fitted and the harness intact, tap CAN channel 1 (pins 55/77, H/L) and channel 2 (pins 87/86, H/L) via breakout or fly leads. (*pinout*, "*CAN — two channels*"; *spec §4*)
 - ▣ 4.2 Log traffic with the ignition on (ECU and GCU powered). Identify PGNs present, source addresses, and update rates.
 - Basic identification/status broadcasts should appear at key-on. **Engine-state PGNs (speed, load) may not appear, or may read zero/stale, until the engine is actually running** — if you need those, that's the "separate supervised activity" from the safety preamble, not a solo add-on to this trip.
 - ▣ 4.3 Compare the captured PGN set against whatever the KG640C is found to expect (from the log itself, and cross-check against the KG640 manual if it surfaces). This is the practical closure the brief couldn't get from documents — the log is the source, not a spec. (*spec §8 U4*)
-

What this closes

#	Unknown	Closed by
U1	Connector family	3.4, 3.5, 3.6, 3.7, 3.8 (photographic — narrows the ranked shortlist in memo 03, may not fully settle it without a logo hit)
U2	Boost air-temp topology	1.3 — closes outright
U4	KG640C PGN set	Group 4 — practical answer via live log
U6	Pin 09 function	1.2, 2.3
U7	Pins 04/06 sense vs. load	2.2
U8	Intake throttle on ECU?	3.11
U9	Catalyst sensor routing	3.12
U10	Which signal is SENT	3.14 — narrows, may not fully close
U11	Engine identity	3.1 — closes outright if plate matches
U12	Ignition relay: power-cut or signal-only	2.1 — the safety-gate item; closes §3's open question either way
—	Pin 01/02 ground hypothesis	1.1, 3.8

Ten register entries touched, two (U2, U11) closeable outright on the spot, one (U12) resolves the live safety-argument branch either way, the rest narrowed with a clear next step recorded regardless of outcome.

Research Memo 09: Resistive Temperature Front-Ends — DSE and SEDEMAC Prior Art

UPDATED 18 Sep 2026 — DSE is not just prior art here, it is the fitted controller

This memo studied Deep Sea Electronics and SEDEMAC as **prior art** — other vendors' answers to a problem we share. It opened by noting that both are genset-controller class devices, "the same class of device as the KG640C we talk to", and was careful that findings from them might not transfer.

The controller on this machine has since been photographed. It is a **Deep Sea Electronics DSE4522 MKII AMF**. The DSE practice described below is therefore not analogous guidance from a comparable product — **it is the behaviour of the device this ECU actually has to work alongside.**

That strengthens §3.1's ground-offset finding considerably. DSE's own manual for the fitted module's siblings (057-260) is emphatic about it: *"It is VERY important that terminal 10 (sensor common) is connected to an earth point on the ENGINE BLOCK, not within the control panel ... This connection MUST NOT be used to provide an earth connection for other terminals or devices."* The differential front-end recommendation in §4 has been adopted into spec §4.

§3.4's scaling caveat also held up under its own test:

`sim/blocks/ntc_frontend.cir` found that copying DSE's constant-current topology onto a 670:1 automotive NTC runs out of rail compliance at the cold end. The divider survived.

The SEDEMAC material remains what it was — prior art, snippet-sourced, and still not verified against a retrieved manual.

Task: U2 supporting research — how two established genset-controller vendors build resistive temperature sensor inputs. Consumes memo 02 (KG640C is the same class of device). Feeds the sheet 3b analog front-end and a re-examination of the already-simulated `sensor_ratiometric` block. **Date:** 2026-09-14 **Author:** Follow-up research, phase 1 exit

Verdict (read this first)

Neither vendor uses a plain pull-up divider to a fixed rail — the topology this project had planned for the U2 NTC branch.

- **Deep Sea Electronics** forces a **fixed current** through the sender and measures the voltage across it with a **differential** input, specified for ± 2 V of **common-mode** (ground offset) rejection. (*Confirmed — read directly from two DSE operator manuals.*)
- **SEDEMAC** uses **ratiometric** resistive sensing with a dedicated **sensor common point** terminal bonded to the engine body. (*Secondary — see the sourcing caveat in §2; the manual itself could not be retrieved.*)

This does not close U2. Both are genset *controllers* — the same class of device as the KG640C we talk to, not the common-rail ECU we are replacing. They read VDO/Datcon-style senders bolted into the block; U2 concerns a charge-air temperature element inside a 4-pin P&T combo on a common-rail engine. This is prior art on how to build the resistive branch **well**, not evidence about which branch this engine is on. Brief task 1.3's resistance measurement remains the only thing that settles U2.

1. Deep Sea Electronics — constant current, differential

Two generations of the same controller, both read directly from the operator manuals via text extraction. **Confirmed.**

DSE8610, Operator Manual Issue 5, document 057-115

Channel	Excitation current	Full scale	Over range / fail
Oil pressure	15 mA	240 Ω	270 Ω
Coolant temperature	10 mA	480 Ω	540 Ω
Flexible sensor (×2)	10 mA	480 Ω	540 Ω

DSE8610 MKII, document 057-254 Issue 3

Unified every channel on 15 mA, 480 Ω full scale, 600 Ω over range/fail, and made all four inputs selectable between resistive, 0–10 V and 4–20 mA modes.

The wording that matters

Identical across every channel of both generations:

Measurement type: Resistance measurement by measuring voltage across sensor with a fixed current applied **Arrangement:** Differential resistance measurement input **Max common mode voltage:** ± 2 V

Accuracy: $\pm 2\%$ of full scale resistance, excluding sensor error

Sources (both downloaded and text-extracted, not read from search snippets): - <https://www.reactpower.com/wp-content/uploads/2020/07/DSE8610-Control-Panel.pdf> - <http://www.davidsonsalessshop.com/catalog/files/Products/Deep%20Sea/DSE8610MKII%20Manual.pdf>

2. SEDEMAC — ratiometric, with a dedicated sensor common point

Sourcing caveat, stated plainly: every attempt to retrieve a SEDEMAC manual failed. ManualsLib, Manualzz and Scribd all returned HTTP 403 to direct fetch, and one aggregator link redirected to an unrelated cryptocurrency domain and was not followed. The figures below come from **search-result snippets**, consistent across two independently phrased queries, and are recorded as **secondary, unverified against the document itself**. They should not be used as design numbers without retrieving the manual.

GC1200 series, analog resistive sensor inputs on terminals 24 / 25 / 26 (engine temperature, fuel level, lube oil pressure):

Property	Value
Type	Ratiometric sensing
Range	10 Ω to 1000 Ω
Open-circuit detection	Above 5.5 k Ω
Accuracy	$\pm 2\%$ of FSD up to 1000 Ω

Separately, terminal 41 is a **sensor common point (SCP)**, specified to bond directly to an electrically sound point on the engine body, serving as the shared reference for all analog senders.

Sources (snippet-level only): - <https://www.scribd.com/document/558526157/sedmac-controller-manual-1202> - <https://www.manualslib.com/manual/1474533/Sedemac-Gc1100.html>

3. What transfers to this design

3.1 The differential input is the load-bearing finding

DSE specifies ± 2 V common-mode rejection on every sender channel, and SEDEMAC solves the same problem from the other direction by insisting on a dedicated sensor common point bonded to the engine body. Both are addressing **ground offset between the sensor's return and the controller's reference**.

This project has that exact problem, and it is visible in the OEM wiring: **pin 34 is a shared sensor ground, spliced between the boost P&T sensor and the coolant sensor** (per `refs/ecu-pinout-extracted.md`, splice V3). A shared return carries the other sensor's current, developing an offset across the harness resistance. A differential input rejects that offset; a single-ended input reads it as signal.

`sim/blocks/sensor_ratiometric.cir` is single-ended. It passes its checks, but those checks never modelled a ground offset, so passing does not mean the topology is right for a shared return. This is a gap in the *check set*, not a failure of the simulation — and it is exactly the kind of thing the simulation programme exists to catch. It should be re-examined before sheet 3b is drawn.

Worth noting the OEM's own hierarchy here: the rail-pressure sensor gets a **dedicated** ground (pin 08, not spliced), while boost and coolant share pin 34. Whoever designed that harness ranked the channels by how much ground offset each could tolerate. That ranking is information, and it agrees with SEDEMAC's SCP discipline.

3.2 Fault detection belongs in the range, not bolted on

Both vendors reserve a band above full scale as the fault region — DSE $480\ \Omega \rightarrow 600\ \Omega$, SEDEMAC anything above $5.5\ \text{k}\Omega$. An open or shorted sender lands somewhere that is unambiguously **not a temperature**, rather than pinning to a rail where it could be read as a plausible cold or hot value. Front-end component values should be chosen so both failure modes land outside the signal band, and firmware should treat that band as a distinct state.

3.3 Constant current linearises the readout

With a pull-up divider, the measured voltage is $V_{\text{ref}} \cdot R / (R + R_{\text{pu}})$ — non-linear in R , so resolution varies across the range and the pull-up's tolerance enters the result directly. With a forced current it is $I \cdot R$, linear, with the current source's accuracy as the only gain error. That is why DSE does it, and it costs a current source per channel.

Ratiometric sensing (SEDEMAC's approach, and the one already used on this project's pressure channels) does not linearise, but it does cancel supply drift by making the ADC reference track the excitation. The two approaches solve different halves of the problem.

3.4 The current values do NOT transfer — a scaling caveat

These controllers expect senders in the **tens to hundreds of ohms** (VDO/Datcon range). This project's own inspection brief predicts "a few $\text{k}\Omega$ " for the NTC case at task 1.3. Forcing DSE's 15 mA through a few $\text{k}\Omega$ would demand tens of volts and is impossible on a 5 V sensor rail.

If the constant-current topology is adopted, excitation must be scaled to roughly the 100 μA class for a $\text{k}\Omega$ -range NTC. Copying DSE's numbers directly would be a straightforward and expensive mistake. **Inferred** from the resistance ranges involved, not sourced.

4. Recommendation

1. **Upgrade the U2 mitigation.** The register's original hedge was a do-not-populate pull-up so either sensor type would work. That hedge assumed the NTC branch is a divider. Design the front-end so the branch can be either a scaled current source or a divider, and the choice remains a populate option after task 1.3 answers U2.
 2. **Add a ground-offset check to `sensor_ratiometric`** before drawing sheet 3b, and reconsider a differential front-end for the two channels that share pin 34. The block's current PASS verdict does not cover this case.
 3. **Retrieve a SEDEMAC manual properly** before using any figure in §2 as a design number. Everything there is snippet-level.
-

5. Negative results, recorded

- No DSE application note on wiring resistive sensors was retrieved, though search results reference one existing on the DSE site. Its guidance on common wiring mistakes would likely be relevant and was not obtained.
- No SEDEMAC document was retrieved in full. Three separate hosts refused.
- Neither vendor publishes a schematic of the input stage — only its specification. The internal topology in §1 is read from the specification's own wording ("fixed current applied", "differential"), which is explicit enough to be treated as confirmed, but no circuit diagram was seen.

Phase 1 — Research Implementation Plan

For agentic workers: REQUIRED SUB-SKILL: Use `superpowers:subagent-driven-development` (recommended) or `superpowers:executing-plans` to implement this plan task-by-task. Steps use checkbox (`- [ ]`) syntax for tracking.

Goal: Close the unknowns that block schematic capture, and produce the sourcing and standards groundwork the later phases depend on.

Architecture: Seven independent research memos, one per question, each written to `docs/research/`. Every memo ends with a verdict that either closes a numbered unknown from the spec's §8 register or states precisely why it cannot be closed from desk research and what physical access would close it. Memos are evidence documents: every factual claim carries a citation, and an uncited claim is a defect.

Tech Stack: WebSearch and WebFetch for primary sources; manufacturer datasheets and distributor stock APIs; Markdown output under version control.

Spec: `docs/superpowers/specs/2026-09-13-ecu-reconciled-spec-design.md`

Global Constraints

- **Every factual claim carries a citation** — a URL, a document number, or an explicit "measured on the engine by on ". No claim from memory.
 - **Distinguish confirmed from inferred.** Confirmed means a primary source says it. Inferred means we reasoned to it. Memos mark every non-obvious claim as one or the other.
 - **A closed unknown states what closed it.** "U5 closed — injector is Bosch 0445110xxx per photograph refs/injector-body.jpg" is a close. "U5 probably Bosch" is not.
 - **Negative results are results.** A memo that establishes the KG640C manual is not publicly available has done its job, provided it records where it looked.
 - **No safety-relevant conclusion from a secondary source.** Anything feeding the §3 overspeed gate needs a manufacturer document or a measurement, not a forum post.
 - **Currency for sourcing: INR**, with availability assessed for **India** (Mouser/DigiKey India, Element14, Robu, local distributors).
 - Memos live at `docs/research/<NN>-<topic>.md`, numbered to match task order.
-

Task 1: Engine identity

Files: - Create: docs/research/01-engine-identity.md

Interfaces: - Consumes: nothing. - Produces: engine make/model/displacement/cylinder count/rated speed/emissions tier, used by Tasks 4 and 6 and by the injector and aftertreatment scope decisions.

Closes: U11. Informs U8, U9.

▣ Step 1: Establish what "GK550" designates

The user reports the engine as "GK550". This does not match a Kirloskar engine designation. Determine whether it is a genset model, an engine model, a Kirloskar Green product code, or a distributor SKU.

Search: Kirloskar GK550 , Kirloskar Green GK 550 genset , KOEL GK550 25 kVA .

▣ Step 2: Work backwards from the manual number

The wiring diagram is 08-GP3-60-001 , file GP3.8703.C4.pdf , hosted at ikonnnect.kirloskar.com/KOEL/Services/TroubleShootingWiringDiagram . Determine what the GP3 family designates. Try fetching the parent index of that URL path for a manual list that maps families to engine models.

▣ Step 3: Cross-check against the hardware evidence

The wiring diagram independently constrains the engine. Any candidate must match all of: 3 cylinders (three injectors, cylinders 1/2/3); common-rail (rail pressure sensor plus fuel metering unit); EGR fitted; a catalyst temperature sensor present but no SCR or DEF; 12 V electrical system. Reject any candidate that fails one.

▣ Step 4: Determine the emissions tier

A 25 kVA genset in India with common-rail, EGR and a DOC but no SCR points to CPCB IV+. Confirm against CPCB IV+ requirements for the 19–56 kW band and record what that implies for aftertreatment scope.

▣ Step 5: Write the memo

Record: engine make, model, displacement, bore/stroke if available, cylinder count, aspiration, rated speed (expected 1500 rpm for 50 Hz), rated output, emissions tier, and the OEM ECU part number if discoverable. Mark each confirmed or inferred. If identity cannot be established, state exactly what photograph of the engine rating plate would close it.

▣ Step 6: Commit

```
git add docs/research/01-engine-identity.md
git commit -m "research: engine identity and emissions tier"
```

Task 2: KG640C genset controller

Files: - Create: docs/research/02-kg640c-gcu.md

Interfaces: - Consumes: Task 1's engine identity (the GCU is sold against engine families). - Produces: the J1939 PGN set the ECU must transmit and receive, and a yes/no on the MPU input that the spec's §3 safety gate turns on.

Closes: U3, U4. U3 is the safety gate — treat this task as the highest priority in the phase.

▣ Step 1: Locate the KG640C manual

Search: Kirloskar KG640C manual, KG640C genset controller datasheet, KOEL KG640C wiring. Also try the ikonnect.kirloskar.com path that served the wiring diagram, substituting the controller's manual number if a list is discoverable.

The KG640C may be a rebadged third-party controller — Deep Sea, ComAp, Datakom and SmartGen all OEM for Indian genset makers. If the manual is unavailable under the Kirloskar name, identify the underlying product and use its documentation, recording the identification basis.

▣ Step 2: Answer the MPU question explicitly

Determine whether the KG640C has a magnetic pickup input terminal. The spec's §3 gate depends on this and nothing else.

Three possible verdicts, all acceptable outputs: - **Yes** — record the terminal designation and the input's electrical spec. The design wires an MPU to it. - **No** — the design needs a standalone overspeed trip module in the panel. Identify two candidate modules with part numbers and prices. - **Cannot determine** — state what would: a photograph of the KG640C terminal strip, or the physical unit.

Do not infer from the wiring diagram's silence. The diagram shows terminals J1–J8 in use; an unused MPU terminal would not appear.

▣ Step 3: Extract the J1939 expectations

Determine which PGNs the controller expects from an engine node, and which it transmits. At minimum establish whether it uses: EEC1 (61444) for speed, ET1 (65262) for coolant temperature, EFL/P1 (65263) for oil pressure, VEP1 (65271) for battery voltage, DM1 (65226) for active faults, and TSC1 (0) for speed commands to the engine.

Record baud rate (expected 250 kbit/s) and whether it expects the engine at source address 0x00.

▣ Step 4: Record the digital I/O contract

From the wiring diagram, terminals J1/J2 carry D+CHG ALT , DIG I/P , DIG O/P , IGNITION RELAY , START RELAY . Establish which of these the ECU interacts with and in which direction, since start/stop sequencing is split between the GCU requesting and the ECU executing.

▣ Step 5: Write the memo

Lead with the U3 verdict — it is the reason this memo exists. Then the PGN tables, the digital I/O contract, and a list of what remains unknown.

▣ Step 6: Commit

```
git add docs/research/02-kg640c-gcu.md
git commit -m "research: KG640C controller, J1939 contract and MPU verdict"
```

Task 3: ECU connector identification

Files: - Create: docs/research/03-ecu-connector.md

Interfaces: - Consumes: Task 1's engine identity. - Produces: candidate connector part numbers and the mating-half sourcing that the adapter-harness deliverable (spec §7.1) depends on.

Closes: U1, partially — full closure needs the photograph.

▣ Step 1: Characterise the connector from the PDF

PDF page 4 shows the ECU connector with pins 1, 5, 28, 73 and 94 called out. Re-render that page region at high DPI and examine the housing: number of pin rows, whether it is a single housing or ganged blocks, the lever or bolt retention style.

```
pdftoppm -r 400 -f 4 -l 4 -png refs/GP3.8703.C4.pdf /tmp/p4
```

▣ Step 2: Match against standard 94-pin automotive ECU headers

94-pin is a common automotive ECU size. Check the usual families: TE Connectivity AMPSEAL, Bosch's own ECU headers (the Bosch EDC17/MD1 series uses 94-pin variants), Molex CMC, Aptiv/Delphi GT. Record which are 94-pin and how their housings compare to the PDF image.

▣ Step 3: Establish mating-half availability and price in India

For each candidate, find the harness-side connector, contacts and seals, with distributor stock and INR pricing. A connector we cannot buy is not a candidate.

▣ Step 4: Specify the photograph that closes U1

Write explicit instructions for what to photograph on the engine: the ECU connector housing from the mating face, any moulded part number or logo on the housing and on the ECU case, and the ECU's own label. Include what lighting and angle make moulded numbers legible.

▣ Step 5: Write the memo

Ranked candidate list with the evidence for each, sourcing table, and the photograph brief.

▣ Step 6: Commit

```
git add docs/research/03-ecu-connector.md
git commit -m "research: ECU connector candidates and sourcing"
```

Task 4: Injector identification and drive profile

Files: - Create: docs/research/04-injector-drive.md

Interfaces: - Consumes: Task 1's engine identity. - Produces: boost rail voltage, peak and hold current targets, and pulse timing — the inputs to the injector driver sheet and the boost converter design in Phase 2.

Closes: U5.

▣ Step 1: Identify the injector family

From the engine identity, determine the common-rail system supplier. Indian 3-cylinder gensets in this class typically use Bosch CRS or Denso. Record the injector part number if discoverable.

▣ Step 2: Establish the drive profile envelope

Solenoid common-rail injectors share a characteristic profile: a boosted opening pulse, then a lower hold current. Establish typical values for the identified family — boost rail voltage, peak current and its duration, hold current, and minimum controllable pulse width.

Where exact figures are unavailable, record the **envelope** across the family and design the hardware to cover it. A driver that spans 40–100 A/ms slew and 12–24 A peak will drive any of them; the calibration narrows it later.

▣ Step 3: Confirm the bank arrangement against firing order

The spec establishes bank A = cylinders 1+3, bank B = cylinder 2. Verify this is consistent with a 3-cylinder firing order (typically 1-3-2 or 1-2-3) — cylinders sharing a high-side bank must never need simultaneous injection. Record the firing order and show the check.

▣ Step 4: Select candidate driver ICs

Identify integrated common-rail injector drivers with integrated or companion boost control. Record for each: channel count, boost capability, current regulation method, whether it handles the high-side bank switching, package, availability in India, and price. Note explicitly whether each is available in small quantities or is automotive-tier allocation only.

▣ Step 5: State the calibration boundary plainly

Record what this hardware can and cannot do without injector characterisation data. The memo must say, unambiguously, that the driver can be built and bench-verified against a known load, and that making it run *this* engine correctly needs flow-vs-pulsewidth data that is not publicly available. Name the three routes to that data — supplier licence, dyno with rate-of-injection measurement, or OEM calibration extraction — with an honest note on the cost and legality of each.

▣ Step 6: Write the memo and commit

```
git add docs/research/04-injector-drive.md
git commit -m "research: injector identification, drive envelope and calibration boundary"
```

Task 5: MCU confirmation and availability

Files: - Create: docs/research/05-mcu-selection.md

Interfaces: - Consumes: the channel budget in spec §5 — 2 CAN, ~14 ADC, ~10 timer channels. - Produces: the confirmed MCU part number, package and KiCad symbol source for Phase 2.

Confirms: spec §5. Does not close a numbered unknown.

▣ Step 1: Verify the channel budget against the S32K148 datasheet

Fetch the S32K148 datasheet and confirm: FlexCAN instance count, ADC channel count and resolution, FlexTimer instances and channels per instance, LPI2C for the barometric sensor, flash and RAM. Record the specific package under consideration and its pin count.

▣ Step 2: Check real availability and price in India

Check Mouser India, DigiKey India and Element14 for S32K148 stock in the chosen package, in single-digit and hundred-unit quantities, with INR pricing and lead time. Repeat for the S32K146 fallback named in

the spec.

An MCU with a 40-week lead time fails this task regardless of how good the datasheet is. If both S32K parts fail on availability, evaluate alternatives against the same channel budget and recommend one — do not silently proceed with an unobtainable part.

▣ Step 3: Confirm KiCad symbol and footprint provenance

Determine whether a symbol and footprint exist in KiCad's stock libraries, in NXP's published EVB design files, or on SnapEDA/Ultra Librarian. Record the source and whether pad pitch was verified against the datasheet. Per the PCB Roadmap's warning, a hand-sourced footprint is the single most likely place to introduce a silent error.

▣ Step 4: Confirm the toolchain

Record: compiler (S32 Design Studio / GCC ARM), debug probe compatible with the chosen part and its price in India, and whether an evaluation board exists that would let firmware work start before the PCB is fabricated.

▣ Step 5: Write the memo and commit

State the verdict in the first line: confirmed, or changed with reasoning.

```
git add docs/research/05-mcu-selection.md
git commit -m "research: MCU confirmation, availability and toolchain"
```

Task 6: Standards and compliance scope

Files: - Create: docs/research/06-standards.md

Interfaces: - Consumes: Task 1's emissions tier. - Produces: the test and design requirements Phase 3 layout must honour, and an honest scope statement on certification.

▣ Step 1: SAE J1939 layers that apply

Summarise what each relevant part requires of this design: J1939-11 (twisted-pair physical layer, termination, stub lengths), J1939-21 (transport protocol, BAM vs RTS/CTS), J1939-71 (the PGN definitions Task 2 identified), J1939-73 (diagnostics, DM1/DM2), J1939-81 (address claiming).

▣ Step 2: ISO 7637-2 transient immunity

Record the pulse definitions that apply to a 12 V system — 1, 2a, 2b, 3a, 3b and load dump 5b — with their amplitudes and the severity levels typically specified. This sets the TVS clamp voltage and the buck's input rating.

▣ Step 3: CISPR 25 conducted emissions

Record the frequency bands and class limits. Note what bench equivalent (LISN plus spectrum analyser) gives useful indication short of a test house, and what it cannot tell you.

▣ Step 4: State the certification position honestly

This is a replacement controller on an emissions-certified engine. Record plainly what that means: modifying the engine controller on a CPCB-IV+ certified genset affects its certification status, and a replacement ECU is not covered by the original certificate. State the regulatory position for a genset in service in India and flag it as a decision for the project owners, not an engineering detail.

▣ Step 5: Write the memo and commit

```
git add docs/research/06-standards.md
git commit -m "research: J1939, ISO 7637-2, CISPR 25 and certification scope"
```

Task 7: Consolidated BOM and sourcing

Files: - Create: docs/research/07-bom-sourcing.md

Interfaces: - Consumes: Tasks 3, 4 and 5 — connector, injector driver and MCU are the three hard sourcing problems. - Produces: a costed BOM skeleton for Phase 2 capture, with every line either sourced or flagged.

▣ Step 1: Build the BOM skeleton from the spec

One line per functional block in spec §4 and §6: EMI filter, reverse-battery controller and P-FET, TVS, buck, sensor-rail LDO and PTCs, 3V3 LDO, supervisor, MCU, crystal, 2× CAN transceiver and protection, VR conditioner, EGR H-bridge, metering-unit low-side driver, injector driver and boost stage, relay-drive FETs, barometric sensor, connectors.

▣ Step 2: Price and check stock for every line

For each: manufacturer part number, distributor, stock quantity, INR unit price at qty 1 and qty 100, lead time. Prefer automotive-qualified (AEC-Q100/Q101) parts and record the qualification status per line.

▣ Step 3: Flag every single-source and long-lead line

Any line with one supplier, or a lead time beyond eight weeks, gets flagged with a named second source or an explicit "no alternative identified". These are the lines that stall a build.

▣ Step 4: Produce a headline cost

Total BOM cost at quantity 1 and quantity 10, excluding PCB fabrication and assembly. Note separately the estimated 4-layer PCB fab cost for the expected board area.

▣ Step 5: Write the memo and commit

```
git add docs/research/07-bom-sourcing.md
git commit -m "research: consolidated BOM with India sourcing and cost"
```

Task 8: Engine inspection brief

Files: - Create: docs/research/08-engine-inspection-brief.md

Interfaces: - Consumes: Tasks 1, 3 and 4 — each contributes an item that only physical access resolves. - Produces: a single checklist someone takes to the machine, closing six unknowns in one visit.

Closes: U2, U6, U7, U8, U9, U10 — none of which desk research can close.

▣ Step 1: Collect every unknown that needs the physical engine

From spec §8: U2 (boost air-temp topology), U6 (pin 09 synchronisation ground), U7 (whether pins 04/06 are sense-only or load feeds), U8 (6-pin intake throttle — on the ECU at all?), U9 (catalyst temperature sensor routing), U10 (which signal uses SENT). Add U1's connector photograph from Task 3 and U11's rating plate from Task 1.

▣ Step 2: Write the photograph list

For each: what to photograph, from what angle, and what detail must be legible. Engine rating plate. ECU case label and connector mating face. Boost P&T sensor body — the part number on it settles U2 outright. Injector body. Intake throttle connector and its cable run. Catalyst temperature sensor and where its harness goes. KG640C terminal strip.

▣ Step 3: Write the measurement list

Measurements need the harness live or disconnected — say which for each, and give the expected reading so an anomaly is recognisable on the spot.

For U2: resistance across boost sensor pins 2 and 1 with the connector unplugged and the engine cold. A few kΩ that falls as it warms means NTC; an open or a fixed high reading suggests a conditioned ratiometric output. This single measurement closes U2.

For U7: voltage on ECU pins 04 and 06 with ignition on, then current draw if it can be measured safely at the fuses. Sense-only pins draw milliamps; load feeds draw amps.

For U6: continuity from pin 09 to battery negative and to sensor ground, engine off, battery disconnected.

▣ Step 4: Write the safety preamble

The brief is for someone working on a genset. Lead with: battery disconnected before any continuity measurement; no measurement inside the alternator output enclosure; the engine does not run during any of this. State that if the set must run for a measurement, that is a separate supervised activity and not part of this checklist.

▣ Step 5: Write the memo and commit

Format it as a checklist that works printed and carried, not as a report.

```
git add docs/research/08-engine-inspection-brief.md
git commit -m "research: engine inspection brief closing six physical unknowns"
```

Phase exit criteria

Phase 1 is complete when:

- ▣ U3 has a verdict. The safety gate in spec §3 is either satisfiable or has a named trip module.
- ▣ U5 has a drive envelope sufficient to design the injector stage, and the calibration boundary is stated in writing.
- ▣ U11 is closed, or the rating-plate photograph is specified and requested.
- ▣ U1 has a ranked candidate list and a photograph brief.
- ▣ The MCU is confirmed or replaced, with stock verified.
- ▣ The engine inspection brief exists and has been handed to whoever has access to the machine.
- ▣ Every unknown that remains open is recorded in an updated spec §8 with what would close it.

Final step: update spec §8 in place so the register reflects reality, and commit that edit with the phase.

What this plan does not cover

Phases 2, 3 and 4 get their own plans, written once Phase 1 closes the unknowns they depend on:

- **Phase 2 — schematic capture.** Can begin on power, CAN, MCU support, speed inputs and relay outputs before Phase 1 finishes. Blocked on U1 for connectors, U2 for the boost air-temp front-end, U5 for the injector stage, U8 for the throttle.

- **Phase 3 — PCB layout.** Blocked on Phase 2.
- **Phase 4 — firmware skeleton.** Unblocked by Phase 1 except for the J1939 layer, which needs Task 2. Could run in parallel.